



Mapping the Invisible City: MetaSUB and the Science of Urban Microbiomes

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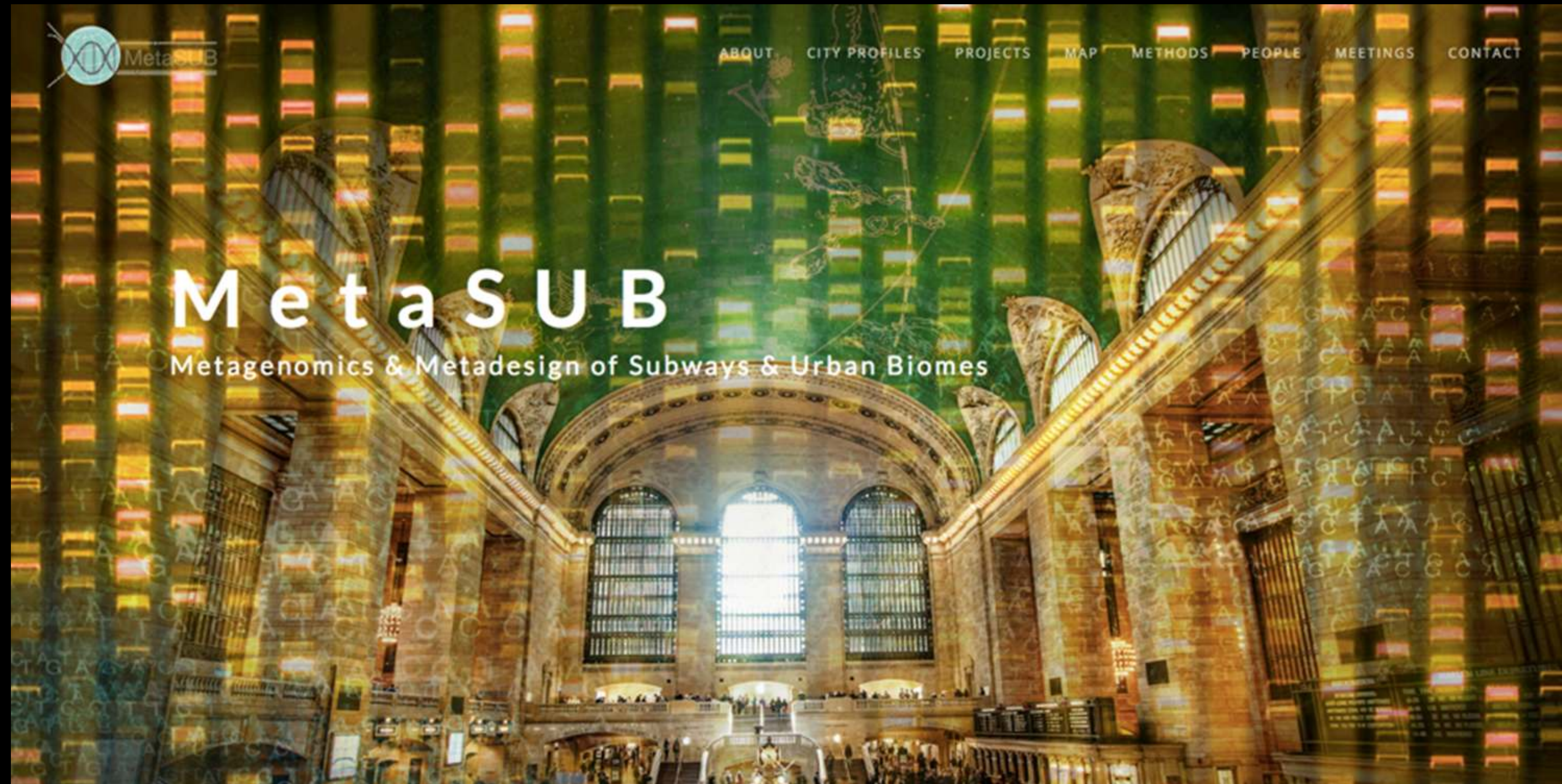
Co-Chair, Bioinformatics Working Group,
NIH Human Virome Program

Executive Director, and Director of Bioinformatics,
The International MetaSUB Consortium



MetaSUB: Global Urban Microbiome Consortium

MetaSUB is an international research consortium mapping the microbial ecosystems of urban transportation systems to understand how urbanization shapes microbial diversity, public health, and planetary biology.



Dr. Christopher E. Mason



MetaSUB: Building a Global Network to Map Urban Microbial Life

*Founded in 2015, the **MetaSUB Consortium** created a worldwide, interdisciplinary network to sample and sequence microbes from transit systems, sewage, hospitals, and other urban environments across more than 100 countries.*

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BACKSTORY · [Volume 25, Issue 11](#), 104993, November 18, 2022 · [Open Access](#)

A history of the MetaSUB consortium: Tracking urban microbes around the globe

[Krista A. Ryon](#)^{4,5} · [Braden T. Tierney](#)^{4,5} · [Alina Frolova](#)^{1,2} · [Andre Kahles](#)^{3,35} · [Christelle Desnues](#)⁶ · [Christos Ouzounis](#)⁹ · [Cynthia Gibas](#)¹⁰ · [Daniela Bezdán](#)^{11,12,13} · [Youping Deng](#)³⁶ · [Ding He](#)¹⁴ · [Emmanuel Dias-Neto](#)¹⁵ · [Eran Elhaik](#)¹⁶ · [Evan Afshin](#)⁷ · [George Grills](#)¹⁷ · [Gregorio Iraola](#)¹⁸ · [Haruo Suzuki](#)¹⁹ · [Johannes Werner](#)²⁰ · [Klas Udekwu](#)²¹ · [Lynn Schriml](#)²² · [Malay Bhattacharyya](#)²³ · [Manuela Oliveira](#)²⁴ · [Maria Mercedes Zambrano](#)²⁵ · [Nur Hazlin Hazrin-Chong](#)²⁶ · [Olayinka Osuolale](#)²⁷ · [Paweł P. Łabaj](#)²⁸ · [Prisca Tiasse](#)²⁹ · [Sampath Rapuri](#)³⁰ · [Silvia Borrás](#)³¹ · [Sofya Pozdniakova](#)³¹ · [Tielu Shi](#)³² · [Ugur Sezerman](#)³³ · [Xavier Rodo](#)³¹ · [Zehra Hazal Sezer](#)³⁴ · [Christopher E. Mason](#)^{7,8}  [Show less](#)

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Show Outline

The MetaSUB Consortium, founded in 2015, is a global consortium with an interdisciplinary team of clinicians, scientists, bioinformaticians, engineers, and designers, with members from more than 100 countries across the globe. This network has continually collected samples from urban and rural sites including subways and transit systems, sewage systems, hospitals, and other environmental sampling. These collections have been ongoing since 2015 and have continued when possible, even throughout the COVID-19 pandemic. The consortium has optimized their workflow for the collection, isolation, and sequencing of DNA and RNA collected from these various sites and processing them for metagenomics analysis, including the identification of SARS-CoV-2 and its variants. Here, the Consortium describes its foundations, and its ongoing work to expand on this network and to focus its scope on the mapping, annotation, and prediction of emerging pathogens, mapping microbial evolution and antibiotic resistance, and the discovery of novel organisms and biosynthetic gene clusters.



Every surface you touched today had microbes on it



You're surrounded by a living, invisible ecosystem of microbes



Cities aren't just concrete and steel, they're biological systems

Outline



Presence

Subways host a living mix of microbes from people, air, soil, and the city itself.



Detection

We use DNA to reveal this invisible ecosystem without sampling people



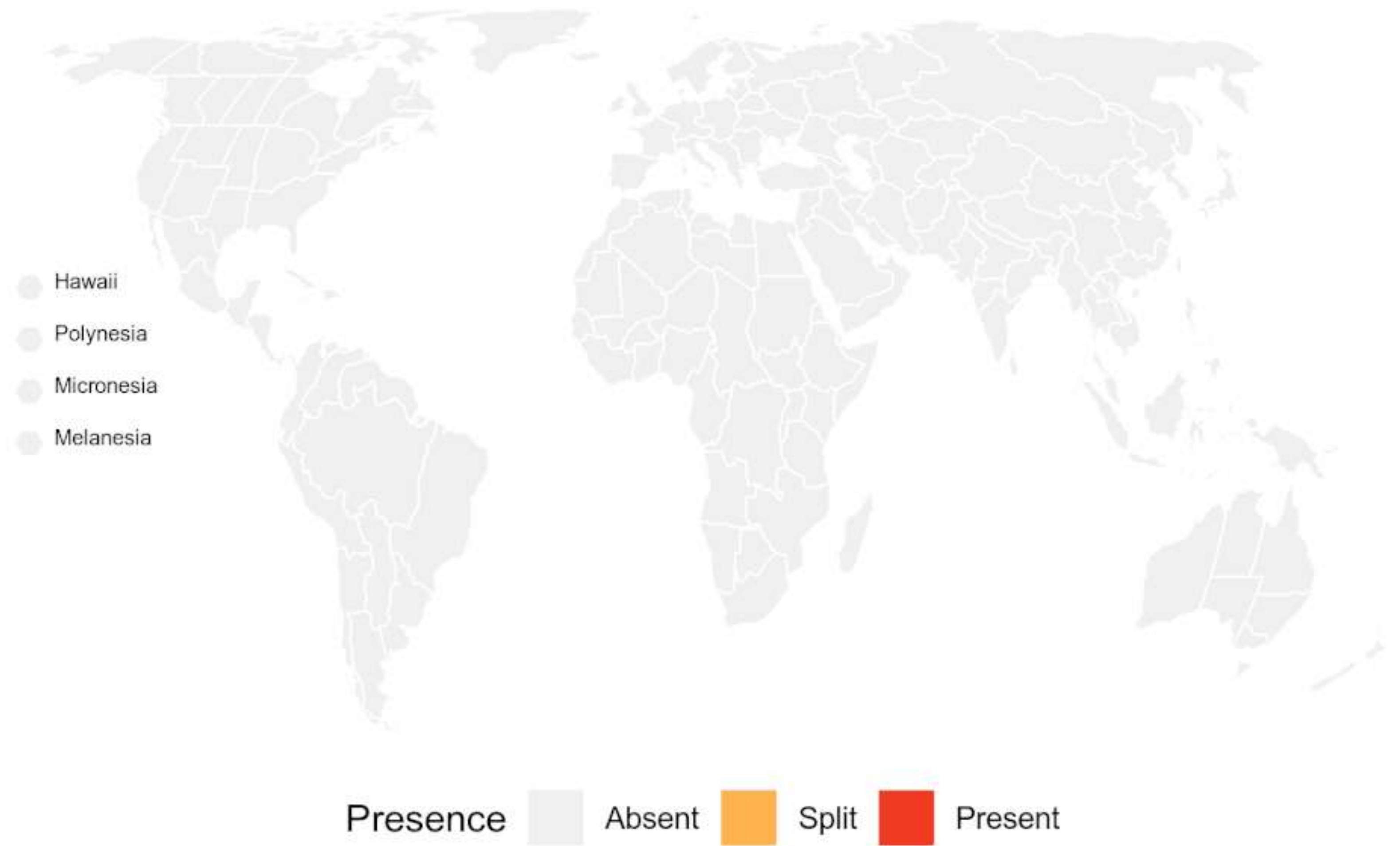
Impact

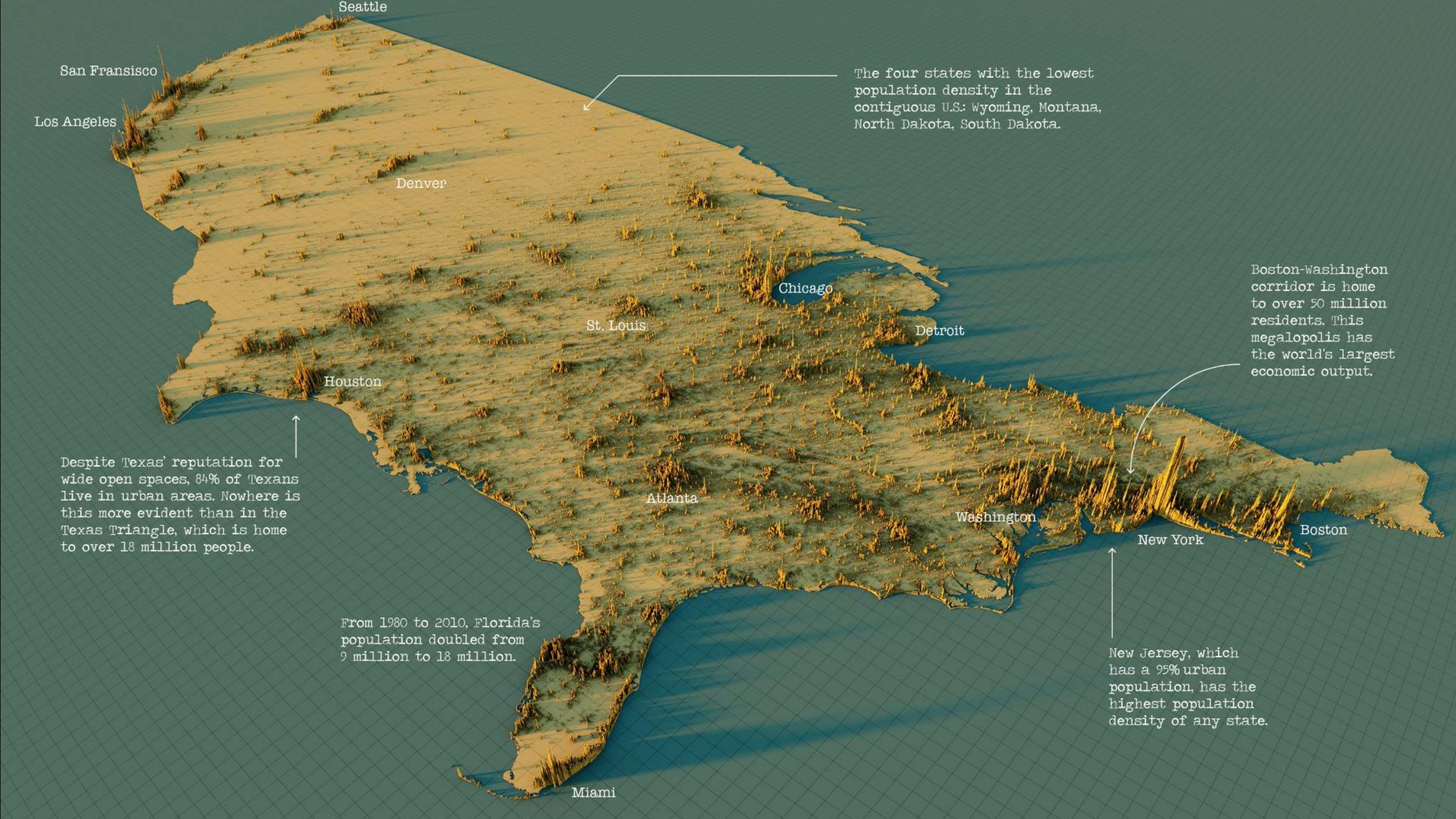
Microbial signals help us monitor health risks and design safer, more resilient cities

The Rise of Global Urban Centers (over 10,000 years)

Human settlements gradually grew into major urban centers across the world over the last 10,000 years, revealing the long-term spread of cities through agriculture, trade, empire, and industrialization.

Urban Centers
10000 years ago





San Fransisco

Los Angeles

Denver

Chicago

St. Louis

Detroit

Houston

Atlanta

Washington

New York

Boston

Miami

The four states with the lowest population density in the contiguous U.S.: Wyoming, Montana, North Dakota, South Dakota.

Boston-Washington corridor is home to over 50 million residents. This megalopolis has the world's largest economic output.

Despite Texas' reputation for wide open spaces, 84% of Texans live in urban areas. Nowhere is this more evident than in the Texas Triangle, which is home to over 18 million people.

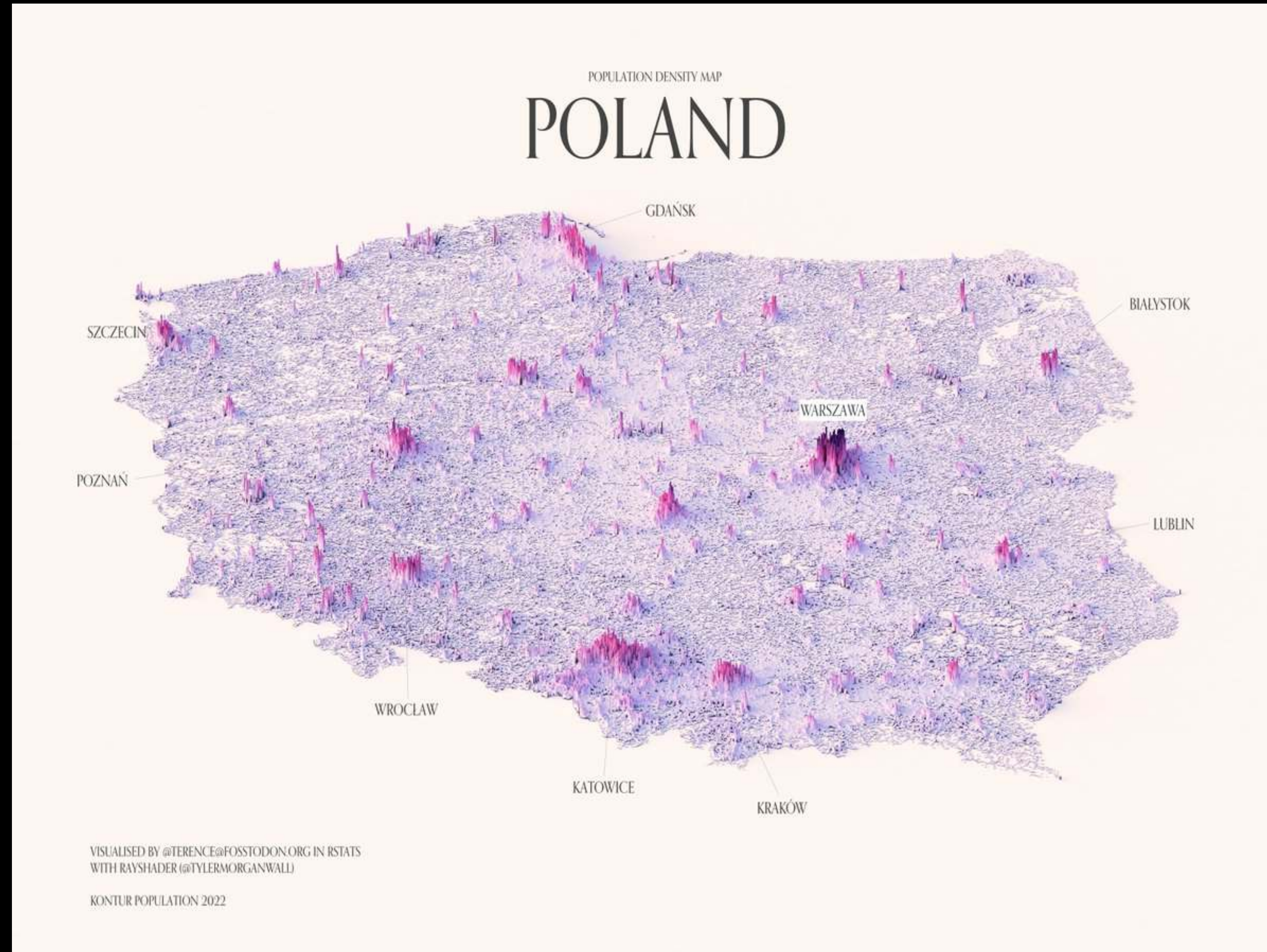
From 1980 to 2010, Florida's population doubled from 9 million to 18 million.

New Jersey, which has a 95% urban population, has the highest population density of any state.

Poland's Population Distribution (2022)

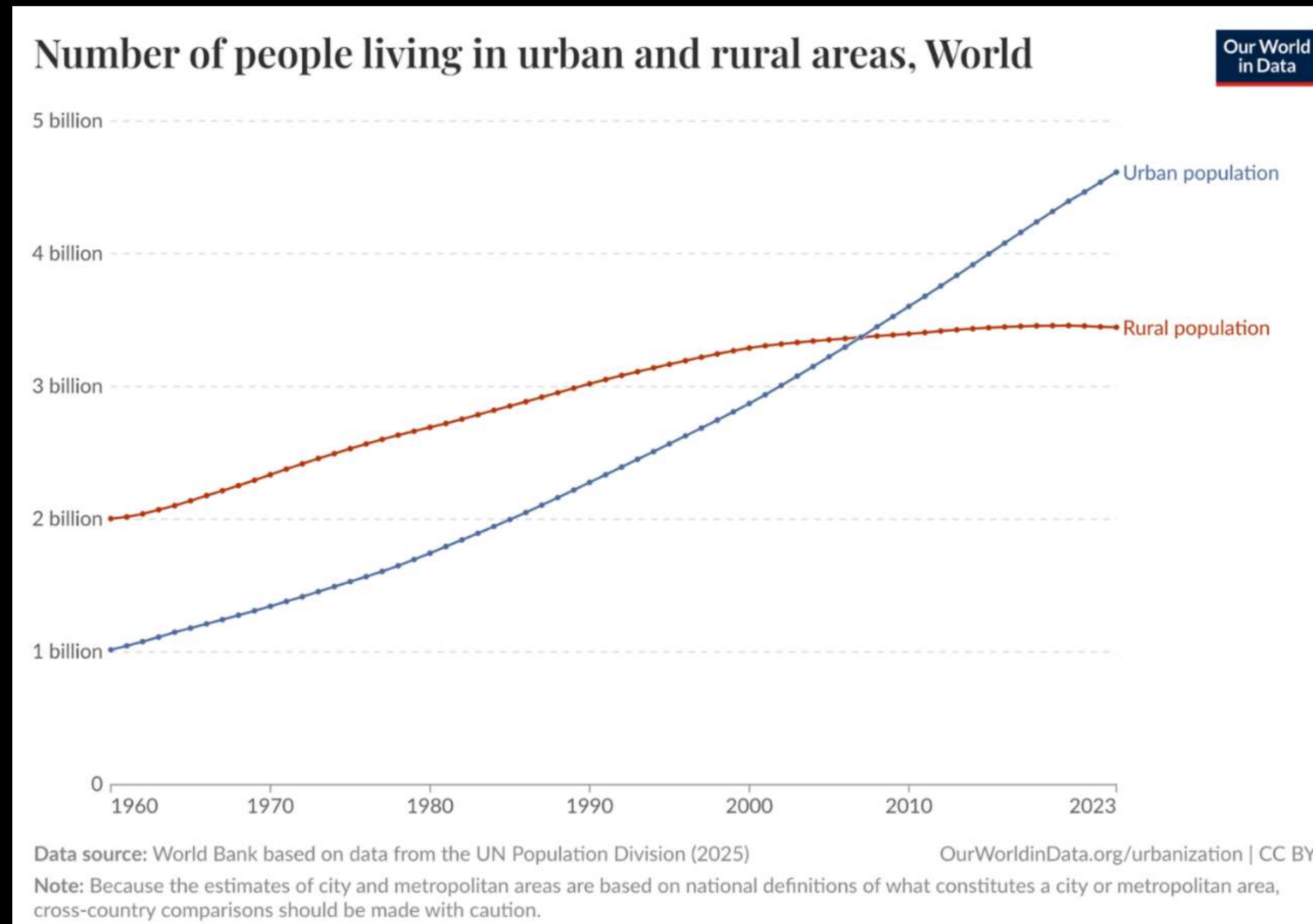
Population density is concentrated around major urban and industrial regions, especially Warsaw, Kraków, Łódź, Wrocław, Poznań, Gdańsk, and the Silesian metropolitan area in the south.

<https://creapills.com/cartographie-densite-population-rayshader-20221230>



Global Urban and Rural Population Over Time (1960–Today)

The world's urban population has grown steadily since 1960 and surpassed the rural population in the modern era, reflecting the global shift toward urbanization.

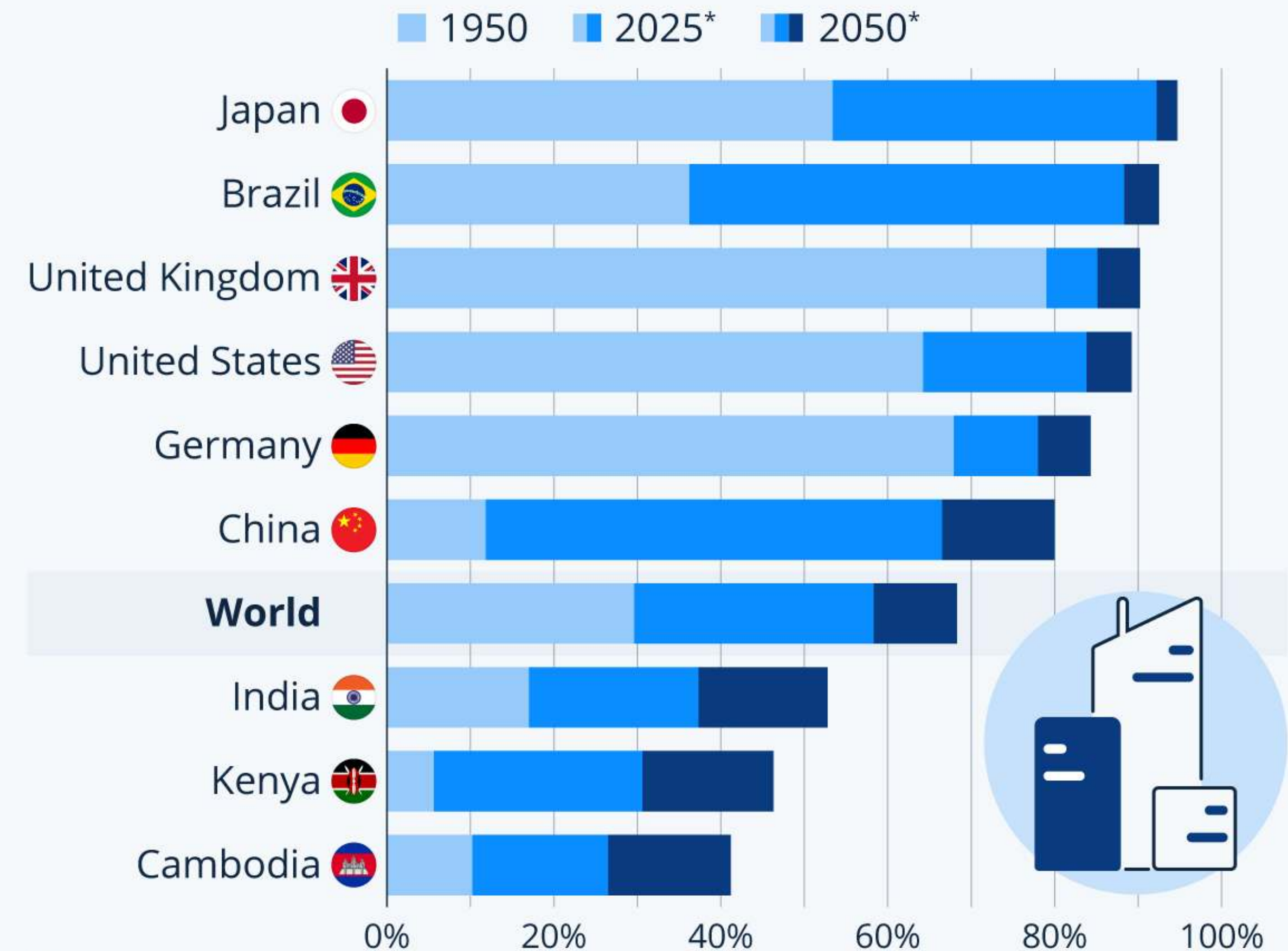


Trends in Urbanization

Accelerating population growth in cities worldwide, driven by economic opportunity, technological connectivity, and climate pressures, is reshaping infrastructure, public health, and ecological systems at unprecedented scales.

7 in 10 People to Live in Urban Areas by 2050

Estimated share of population residing in urban areas at mid-year in 1950, 2025 & 2050 in selected countries



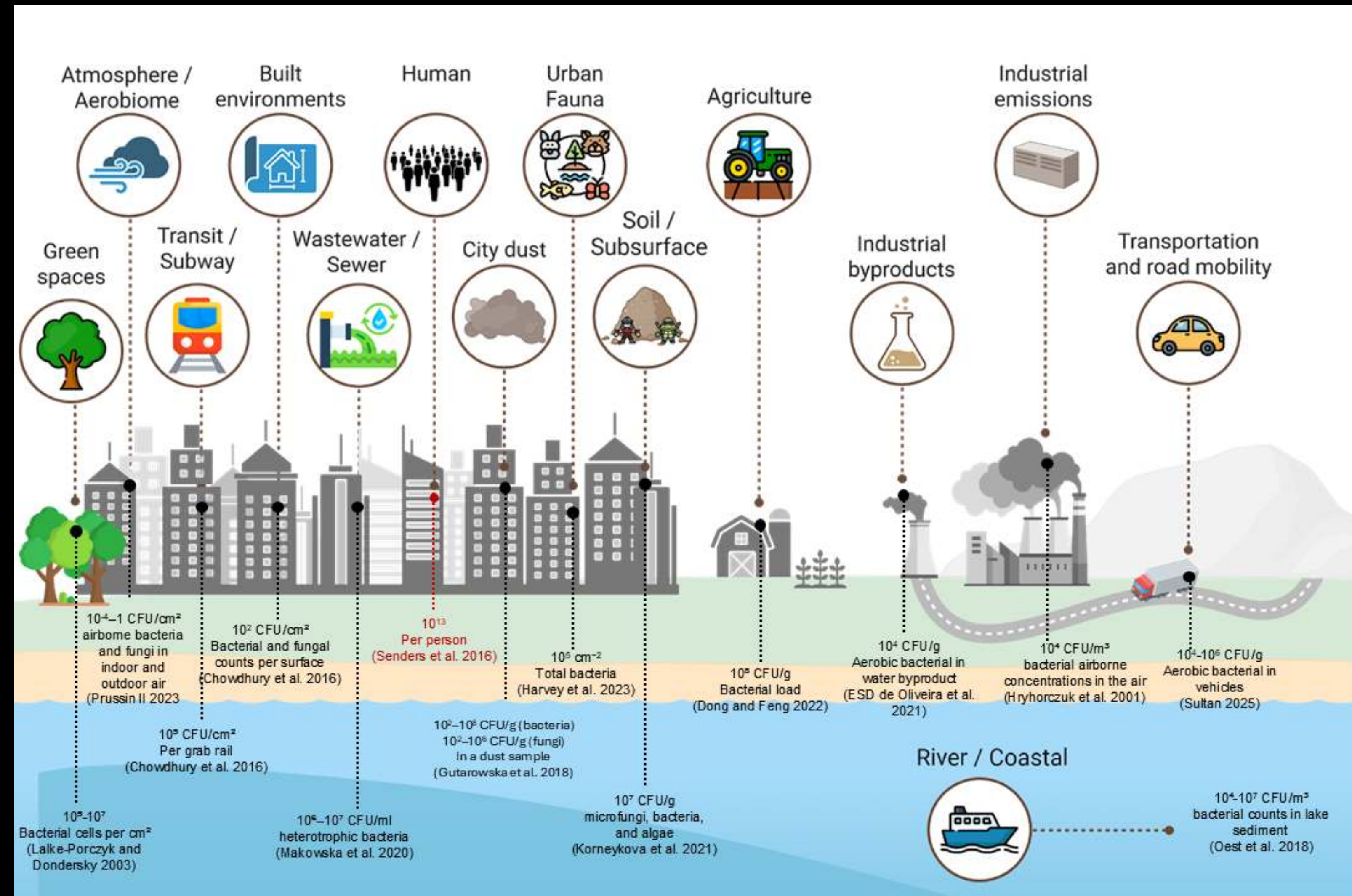
* Projection

Source: UN Population Division



Determinants of Urban Microbial Ecology

The urban microbiome reflects interactions between people, infrastructure, climate, and the environment.



Passenger Volume Across U.S. Subway systems

New York City Subway: ~2.04 billion trips in 2024, making it by far the busiest rapid transit system in the United States.

Washington Metro Metrorail: ~166.7 million trips in 2024 across the Washington, DC metropolitan area.

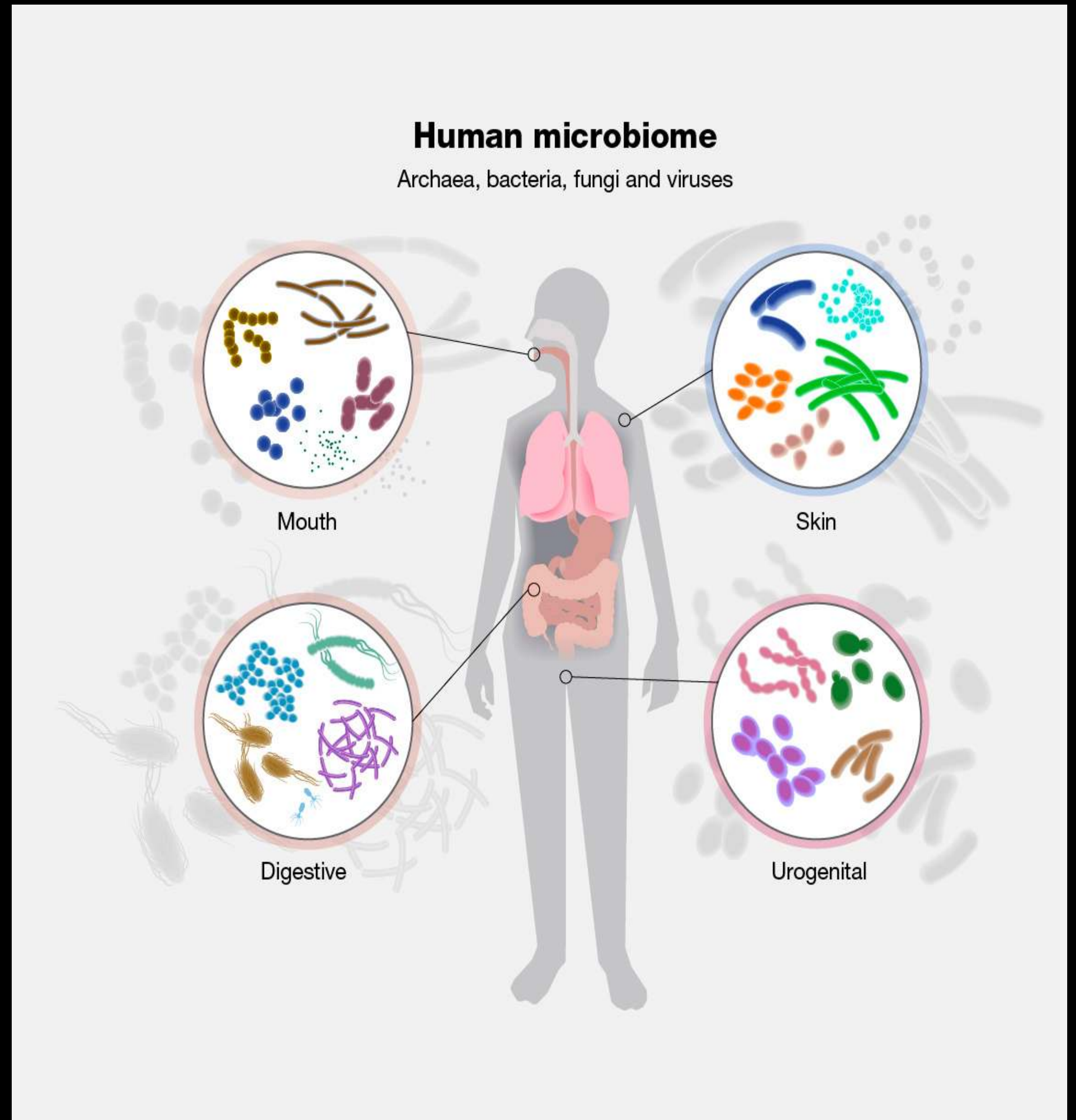
Chicago “L”: ~127.5 million trips in 2024 on its heavy rail network.



What is a microbiome?

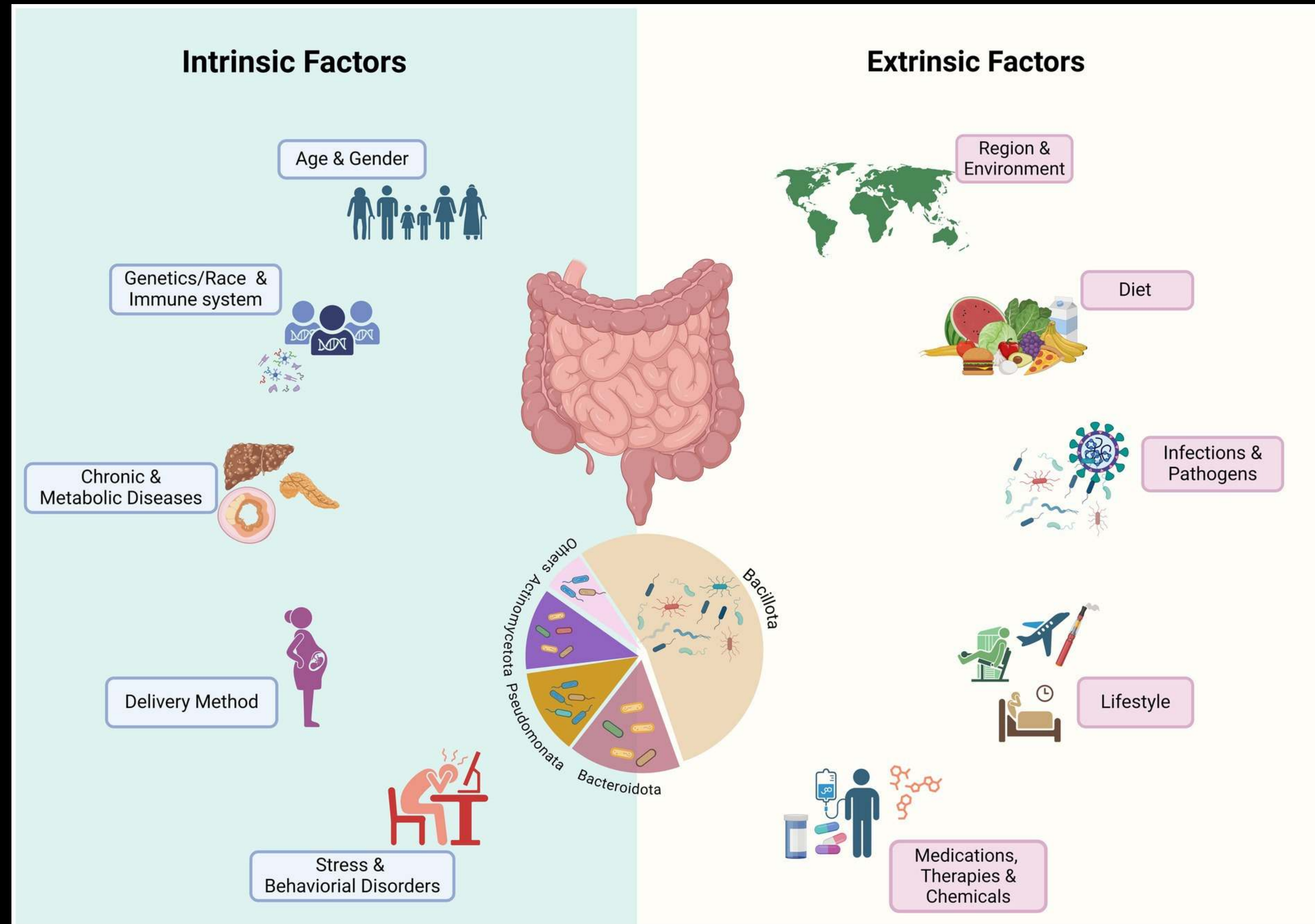
What is a microbiome?

*A **microbiome** is the collection of all microbes, including bacteria, fungi, viruses, and their genes, that inhabit an environment.*



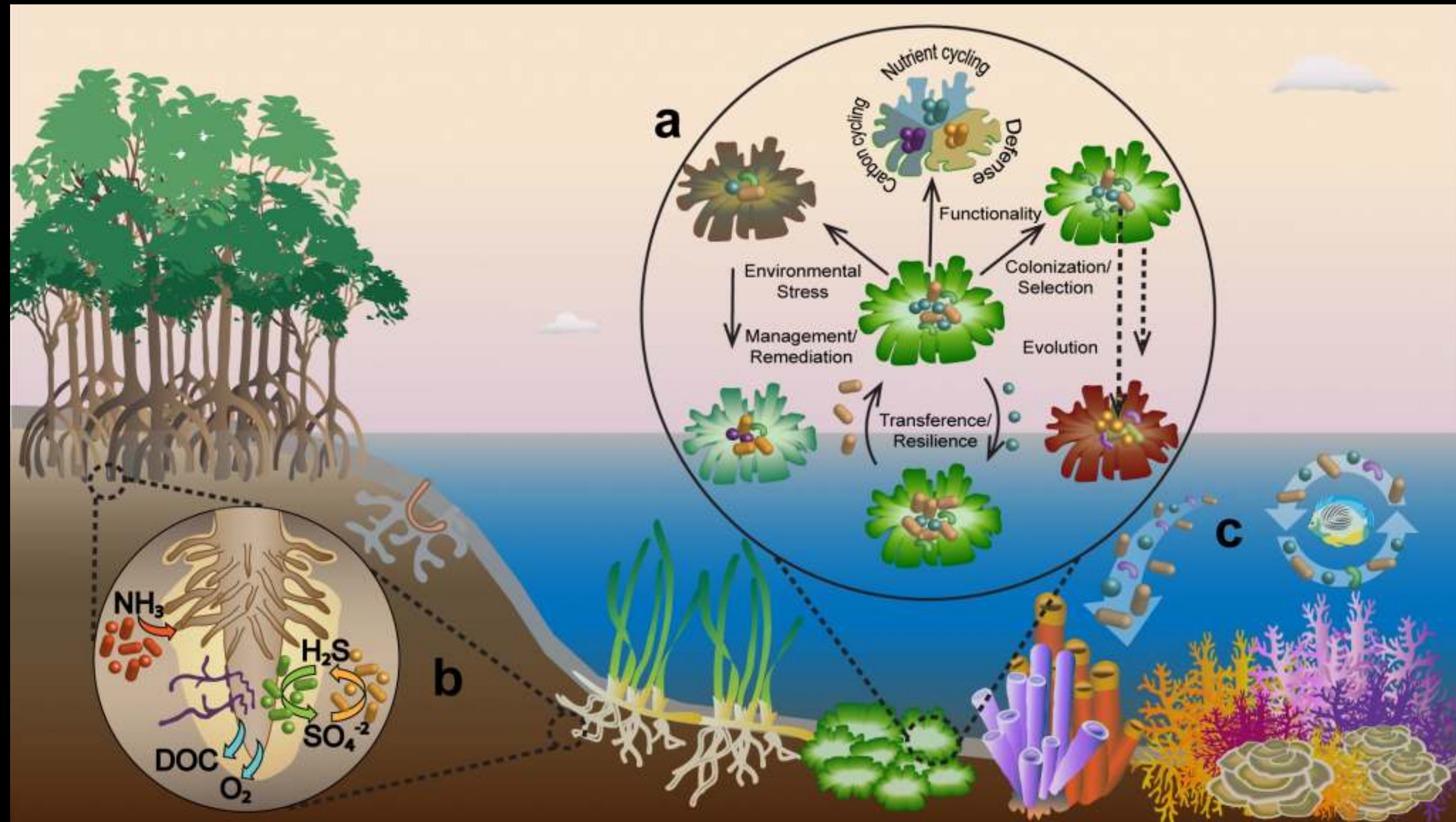
What influences the microbiome?

Dysbiosis is an imbalance of microorganisms in your body, which can affect your health and well-being.



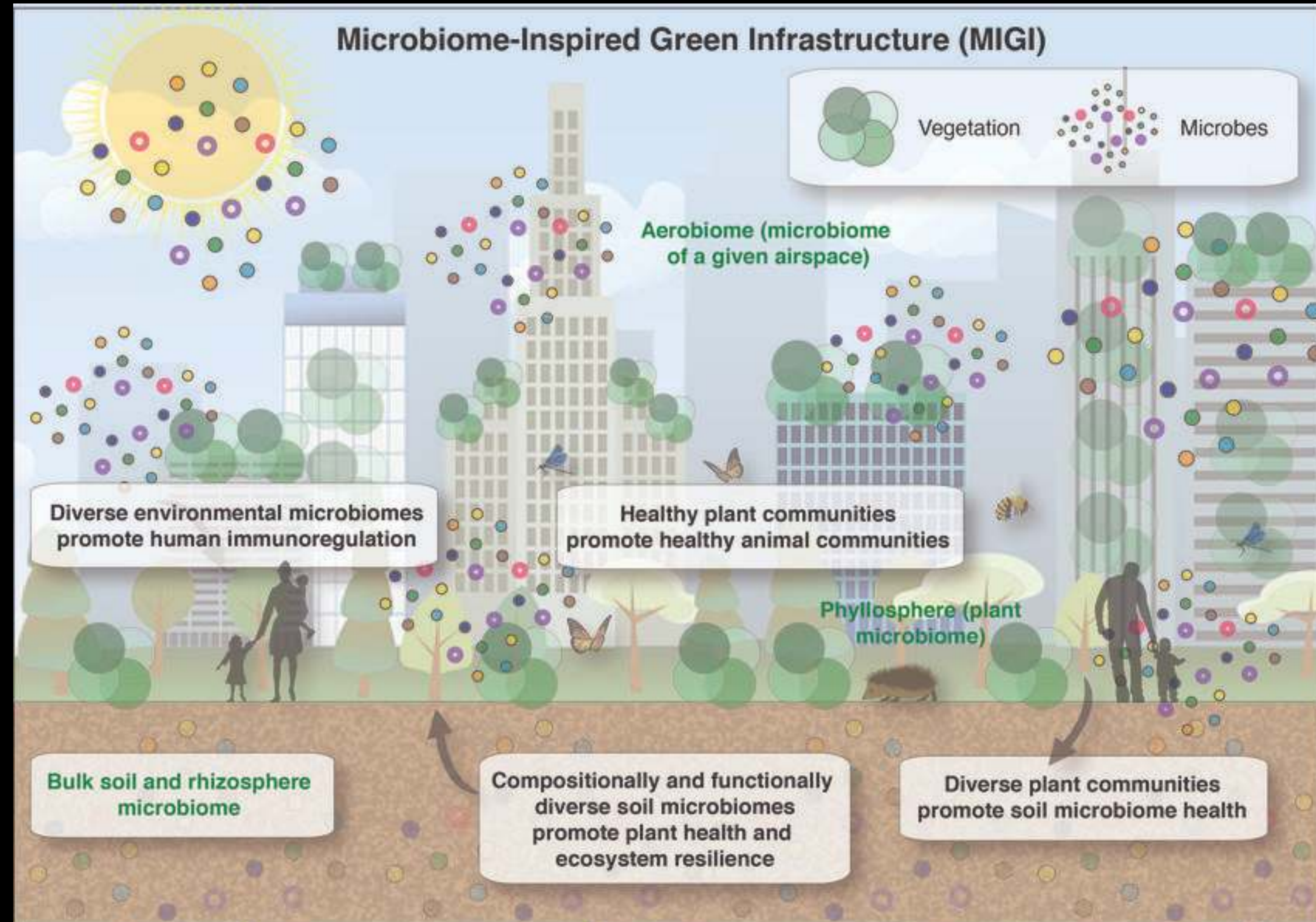
The life around us

The environmental microbiome shapes the air, surfaces, and microbes we encounter.



Urban multispecies health

Environmental microbiomes are the foundations of our ecosystems, and are essential to plant and animal health (including humans)



**Do cities have a
microbial signature?**

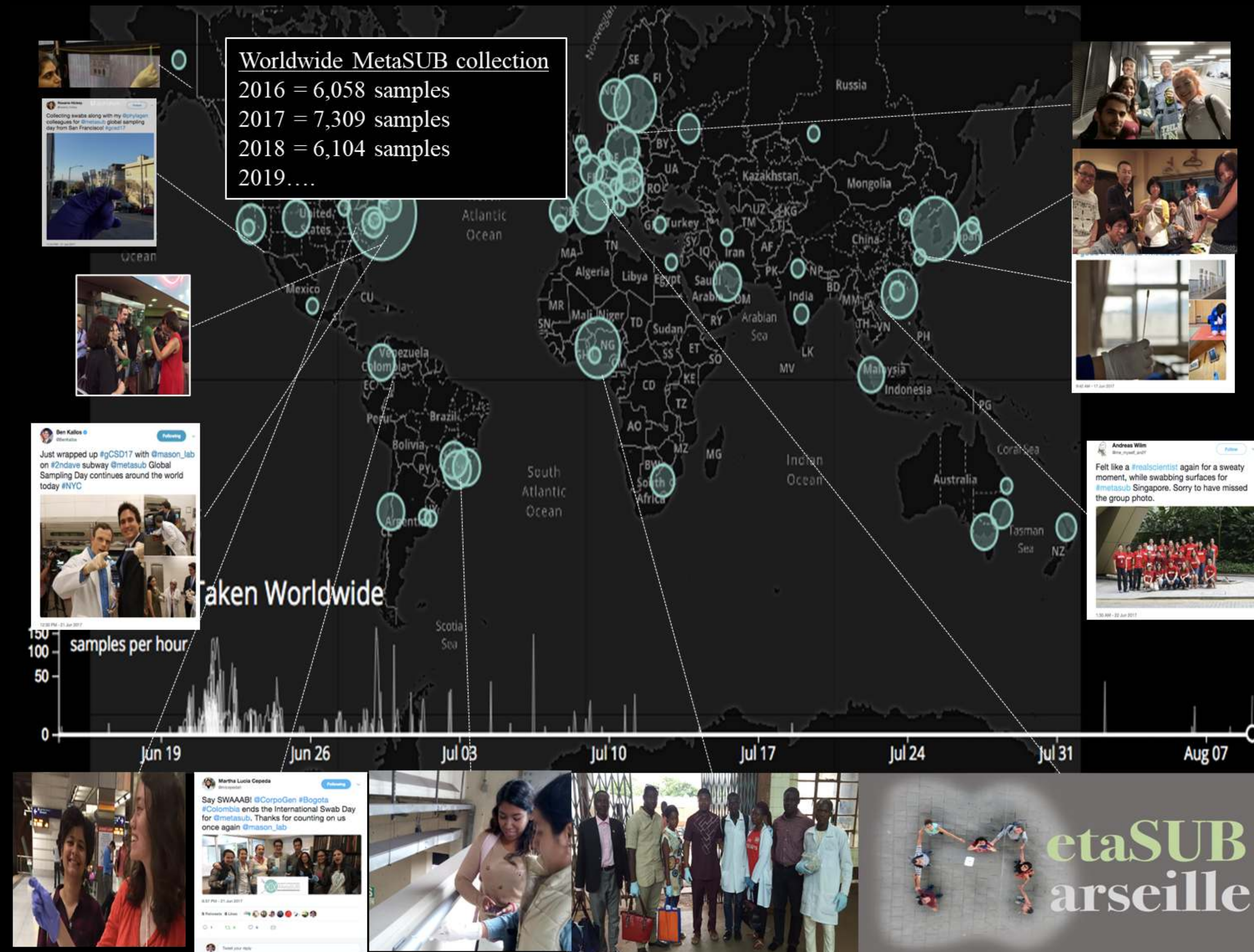


Global City Sampling Day (gCSD)

<http://metasub.org/>

gCSD: Global City Sampling Day

From 2016–2026, MetaSUB conducted coordinated global city sampling to systematically characterize urban microbial communities across major metropolitan transit systems worldwide



-  Dashboard
-  Organizations
-  Projects
-  Public apps
-  Catalogs

[MetaSUB Consortium](#) > [gCSD 2025](#)

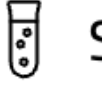
OrganizationProject

 Project name
gCSD 2025 GRN

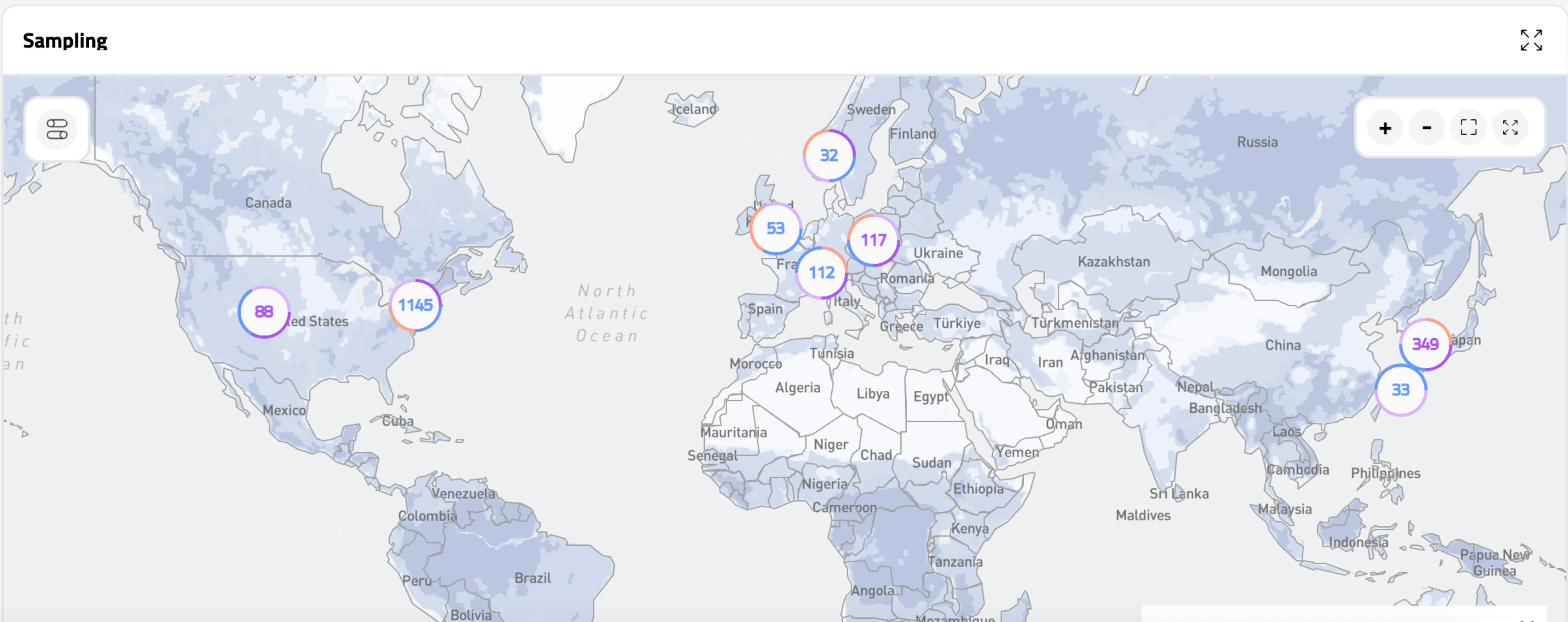
 Privacy level
Public

 Samples
1929

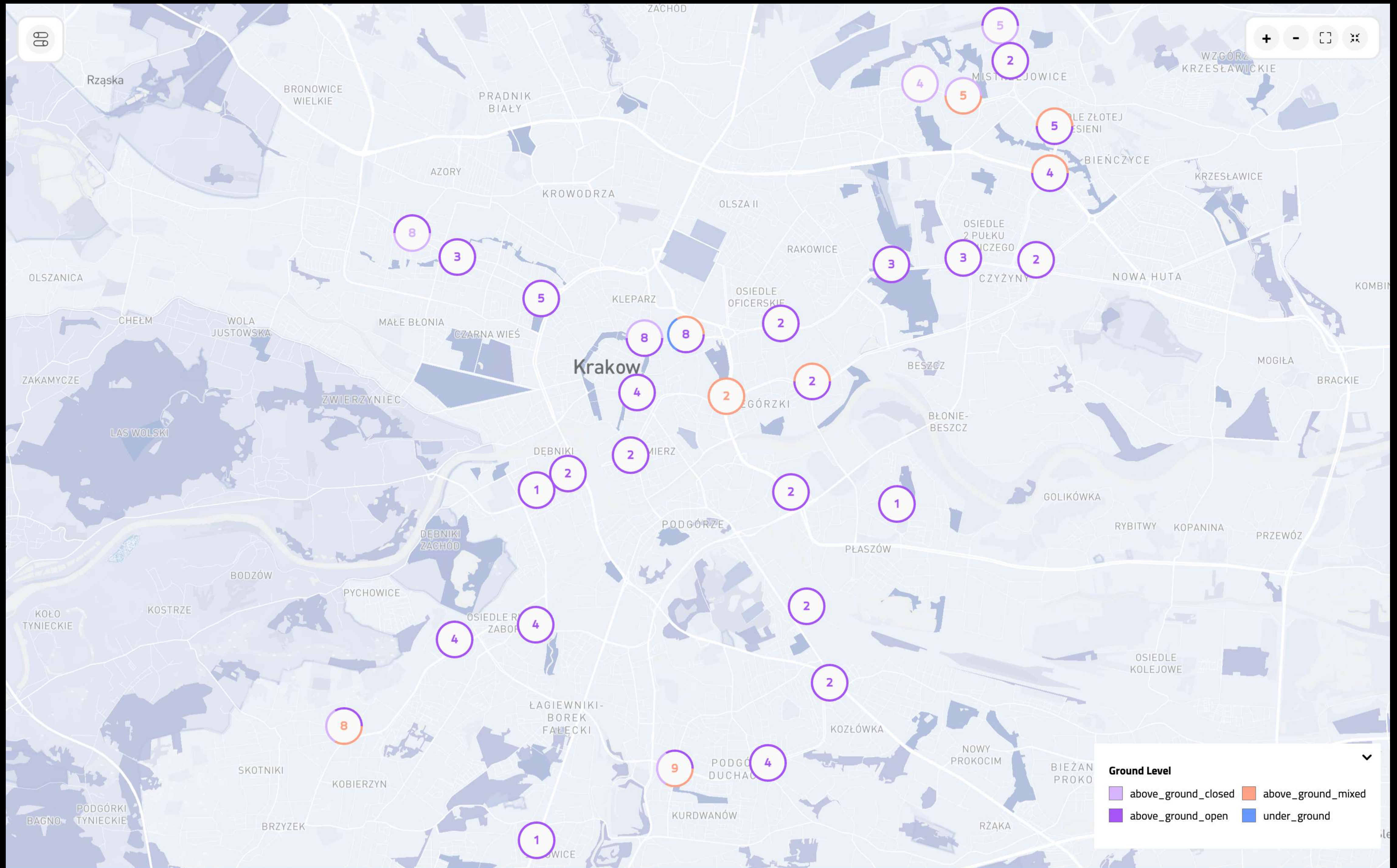
 Global City Sampling Day, June 20 2025

-  Dashboard
-  Samples
-  Project folders
-  Apps
-  Runs
-  Documentation

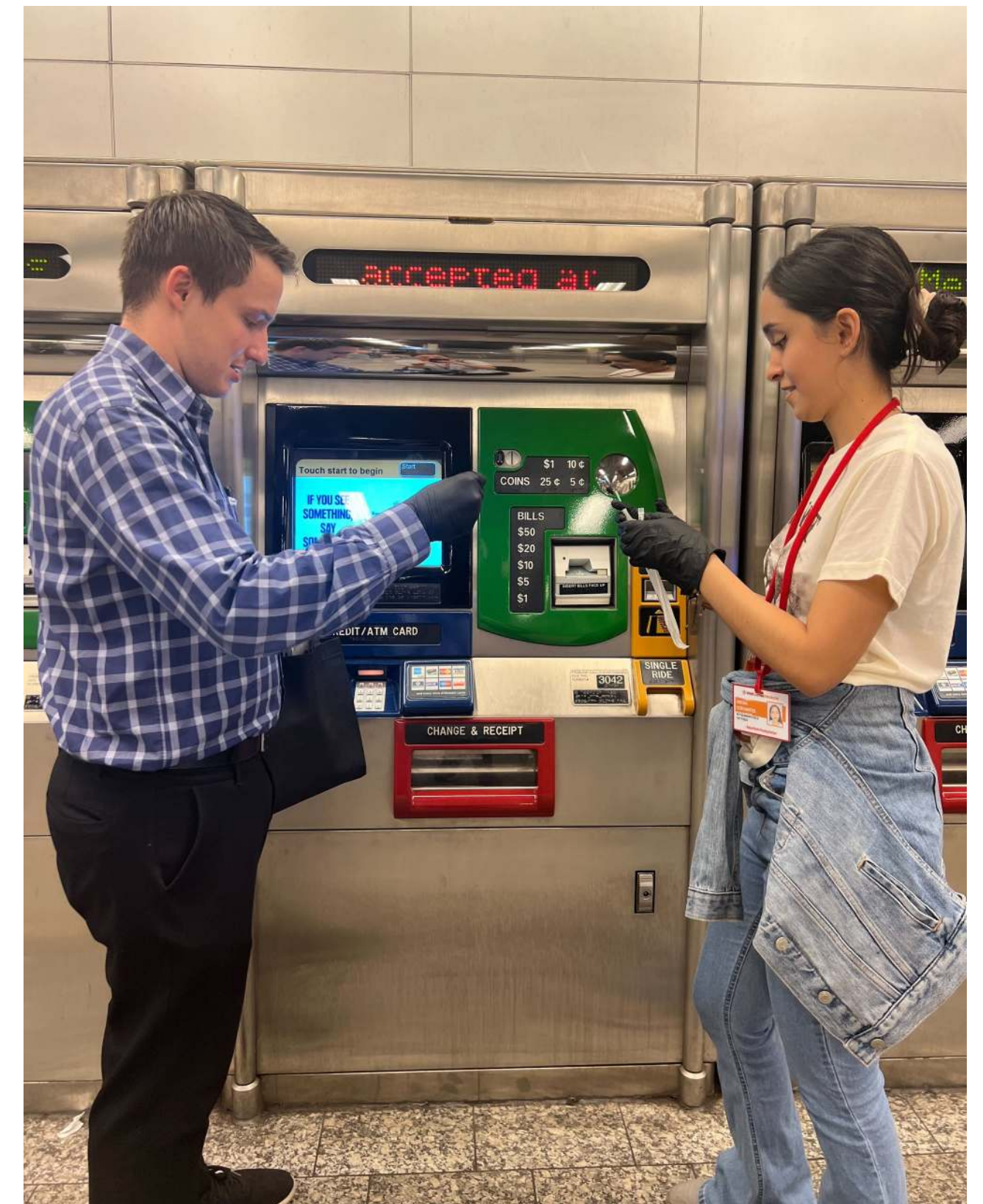
Default dashboard



-  Tutorial
-  Support



Collection Team Assignments



Overview of Global City Sampling Day (gCSD)



Metadata Collection

City
Country
Barcode (Sample ID)
Photo of barcode
Sample type
Location type
Location name
Station line (if at a station)
Sublocation
Setting
Ground level
Object type / Sampling place
Surface material
Traffic
Temperature & Humidity
Geolocation
Timepoint & Date
Additional information (weather etc.)

***Bold** is required information

- **Collection of metadata is critical to sample processing and analysis**
- **Dr. Lucaci will go over KoboToolBox next**



Metadata Collection: KoBoToolbox



KoBoToolbox is a free open-source tool for mobile data collection, available to all. The platform is:

- Free to use
- Compatible with most computer browsers
- Open source



BARCODE NUMBER
Enter your barcode information located on the side of the sample tube under the barcode. This will be a 10 digit code.

If a code is entered that is more or less than 10 digits you will be unable to submit the survey,



*Enter the barcode number listed on the tube:
This should be a 10 number code starting with the number zero. Example: 0236537381

*Select sampling place

none selected

- ☐ Door
- ☐ Escalator handrail
- ☐ Floor
- ☐ Garbage can
- ☐ Ground
- ☐ Handrail

ee.kobotoolbox.org

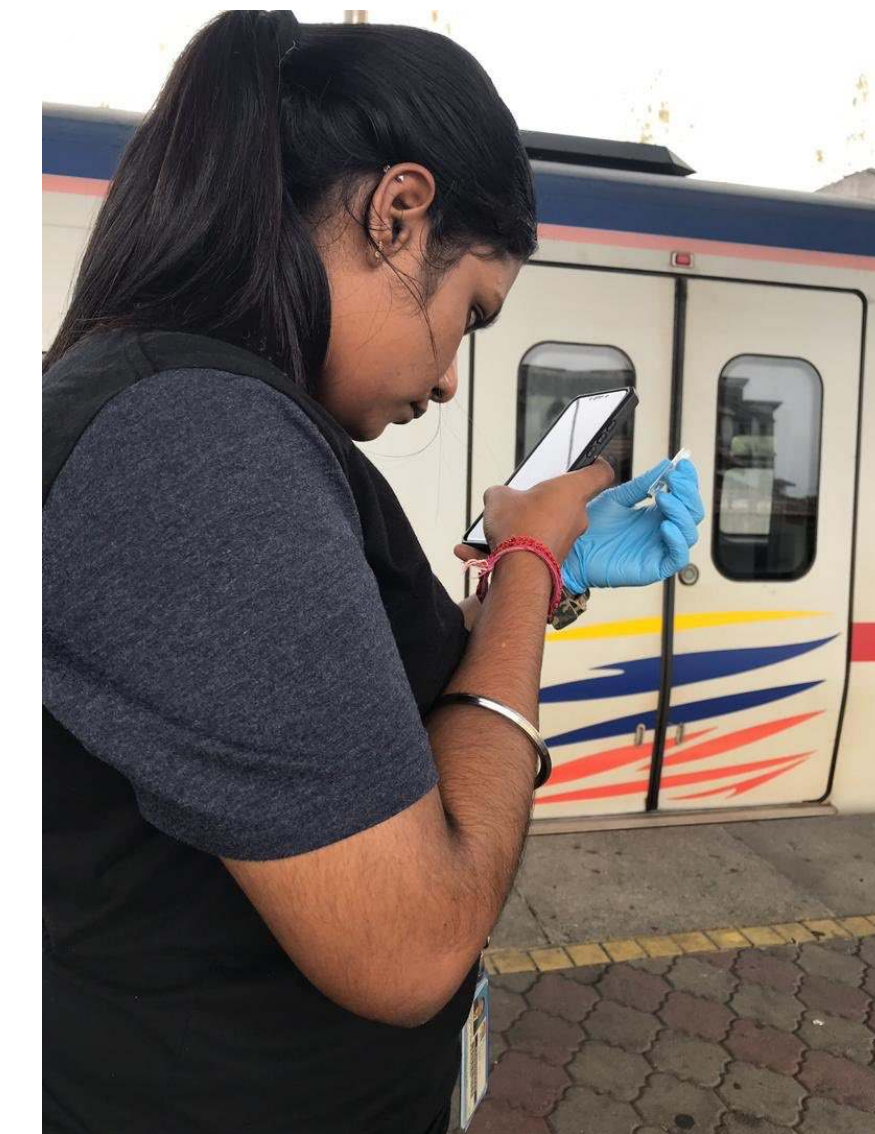
Select geolocation
Use the button to calculate the latitude and longitude of your sampling location:

latitude (x.y °)

longitude (x.y °)

altitude (m)

accuracy (m)



Register for KoBoToolbox!

We will confirm that we have your username and metadata plan!

Metadata Collection: KoBoToolbox



MetaSUB: Metadata protocol (surface sampling)

In this guide, you will find an explanation of the of the online data collection tool KoBoToobox (<https://kf.kobotoolbox.org>) used for data collection by the MetaSUB Consortium.

- I. METADATA.....2
- II. KOBOTOOLBOX.....2
 - i. Collection Survey
 - ii. Registration & Logging in
 - iii. Accessing the project
 - iv. Sharing the project
 - v. Submitting entries
 - i. Offline entries/Draft
 - vi. Following data collection
- III. DATA COLLECTION (BY HAND/EXCEL)9|
- IV. COLLECTION SURVEY OUTLINE.....10
- V. CONTACT.....18

REGISTRATION & LOGGING IN

- Registration form: <https://kf.kobotoolbox.org/accounts/register/>

Follow the directions to register. Use small case for login username only. KoBoToolbox will then send you a **confirmation link** through your email account, which you **must follow** in order to complete the registration.

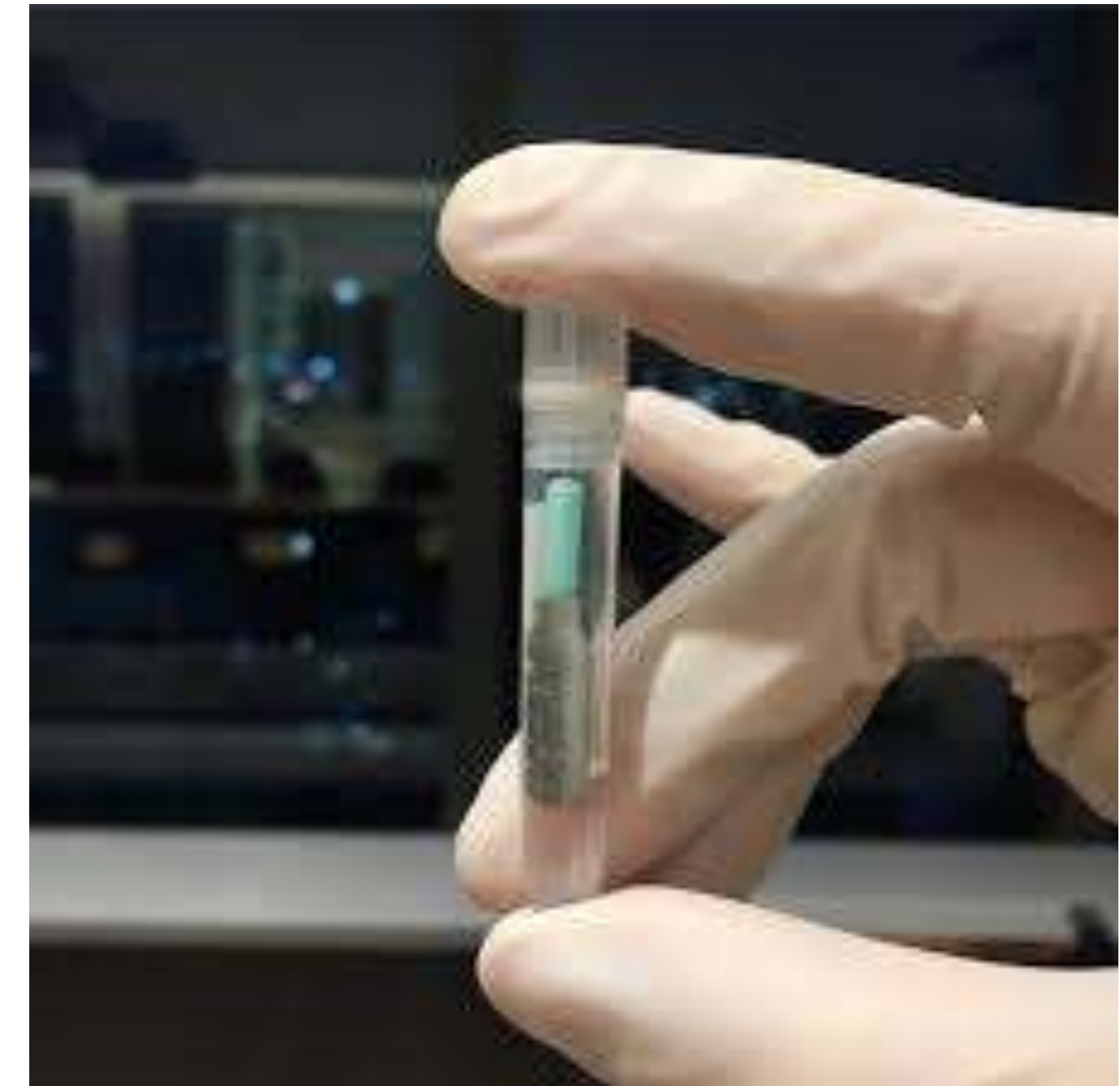
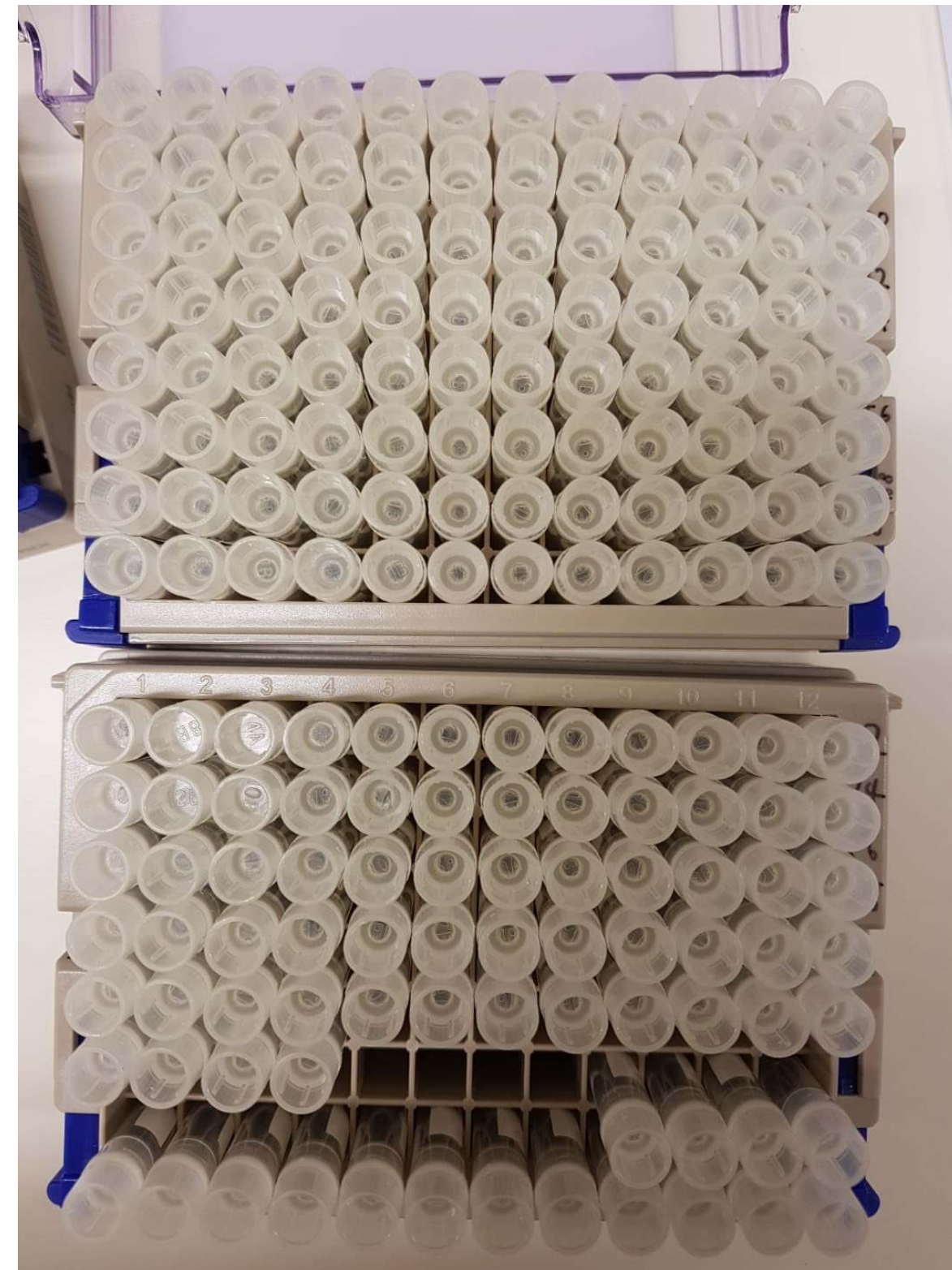
- Logging in

Use the link (<https://www.kobotoolbox.org>) to access KoBoToolbox on a computer. It is supported by most browsers. Click **SIGN UP** in order to find login.



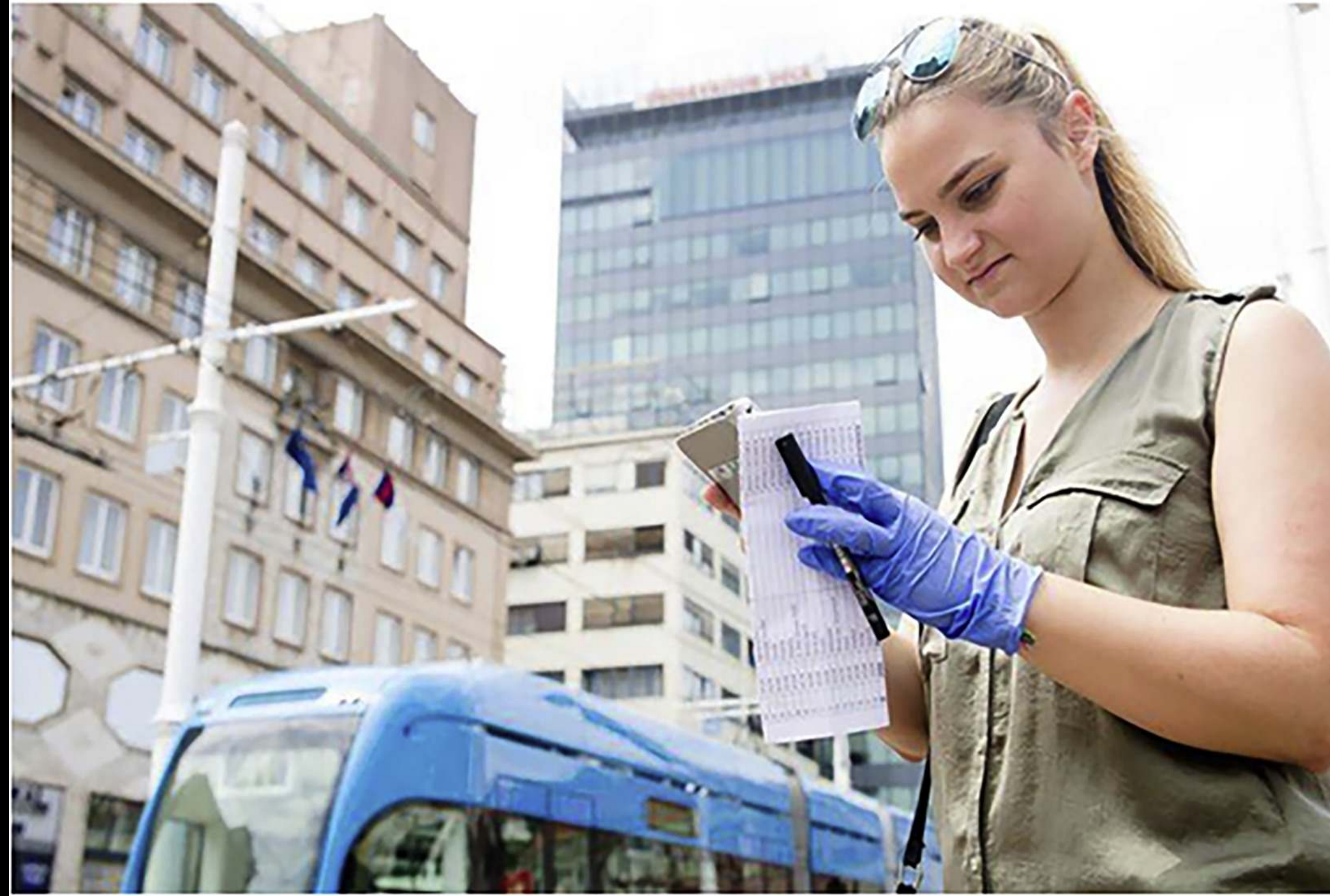
Global City Sampling Day – June 20th

Please ship all samples to NYC before September 1st!









DHS BioWatch

BioWatch (est 2001, ~30 cities) is a U.S. federal government program to detect the release of pathogens into the air as part of a terrorist attack on major American cities.



What have we found?

End-to-End Global Urban Metagenomics

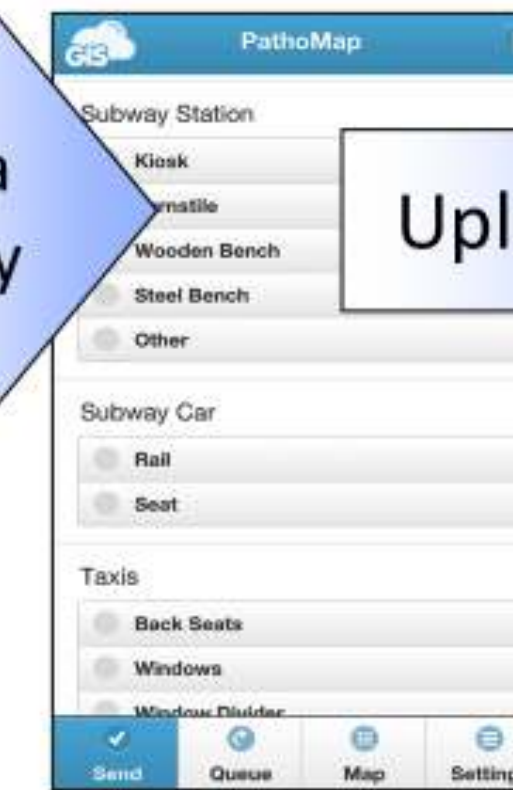
A city-wide survey of New York City's metagenome spanning all five boroughs, with samples collected from 466 subway stations across 24 subway lines.



1. Swab (3 min)



2. Annotate

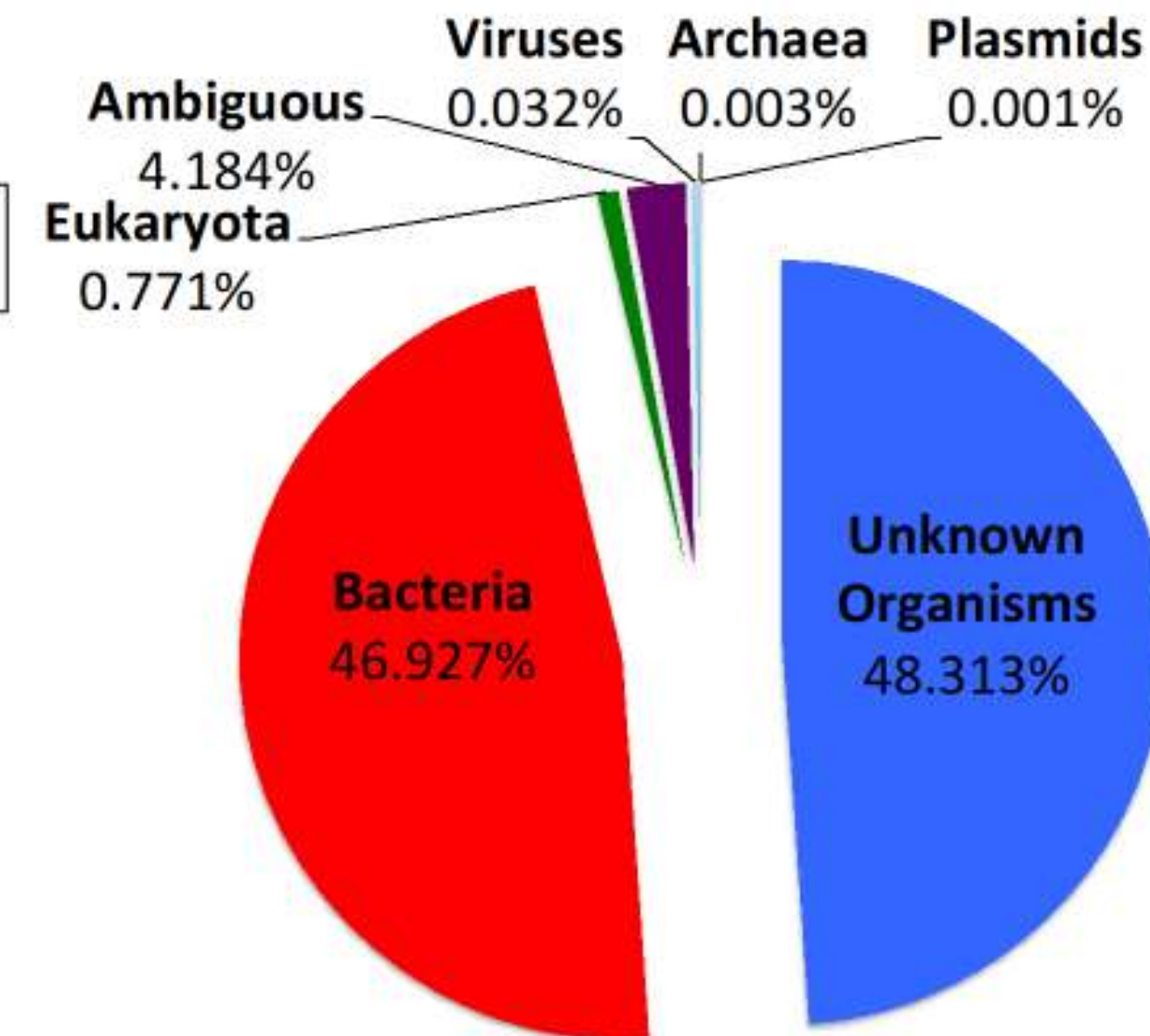
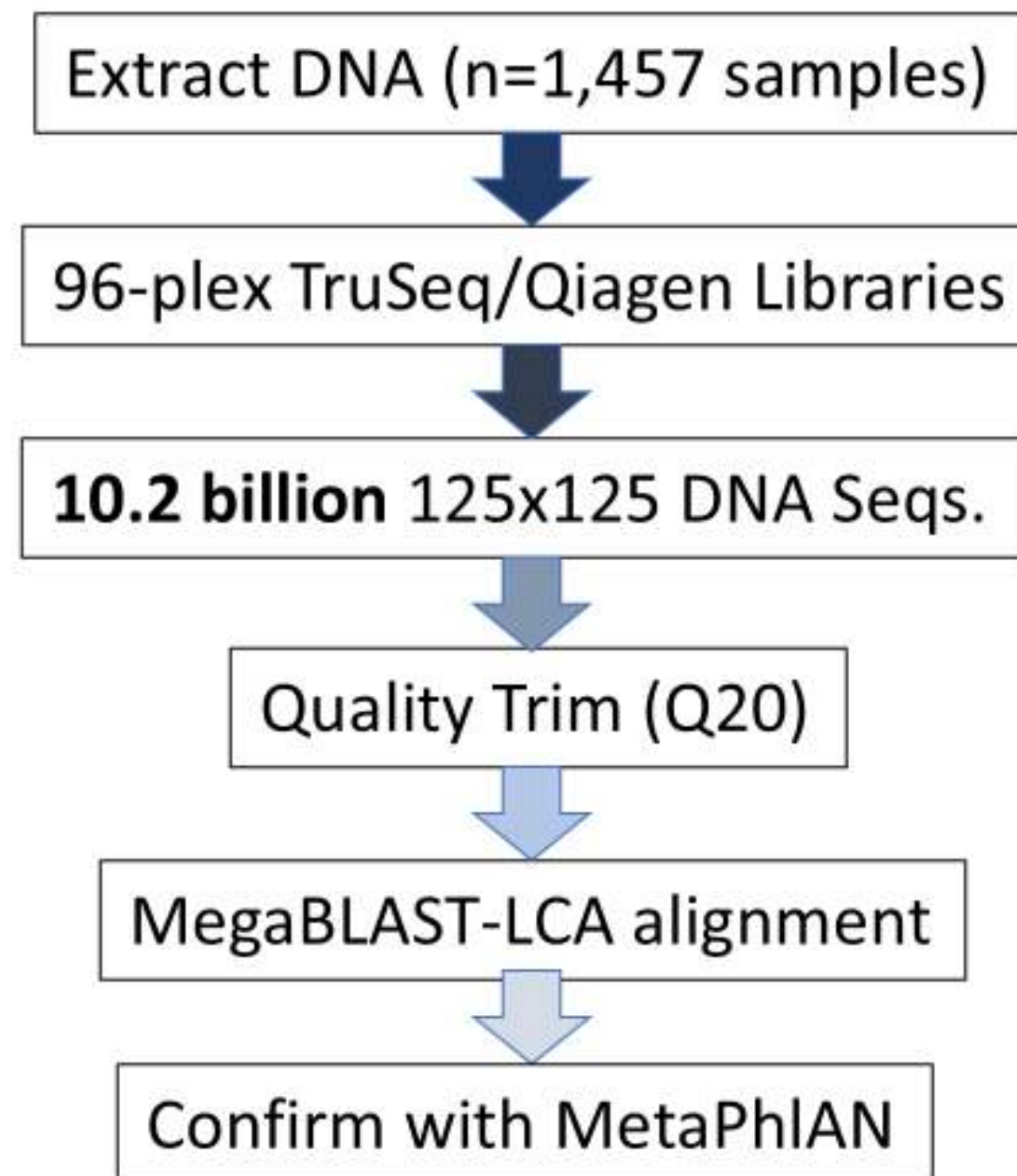


3. GPS-tag/timestamp

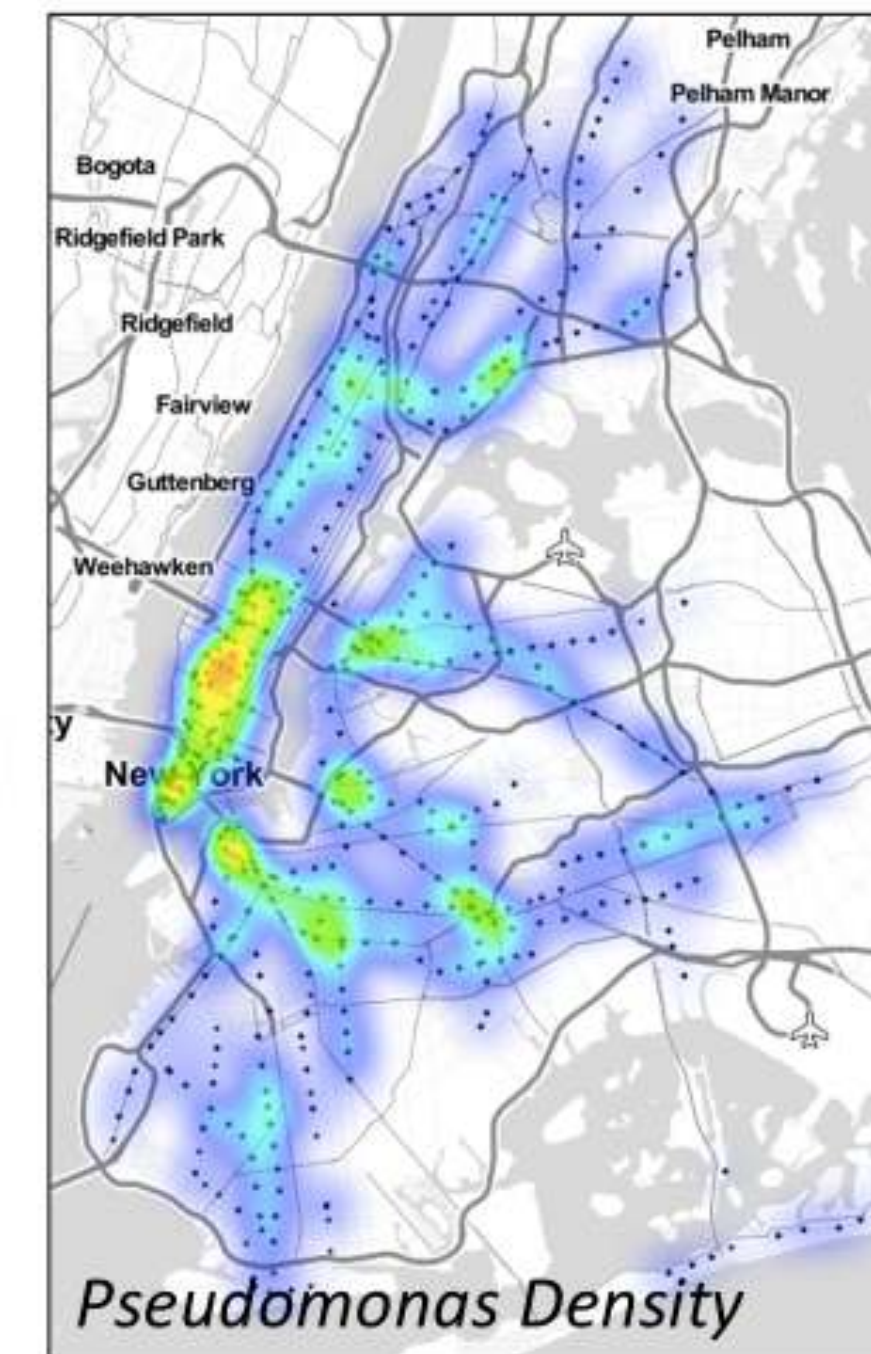


Data
Entry

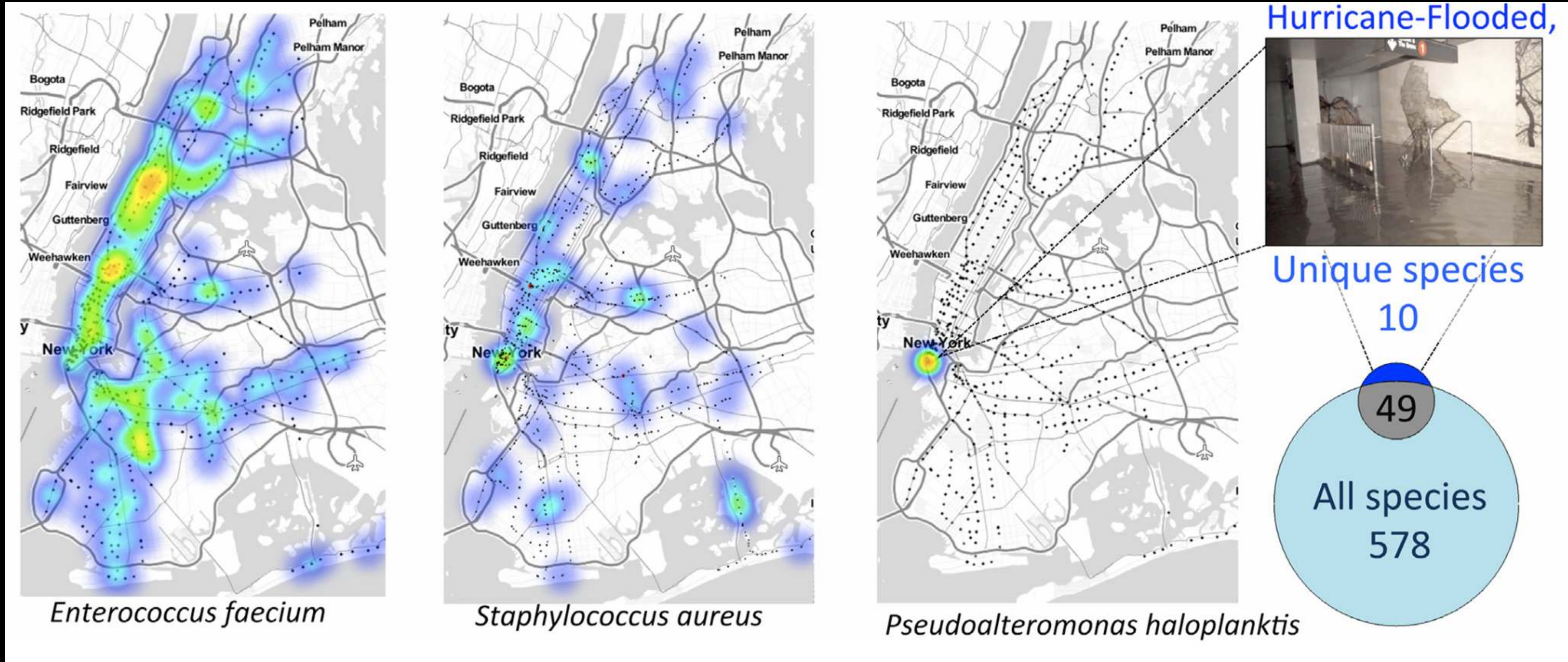
Upload



**Half of the world
under our fingertips is unknown**



Spatial Patterns and Environmental Perturbations in the NYC Subway Microbiome





A persistent “molecular echo” of the cold, ocean water





**Uptown &
The Bronx**



South Ferry
24 hour booth
Elev to mezz

M5
M15
M20



South
Ferry

South Ferry
24 hour booth
Elev to mezz

M5
M15
M20



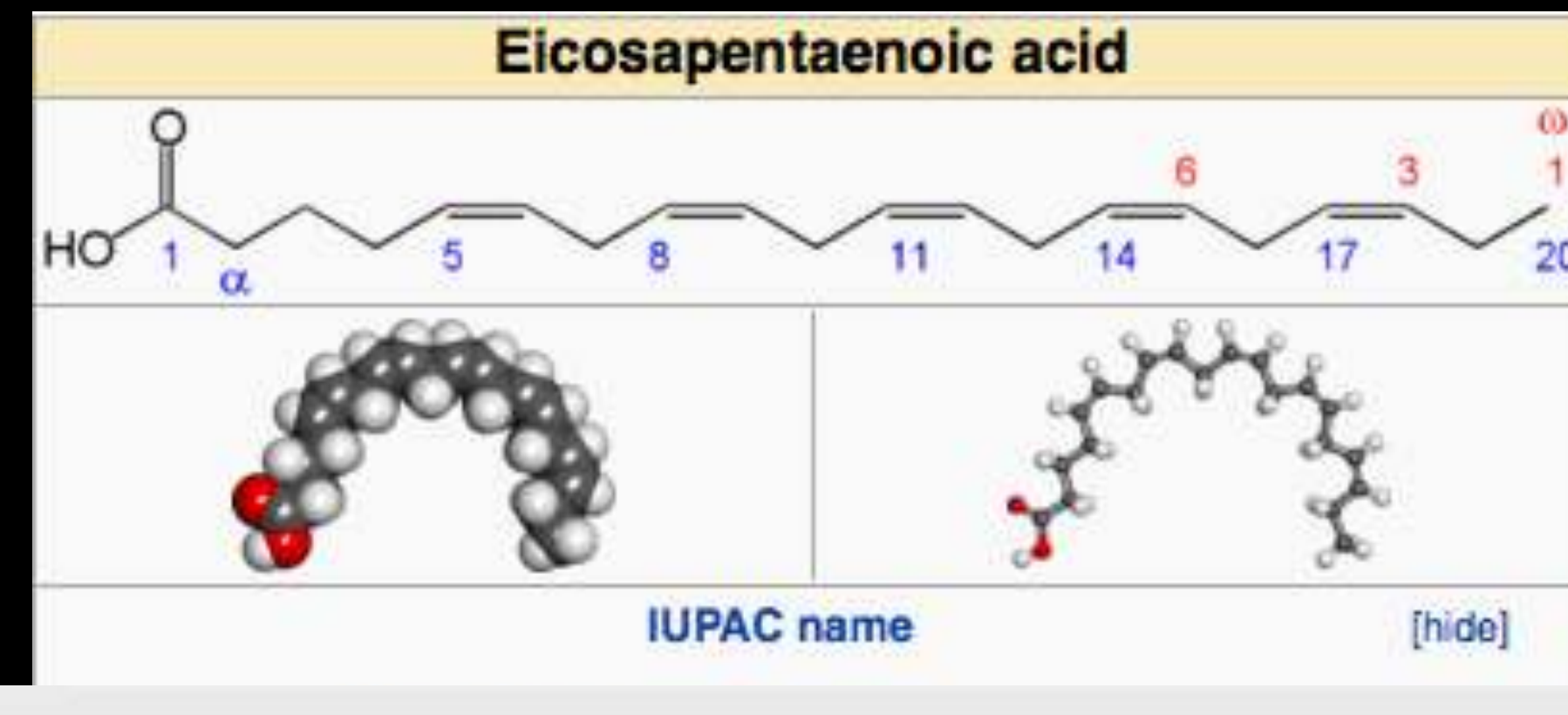
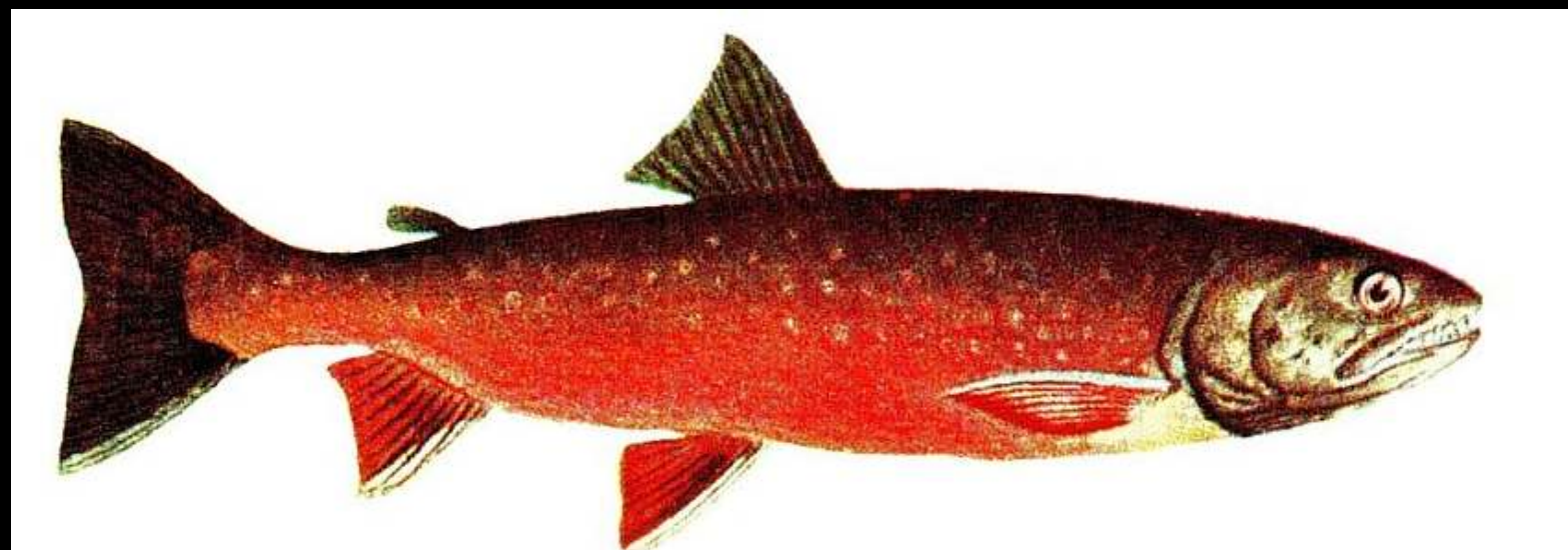
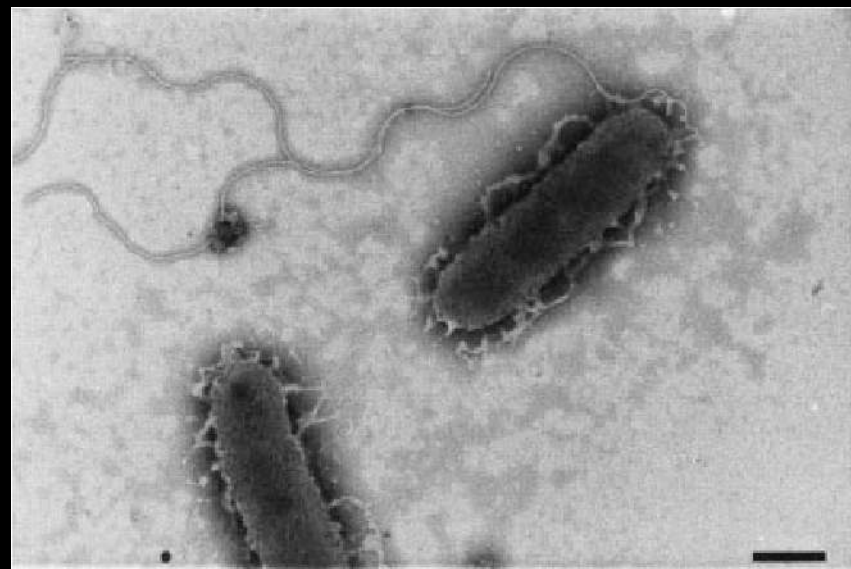
South
Ferry

Shewanella frigidimarina

An Antarctic species with the ability to produce eicosapentaenoic acid (EPA). It grows anaerobically by dissimilatory Fe (III) reduction.[1] Its cells are motile and rod shaped

EPA is obtained in the human diet by eating oily fish or fish oil, e.g. cod liver, herring, mackerel, and salmon.

Shewanella frigidimarina



Biological Psychiatry

A Journal of Psychiatric Neuroscience and Therapeutics

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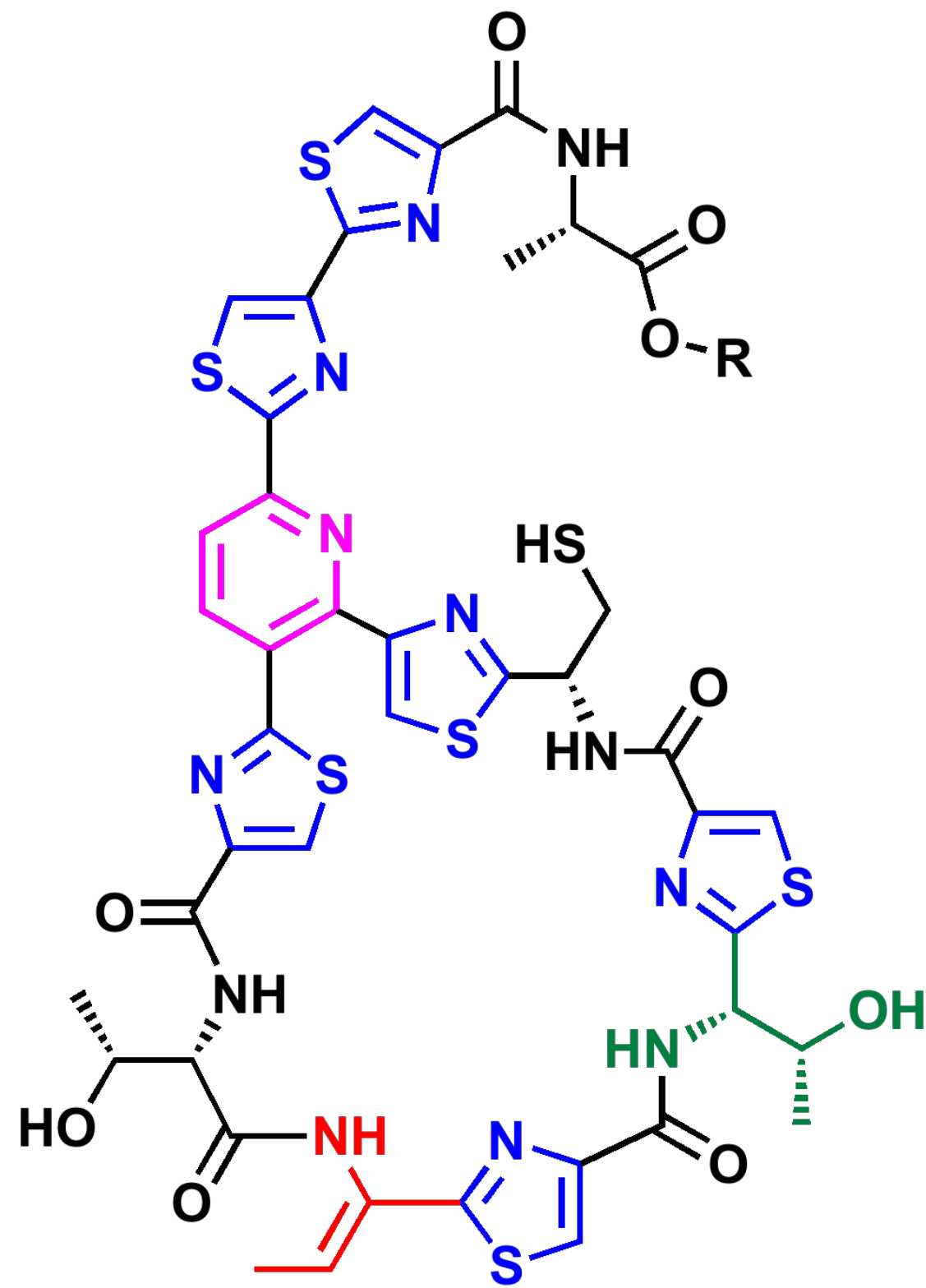
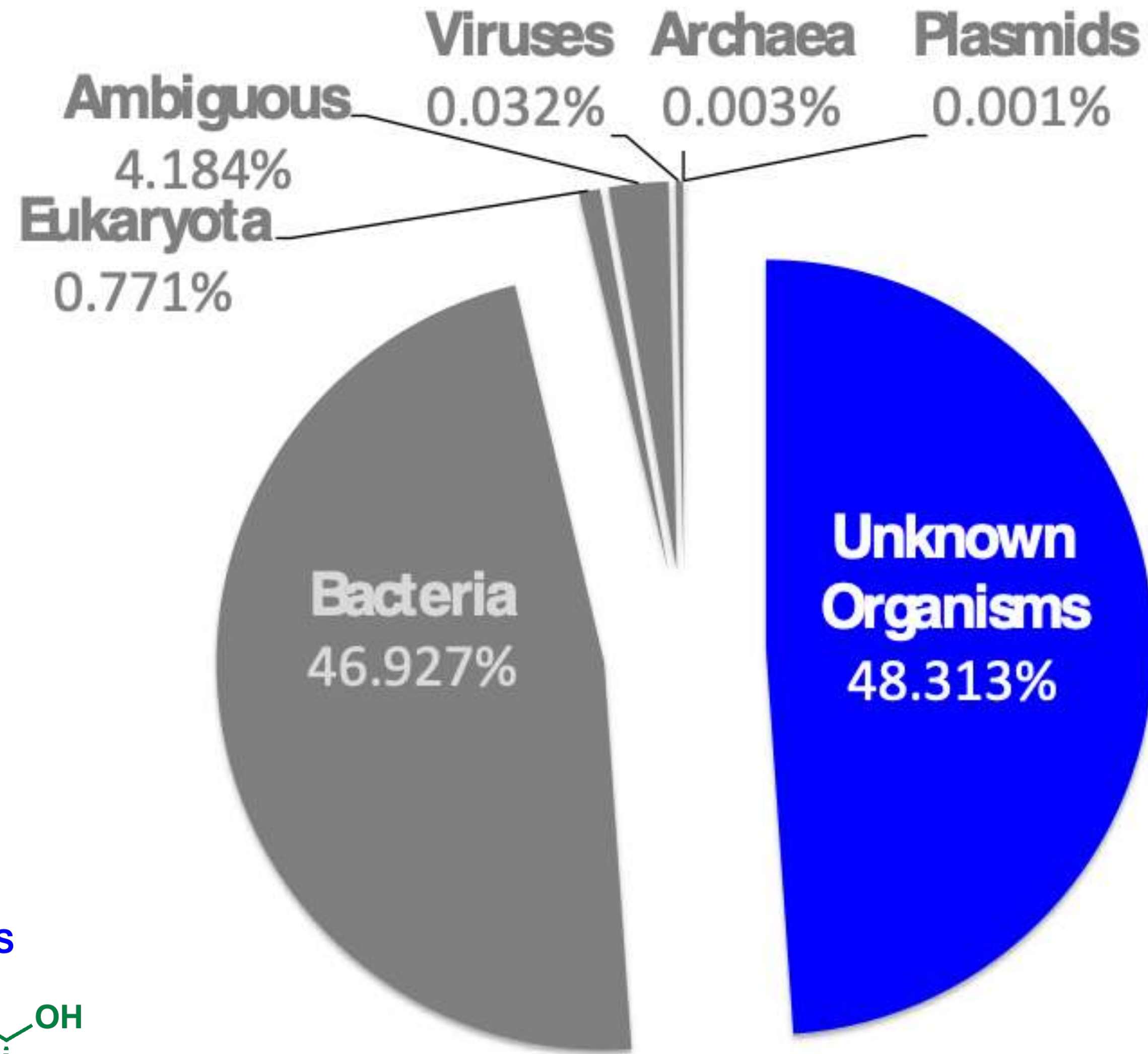
Biological Psychiatry

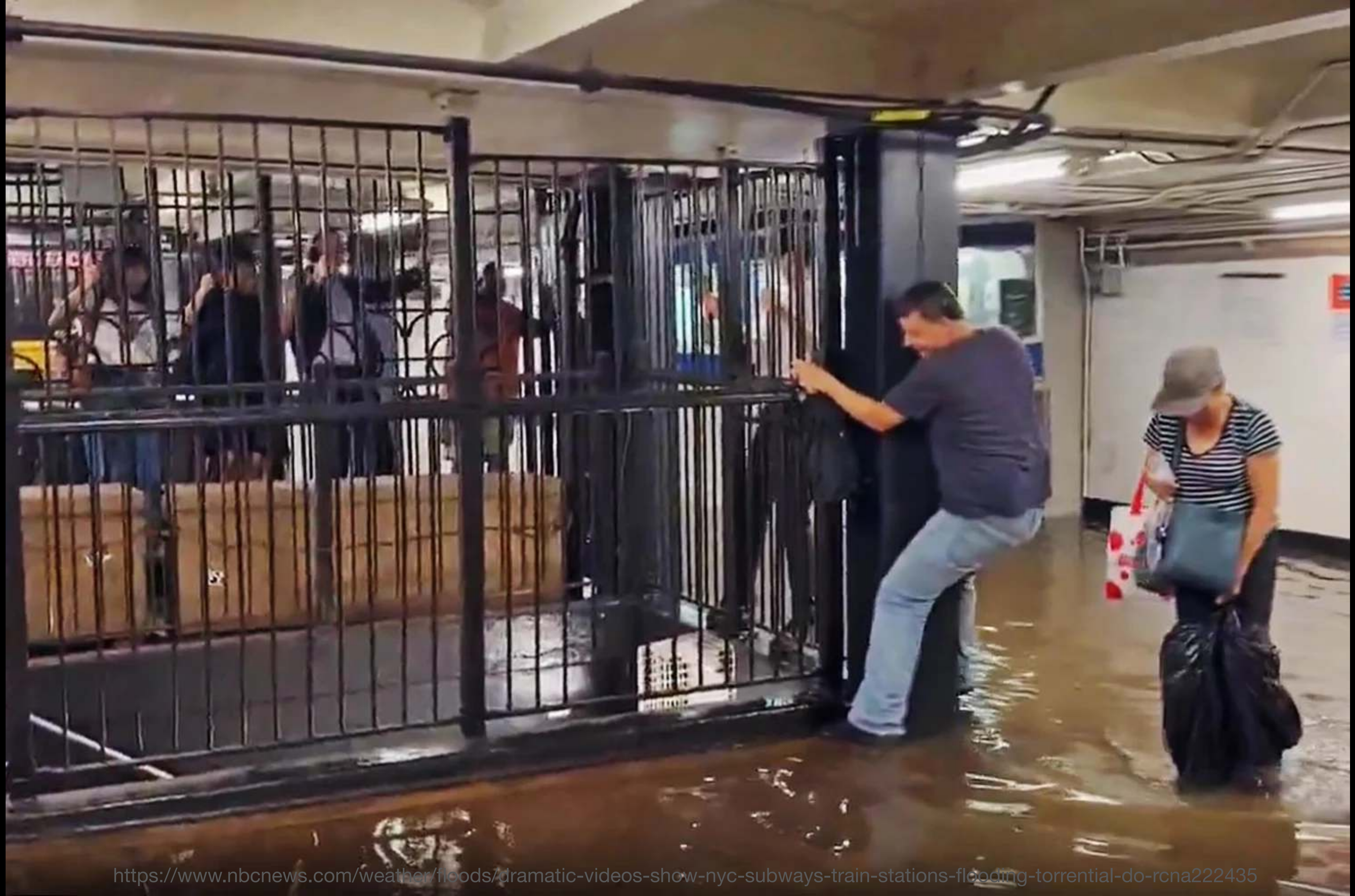
Next Article >

Volume 56, Issue 7, Pages 490–496, October 1, 2004

Suicide attempt and n-3 fatty acid levels in red blood cells: A case control study in china

Who is there, and what are they making?

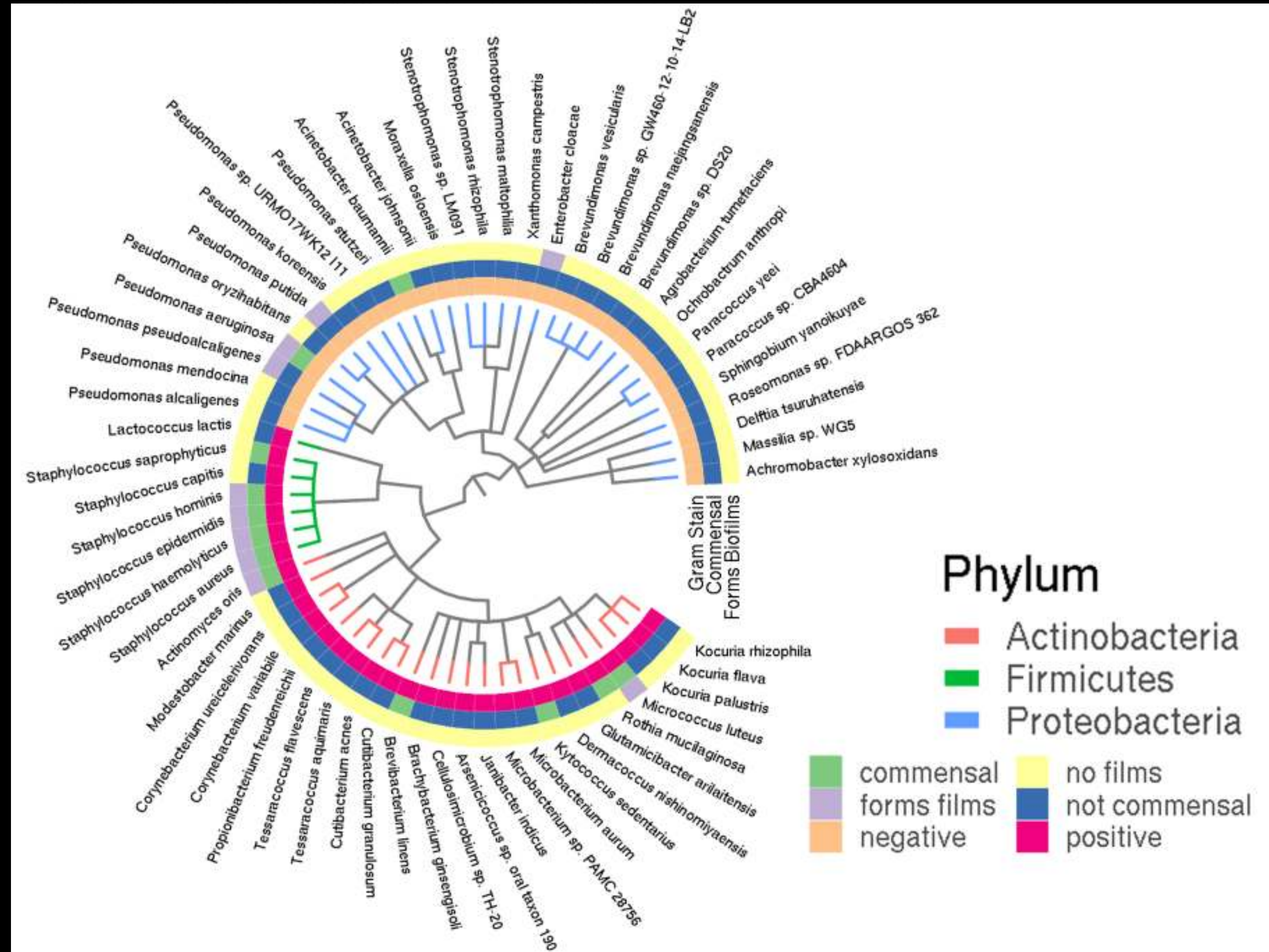






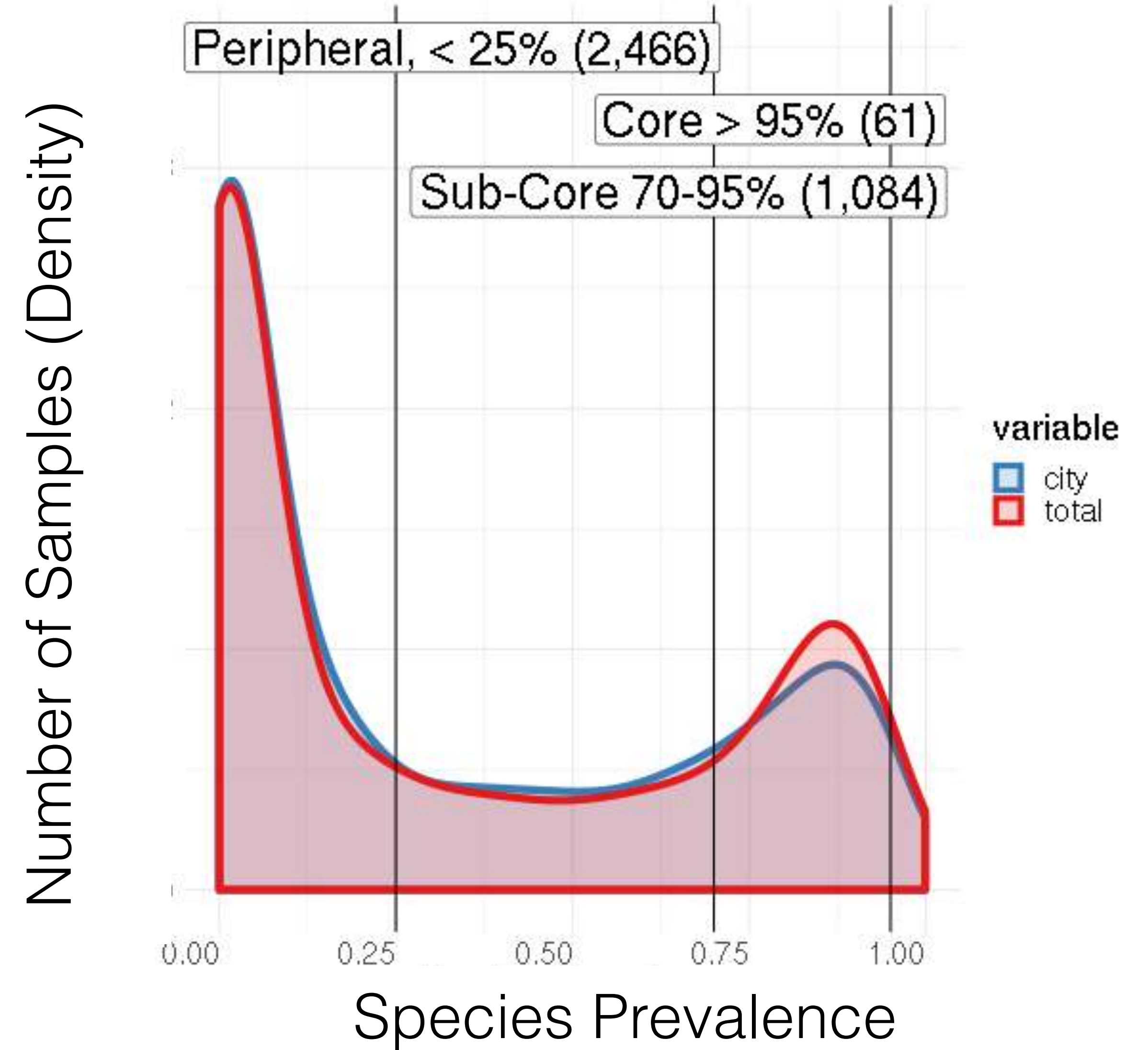
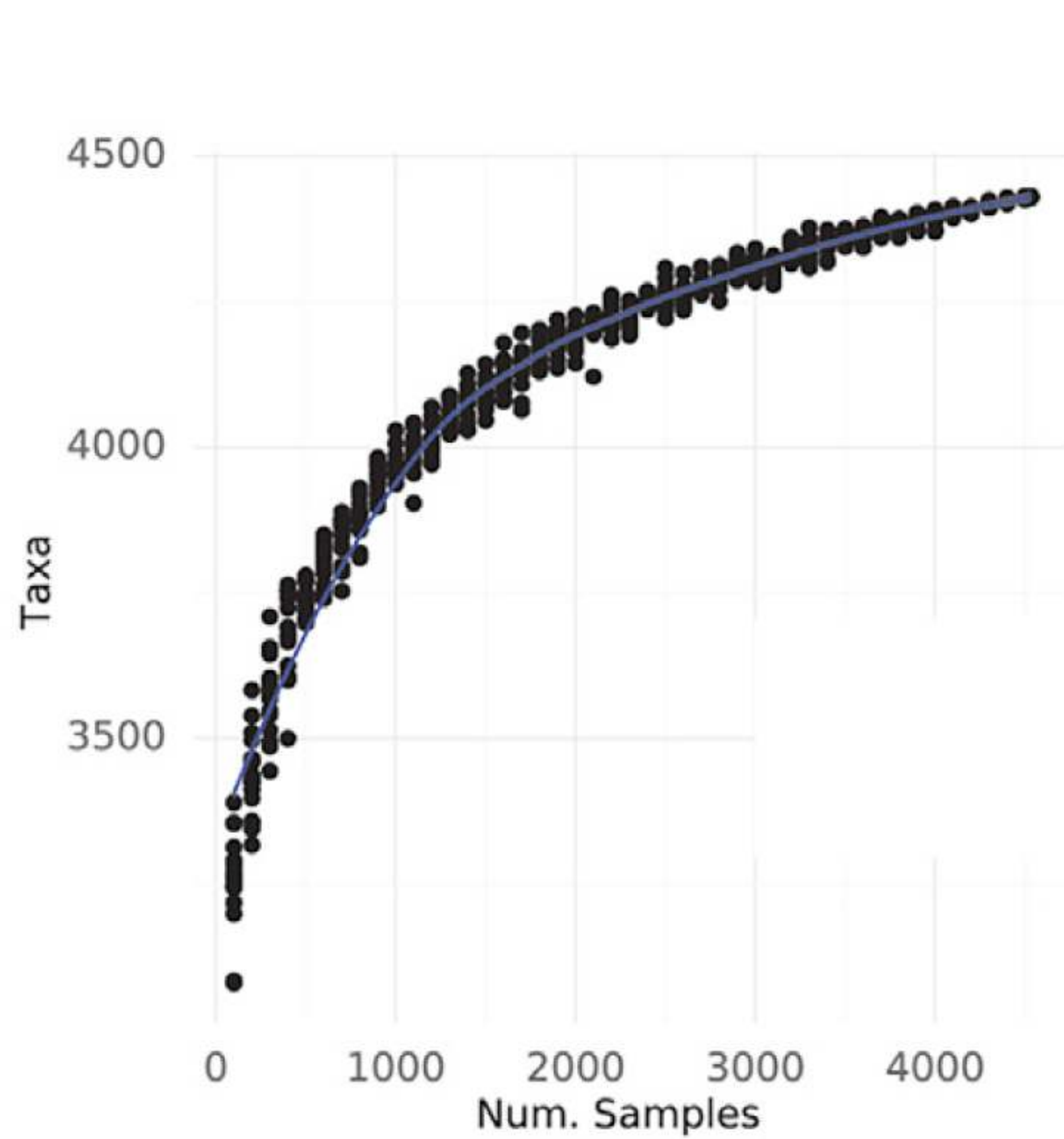
A Universal Microbial Signature of Cities

We identified a core set of 31 microbial species that appeared in >97 % of the 4,728 urban metagenomic samples collected from mass-transit systems across 60 cities worldwide, forming a “urban core” distinct from common human-commensal microbes



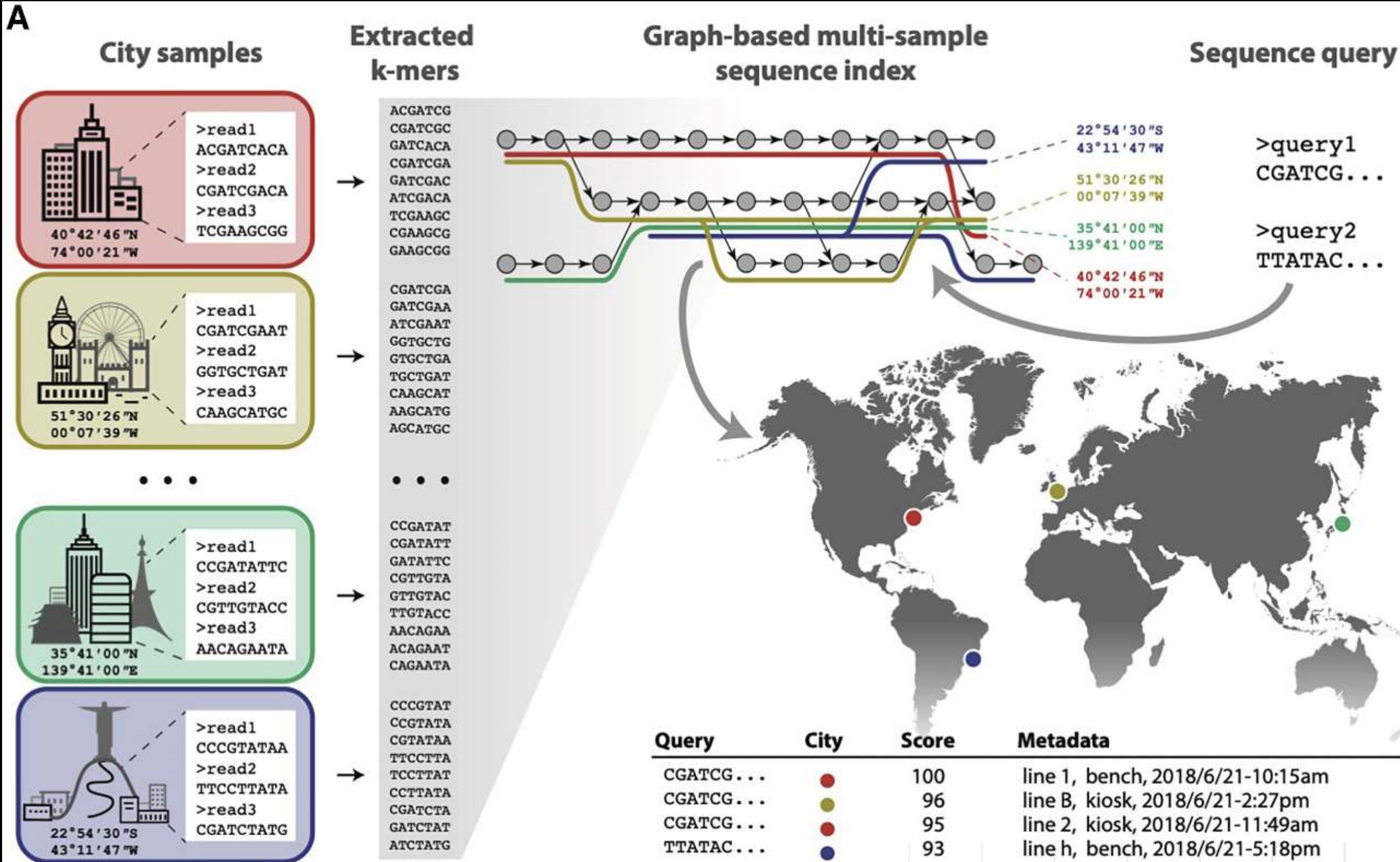
© Danko, Bezdán *et al.*, *Cell*, 2021

A Core Urban Microbiome found across 4,728 samples and 60 cities



Graph-Based Representation of Global Urban Metagenomes

City-scale metagenomes are encoded as a *k-mer graph* that enables rapid sequence matching and global origin mapping



MetaGraph: Making Petabase-Scale Sequence Archives Searchable

This approach makes global biological sequence archives searchable at low cost, opening new opportunities for pathogen discovery, comparative genomics, biomedical mining, and large-scale reuse of existing sequencing data.

nature

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Article | [Open access](#) | Published: 08 October 2025

Efficient and accurate search in petabase-scale sequence repositories

[Mikhail Karasikov](#), [Harun Mustafa](#), [Daniel Danciu](#), [Oleksandr Kulkov](#), [Marc Zimmermann](#), [Christopher Barber](#), [Gunnar Rättsch](#) ✉ & [André Kahles](#) ✉

[Nature](#) **647**, 1036–1044 (2025) | [Cite this article](#)

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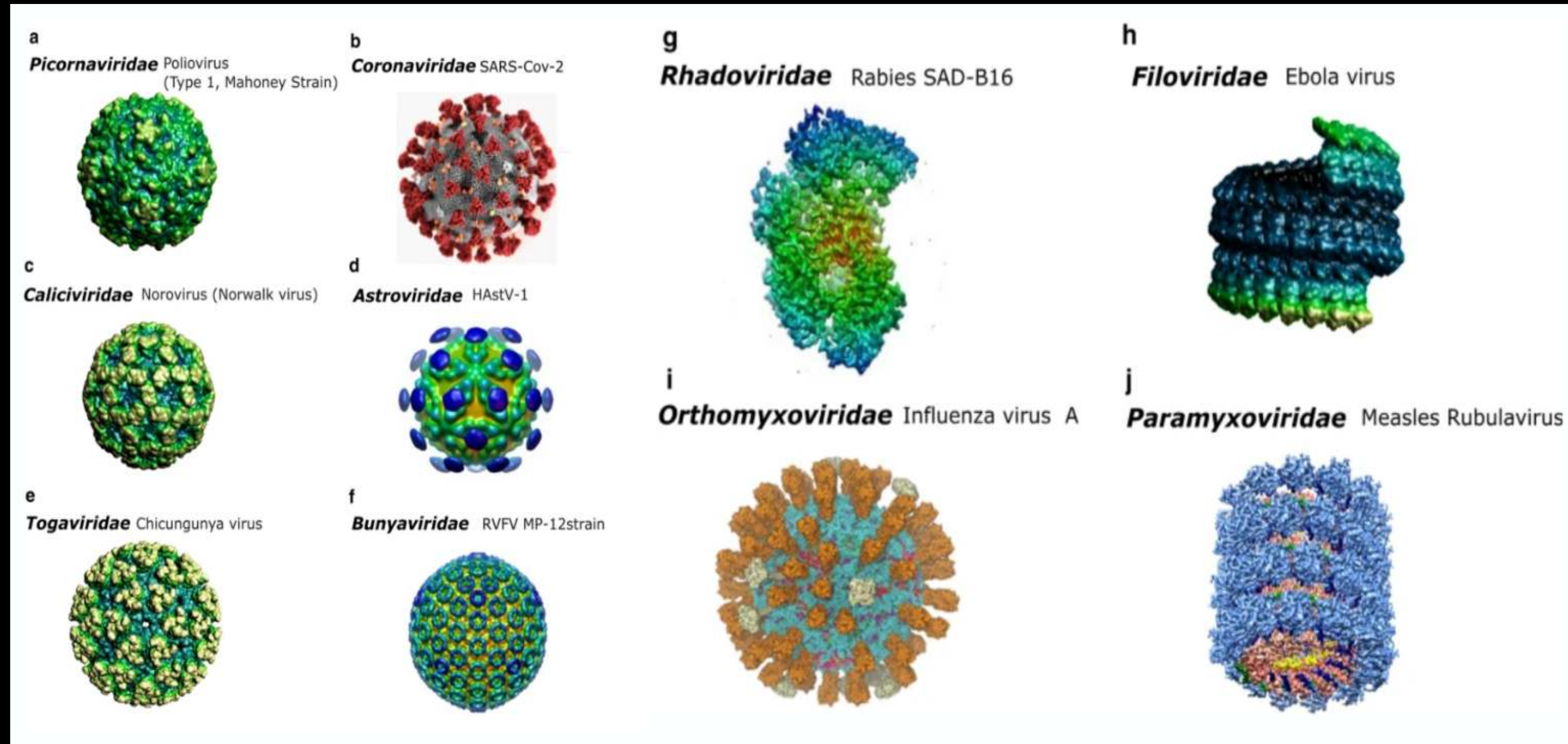
70k Accesses | 15 Citations | 248 Altmetric | [Metrics](#)

Abstract

The amount of biological sequencing data available in public repositories is growing rapidly, forming a critical resource for biomedicine. However, making these data efficiently and accurately full-text searchable remains challenging. Here we build on efficient data structures and algorithms for representing large sequence sets^{1,2,3,4,5,6}. We present MetaGraph, a methodological framework that enables us to scalably index large sets of DNA, RNA or protein sequences using annotated de Bruijn graphs. Integrating data from seven public sources^{7,8,9,10,11,12,13}, we make 18.8 million unique DNA and RNA sequence sets and 210 billion amino acid residues across all clades of life—including viruses, bacteria, fungi, plants, animals and humans—full-text searchable. We demonstrate the feasibility of a cost-effective full-text search in large sequence repositories (67 petabase pairs (Pbp) of raw sequence) at an on-demand cost of around US\$100 for small queries up to 1 megabase pairs (Mbp) and down to US\$0.74 per queried Mbp for large queries. We show that the highly compressed representation of all public biological sequences could fit on a few consumer hard drives (total cost of around US\$2,500), making it cost-effective to use and readily transportable for further analysis. We explore several practical use cases to mine existing archives for interesting associations, demonstrating the use of our indexes for integrative analyses, and illustrating that such capabilities are poised to catalyse advancements in biomedical research.

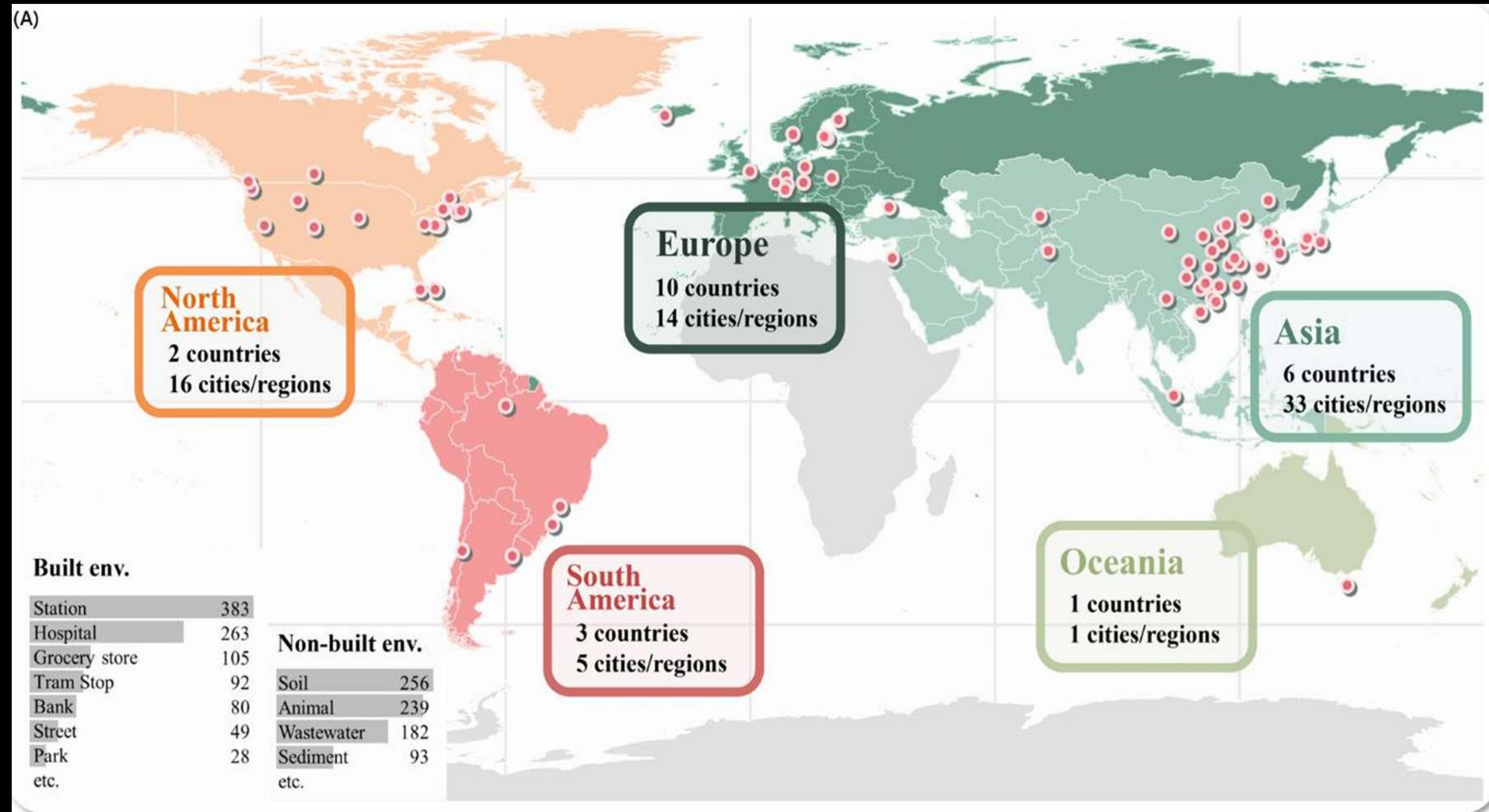
Mapping the Urban RNA Virosphere

RNA viruses matter in cities because they evolve quickly where people, animals, and environments constantly mix, creating early signals of emerging health threats



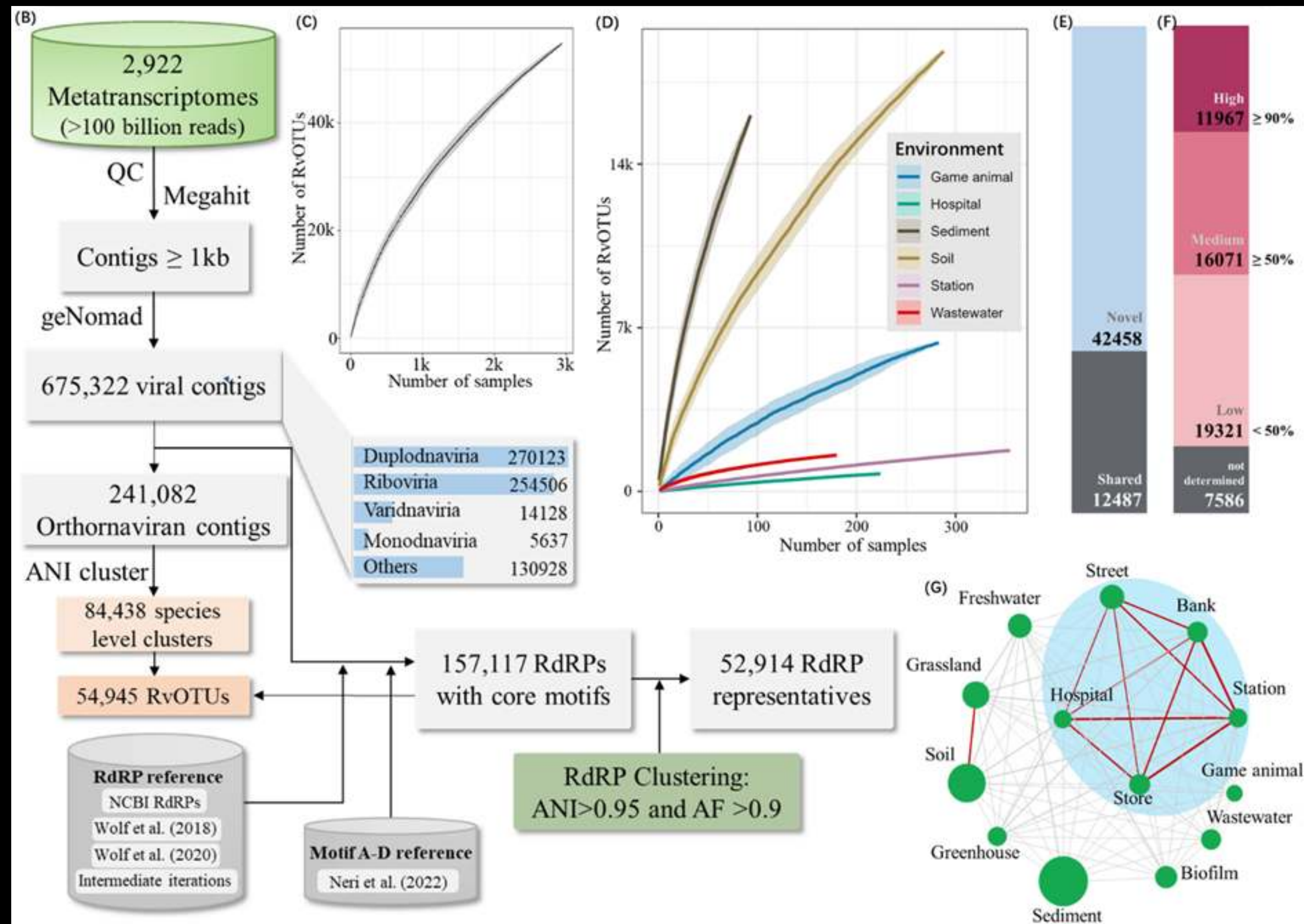
The Urban & Peri-urban RNA Virus Atlas (UPVAtlas)

UPVAtlas maps 54,945 RNA viruses from 2,922 samples (collected 1H 2020) across 102 cities, revealing urban RNA viral diversity was previously unknown.



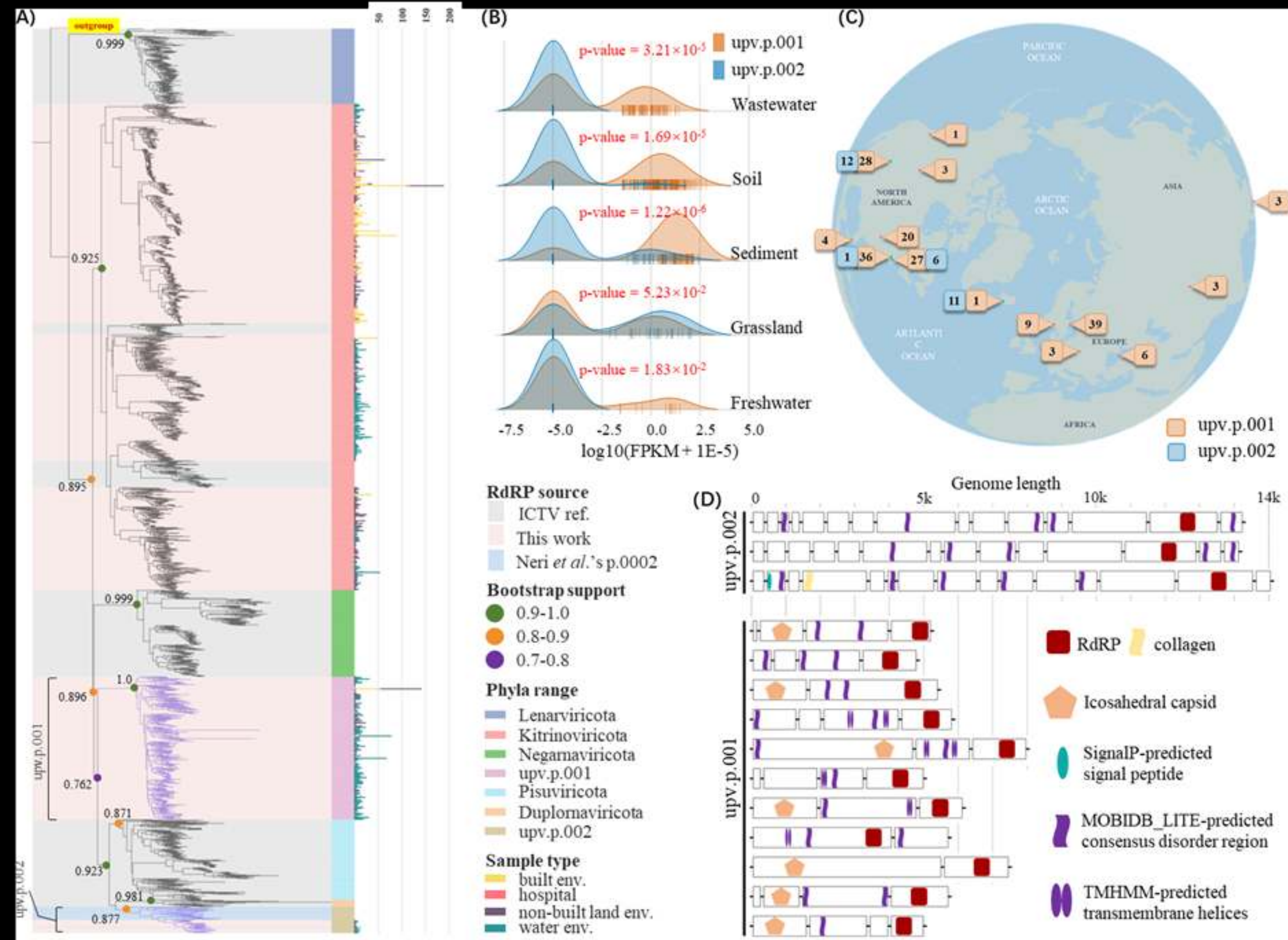
Computational Pipeline for RNA Virus Mining in Metatranscriptomes

Analysis reveals substantial novel urban RNA virus diversity, environment-structured richness, and cross-habitat connectivity.



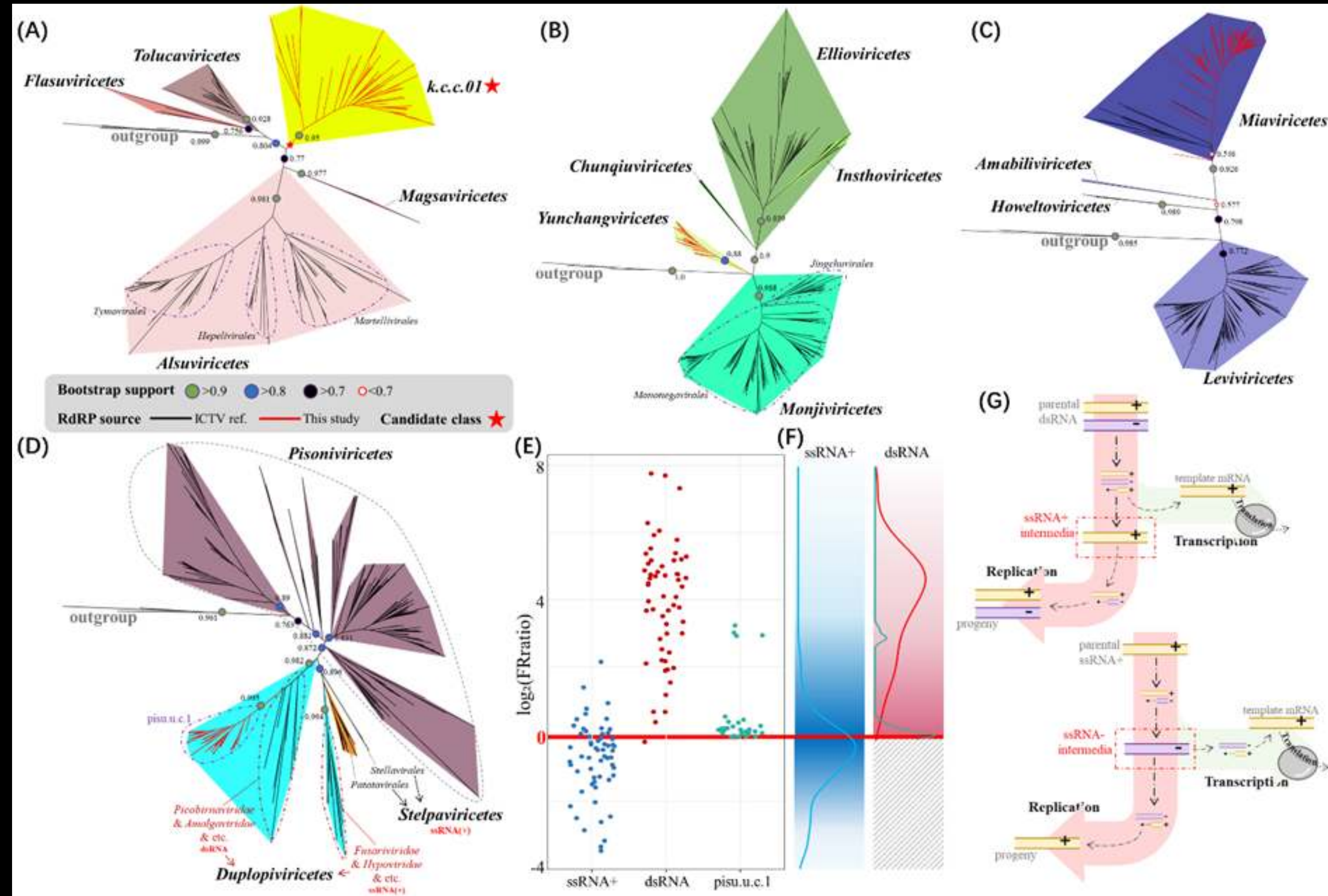
RdRP Core Motifs Enable Phylogenetic Reconstruction of the RNA Virome

Phylogenetic analysis of conserved RdRP motifs reveals two candidate novel RNA virus phyla, their genome architectures, and global environmental abundance patterns.



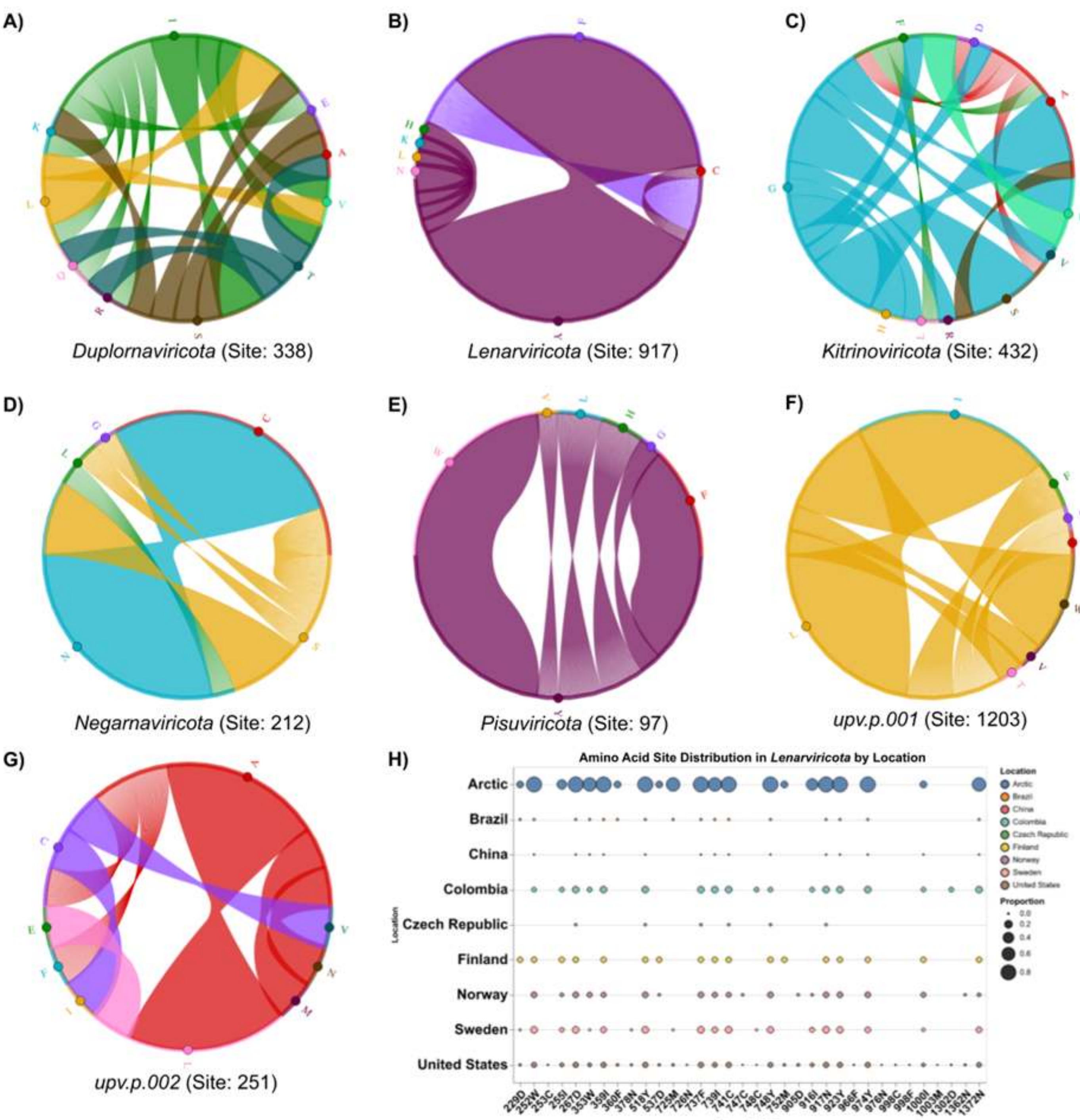
Phylogenetic Reconstruction and Strand-Specific Signatures of Established RNA Virus Phyla

RdRP-based phylogenies resolve class-level structure across major RNA virus phyla, while strand-specific logFRratio analyses distinguish ssRNA+ and dsRNA viruses consistent with their replication biology.



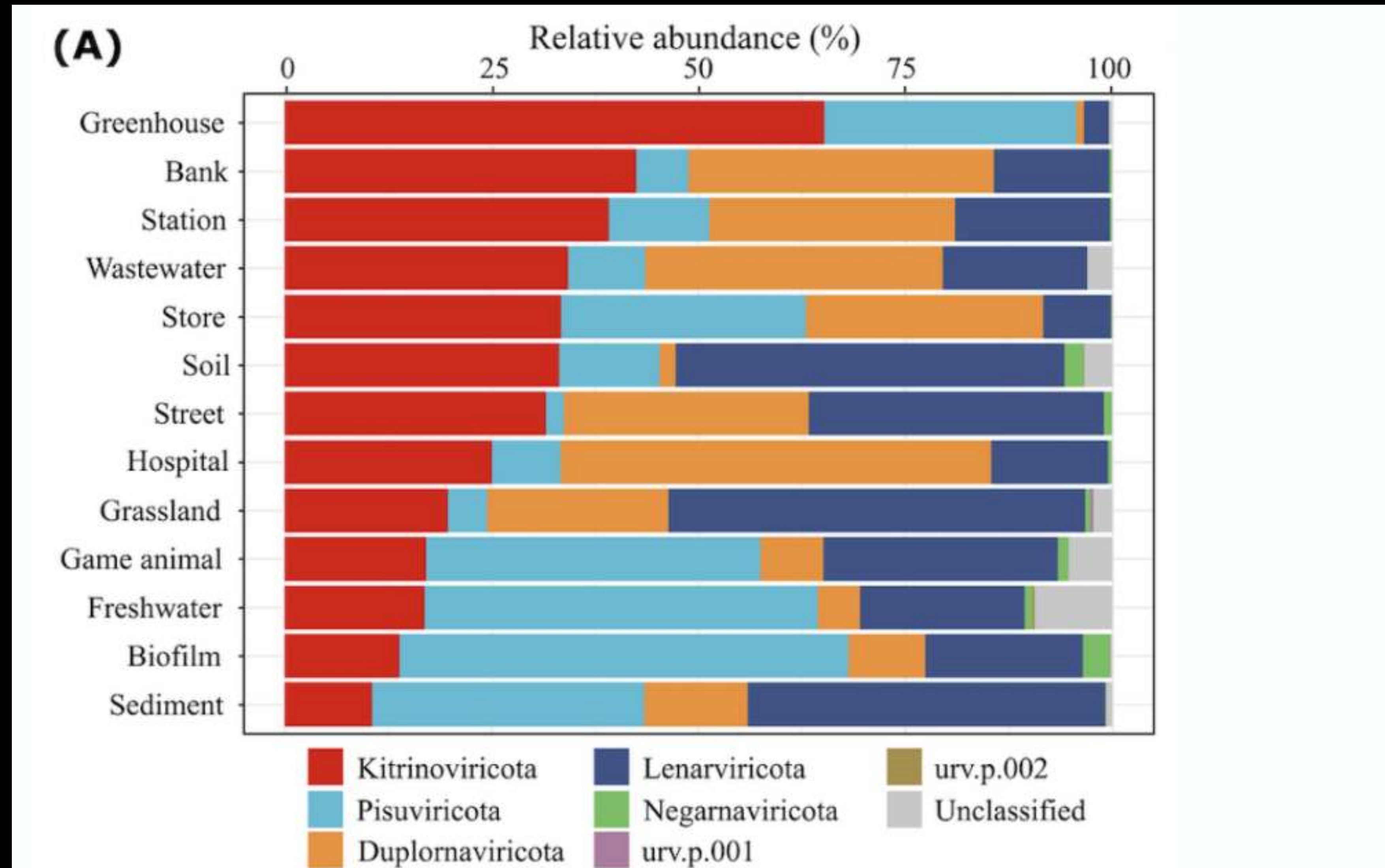
Directional Selection Shapes Amino Acid Exchange Landscapes Across RNA Virus Phyla

Site-specific analyses reveal heterogeneous amino acid exchange patterns consistent with diverse selective pressures, with globally distributed directional selection signals exemplified in *Lenarviricota*.

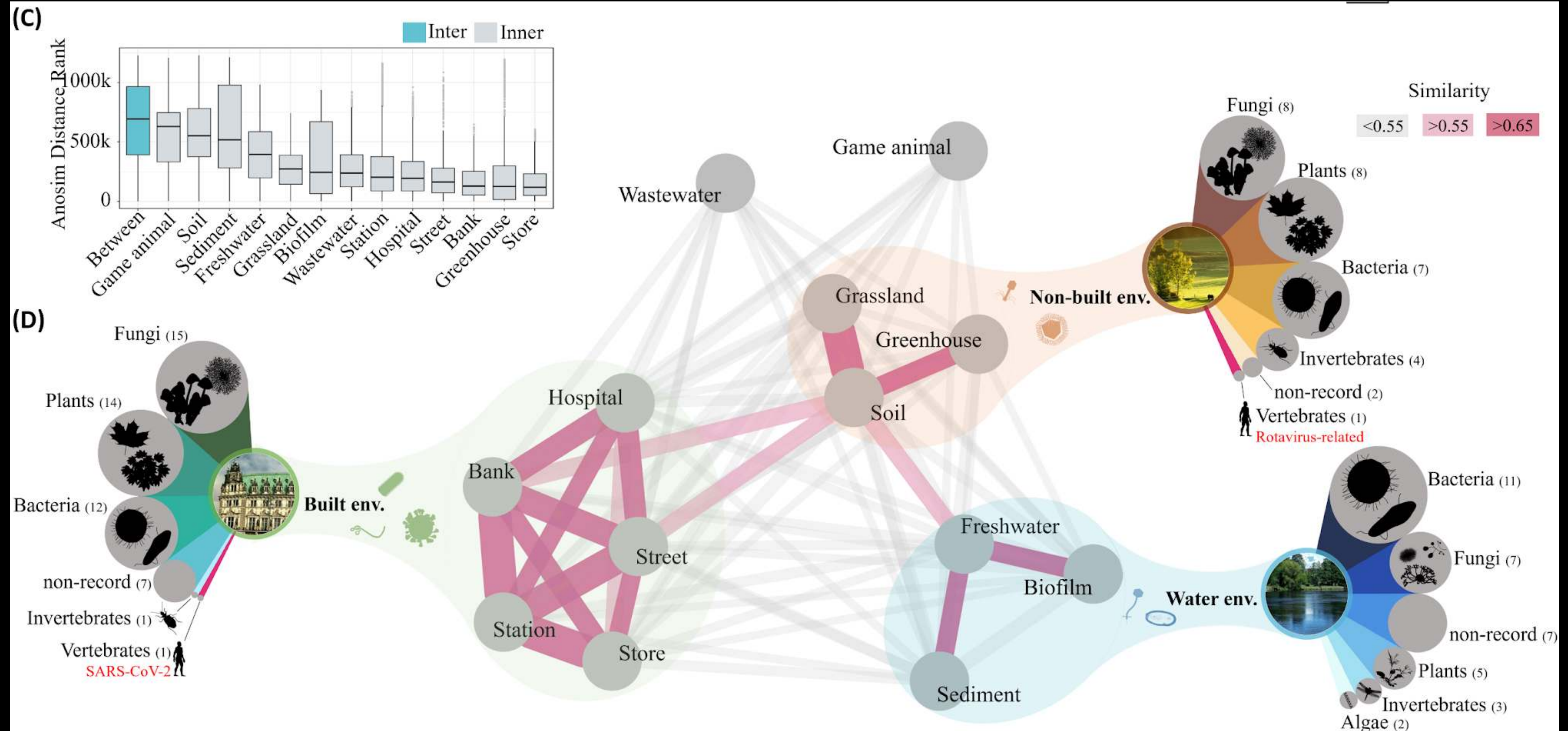


Environmental Distribution of the Urban & Peri-urban RNA Virome

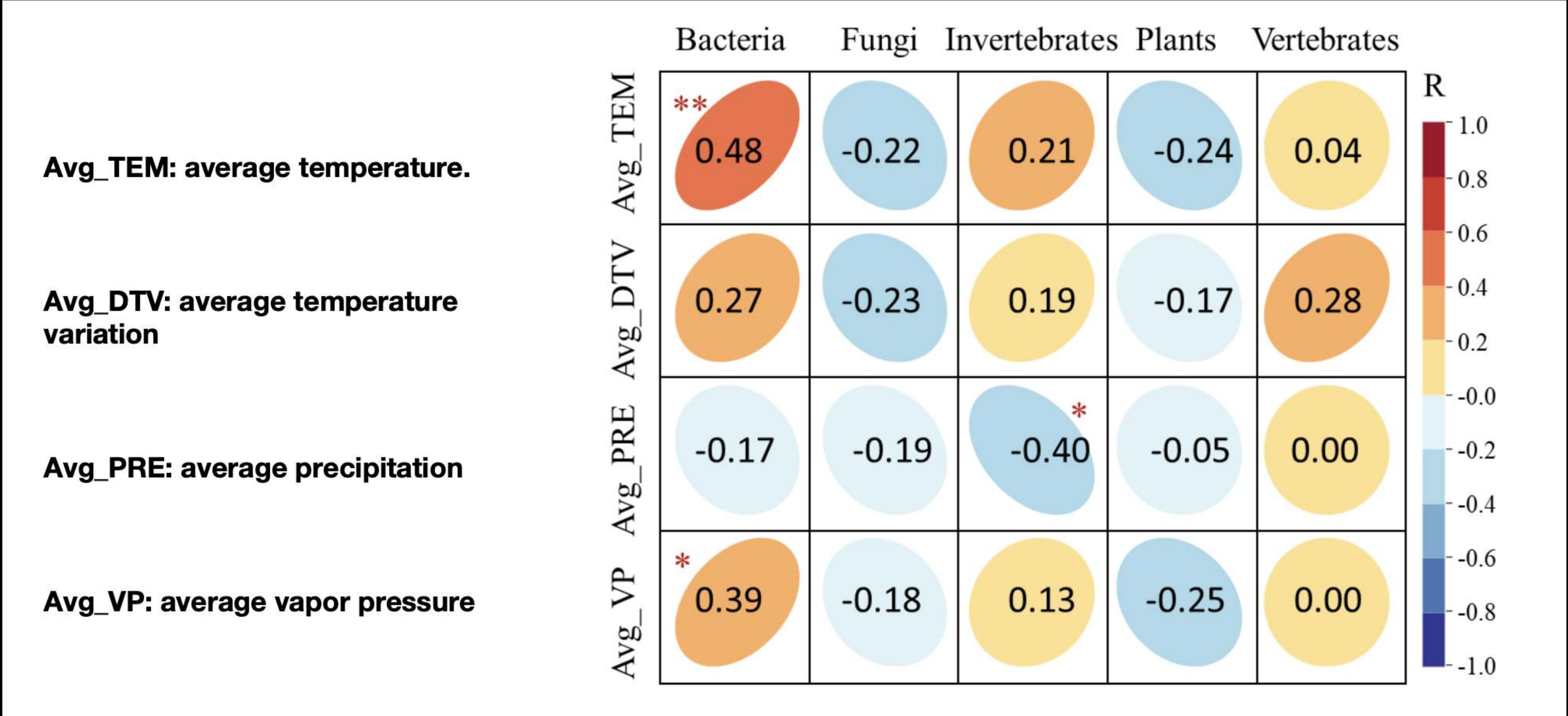
RNA virus phylum composition varies markedly across environmental sample types based on FPKM-normalized relative abundance.



Understanding RNA viruses in Urban Environments



Correlation between the relative abundance of RNA viruses hosted by different organisms and meteorological conditions



UPVAtlas Highlights

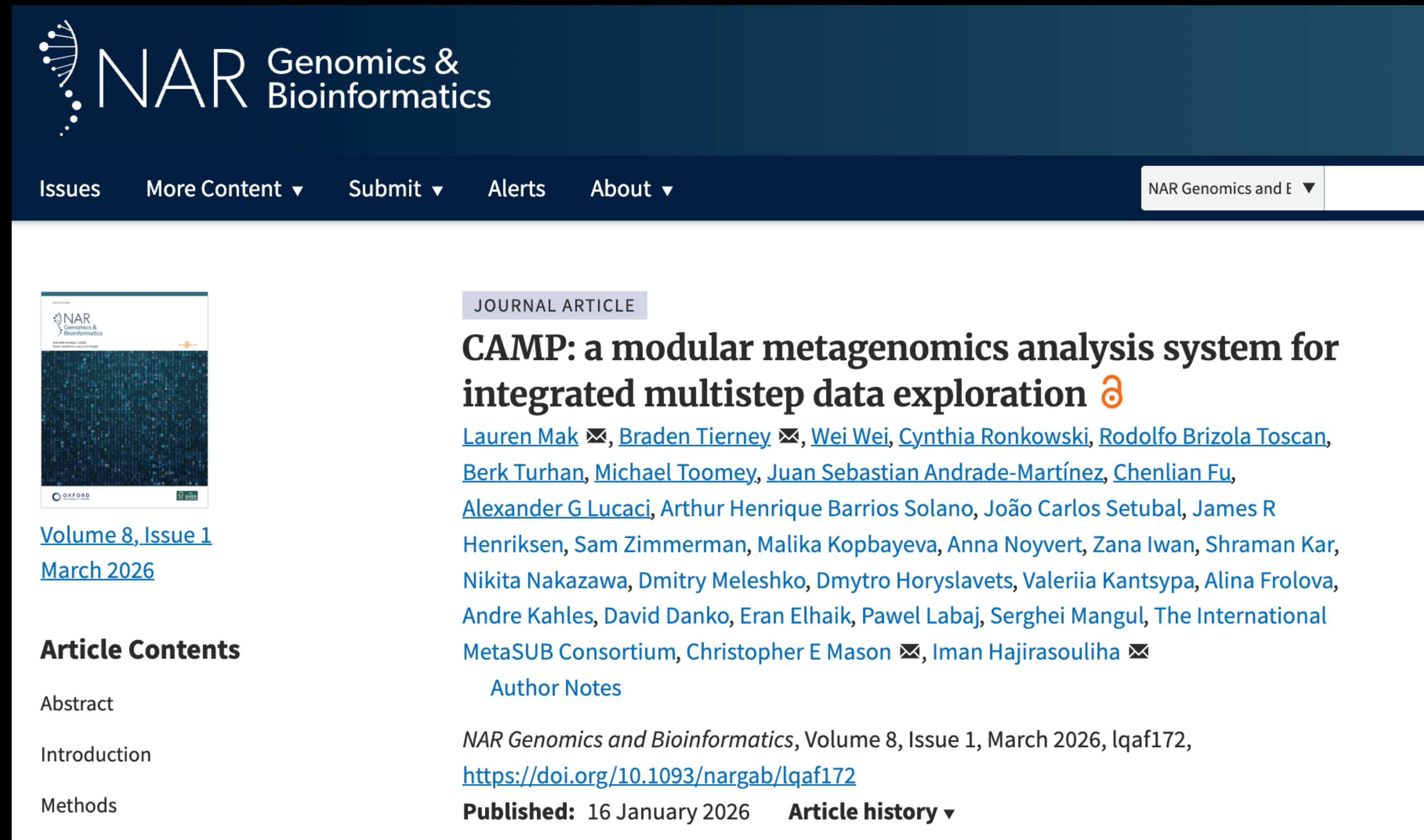


UPVAtlas
Urban & Peri-urban RNA Virus Atlas

- **Deepest metatranscriptomic profiling of global cities to date.**
We examined ~3K samples (collected 1H2020), containing >100B reads to study RNA virus biology in urban environments, their hosts, and pathogenicity.
- **Our analysis uncovered two novel *Duplornaviricota* phyla** supporting the polyphyletic nature of this clade, with support from the published literature (Neri et al., 2022, Zayed et al., 2022).
- **Selection analyses of RdRp identified amino acid sites with directional selective pressures**, this may describe local adaption of viral groups.

CAMP: A Modular Workflow System for Scalable Metagenomics Analysis

- *CAMP provides a Snakemake-based, modular framework for building reproducible metagenomics workflows that can run individual analysis steps independently or connect them into larger multistep pipelines.*



The screenshot shows the NAR Genomics & Bioinformatics journal article page for the article "CAMP: a modular metagenomics analysis system for integrated multistep data exploration". The page features a dark blue header with the NAR logo and navigation links. The article title is prominently displayed in bold, followed by the authors' names and their ORCID iD icons. The article is categorized as a "JOURNAL ARTICLE". The publication details include the volume, issue, and date. The "Article Contents" section lists the Abstract, Introduction, and Methods. The "Published" date is 16 January 2026, and there is a link to the "Article history".

NAR Genomics & Bioinformatics

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JOURNAL ARTICLE

CAMP: a modular metagenomics analysis system for integrated multistep data exploration 

[Lauren Mak](#) , [Braden Tierney](#) , [Wei Wei](#), [Cynthia Ronkowski](#), [Rodolfo Brizola Toscan](#), [Berk Turhan](#), [Michael Toomey](#), [Juan Sebastian Andrade-Martínez](#), [Chenlian Fu](#), [Alexander G Lucaci](#), Arthur Henrique Barrios Solano, João Carlos Setubal, James R Henriksen, Sam Zimmerman, Malika Kopbayeva, Anna Noyvert, Zana Iwan, Shraman Kar, Nikita Nakazawa, Dmitry Meleshko, Dmytro Horyslavets, Valeriia Kantsypa, Alina Frolova, Andre Kahles, David Danko, Eran Elhaik, Pawel Labaj, Serghei Mangul, The International MetaSUB Consortium, Christopher E Mason , Iman Hajirasouliha 

Author Notes

NAR Genomics and Bioinformatics, Volume 8, Issue 1, March 2026, lqaf172, <https://doi.org/10.1093/nargab/lqaf172>

Published: 16 January 2026 **Article history** ▾

Article Contents

Abstract

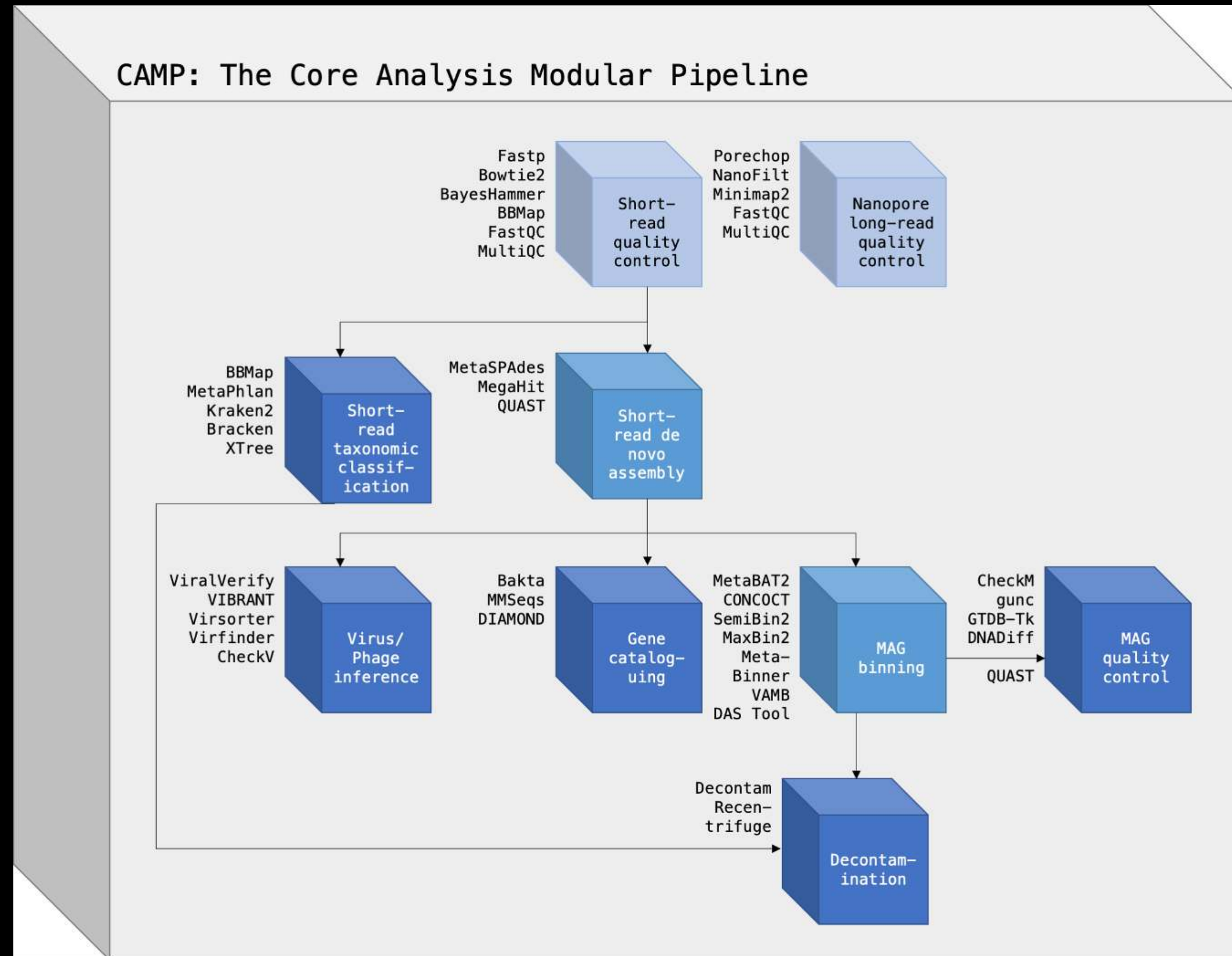
Introduction

Methods

Volume 8, Issue 1
March 2026

CAMP Modules: A Standardized Architecture for End-to-End Metagenomics Workflows

- *The CAMP ecosystem is a collection of modular analysis components that share a common internal structure while wrapping different algorithms for preprocessing, assembly, taxonomic classification, quality assessment, and functional profiling.*



Why This Actually Matters to You

We don't need to sample people to protect people



[Home](#) > [Science Magazine](#) > [8 June 2012](#) > Hood, 336 (6086): 1209

Article Views

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VERSION HISTORY

► [336/6086/1209](#)

Published Online June 6 2012

Science 8 June 2012:

Vol. 336 no. 6086 p. 1209

DOI: 10.1126/science.1225475

EDITORIAL

Tackling the Microbiome

Leroy Hood

36%
of the small molecules
in your blood
transit the microbiome

nature immunology

Proceedings of the National Academy of Sciences of the United States of America

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NATURE IMMUNOLOGY |

Lymphotoxin α enables diet-induced

Vaibhav Upadhyay, Valeriya
Alexei V Tumanov, Ekaterina
Hong Tang & Yang-Xin Fu

[Affiliations](#) | [Contributions](#)

Current Issue > vol. 109 no. 2 > Sharon Greenblum, 594–599, doi: 10.1073/pnas.1116053109

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Performing your original search "microbiome disease links science" in PNAS retrieves 80 additional results.

Metagenomic systems biology of the human gut microbiome reveals topological shifts associated with obesity and inflammatory bowel disease

Sharon Greenblum^a, Peter J. Turnbaugh^b, and Elhanan Borenstein^{a,c,d,1}

[Author Affiliations](#)

- Acne
- Antibiotic-associated diarrhea
- Asthma/allergies
- Autism
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- Depression and anxiety
- Diabetes
- Eczema
- Gastric ulcers
- Hardening of the arteries
- Inflammatory bowel diseases
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- Obesity

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NATURE | LETTER

Diet rapidly and reproducibly alters the human gut microbiome

Lawrence A. David, Corinne F. Maurice, Rachel N. Carmody, David B. Gootenberg, Julie E. Button, Benjamin E. Wolfe, Alisha V. Ling, A. Sloan Devlin, Yug Varma, Michael A. Fischbach, Sudha B. Biddinger, Rachel J. Dutton & Peter J. Turnbaugh

[Affiliations](#) | [Contributions](#) | [Corresponding author](#)

Nature (2013) | doi:10.1038/nature12820

Received 18 April 2013 | Accepted 29 October 2013 | Published online 11 December 2013

learn.genetics.utah.edu/content/microbiome/disease/



Human Microbiome Consortium

Bacterial cells can help tumors

REPORT

Potential role of intratumor bacteria in mediating tumor resistance to the chemotherapeutic drug gemcitabine

Leore T. Geller^{1,*}, Michal Barzily-Rokni^{2,*}, Tal Danino^{3,†}, Oliver H. Jonas^{4,5}, Noam Shental⁶, Deborah Nejman¹, Nancy Gavert¹, Yaara Zwang¹, Zachary A. Cooper^{7,8,‡}, Kevin Shee², Christoph A. Thaiss⁹, Alexandre Reuben⁸, Jonathan Livny², Roi Avraham¹⁰, Dennie T. Frederick¹¹, Matteo Ligorio¹², Kelly Chatman¹³, Stephen E. Johnston², Carrie M. Mosher², Alexander Brandis¹⁴, Garold Fuks¹⁵, Candice Gurbatri¹⁶, Vancheswaran Gopalakrishnan⁸, Michael Kim⁸, Mark W. Hurd¹⁷, Matthew Katz⁸, Jason Fleming⁸, Anirban Maitra¹⁸, David A. Smith², Matt Skalak³, Jeffrey Bu³, Monia Michaud¹⁹, Sunia A. Trauger¹³, Iris Barshack^{20,21}, Talia Golan^{21,22}, Judith Sandbank²¹, Keith T. Flaherty¹², Anna Mandinova^{2,23}, Wendy S. Garrett^{2,19,24}, Sarah P. Thayer²⁵, Cristina R. Ferrone²⁶, Curtis Huttenhower^{2,27}, Sangeeta N. Bhatia^{2,28,29,30,31,32,33}, Dirk Gevers^{2,§}, Jennifer A. Wargo^{7,8}, Todd R. Golub^{34,35,36,¶}, Ravid Straussman^{1,¶,¶}

“Certain bacteria express enzymes capable of metabolizing the cancer chemotherapeutic drug *gemcitabine* into an inactive form...an effect that was reversed by antibiotic treatment in mice. A high percentage of human pancreatic ductal adenocarcinomas contain the culprit bacteria (gammaproteobacteria). ...Efficacy of an existing therapy for this lethal cancer might be improved by co-treatment with antibiotics.”



24% of the drugs with human targets, including members of all therapeutic classes, inhibited the growth of at least one strain *in vitro*

Extensive impact of non-antibiotic drugs on human gut bacteria

Lisa Maier, Mihaela Pruteanu, Michael Kuhn, Georg Zeller[✉], Anja Telzerow, Exene Erin Anderson, Ana Rita Brochado, Keith Conrad Fernandez, Hitomi Dose, Hirotada Mori, Kiran Raosaheb Patil[✉], Peer Bork[✉] & Athanasios Typas[✉]

Nature **555**, 623–628 (29 March 2018)

doi:10.1038/nature25979

[Download Citation](#)

Received: 26 March 2017

Accepted: 08 February 2018

Published: 19 March 2018

Many organisms work on your behalf!

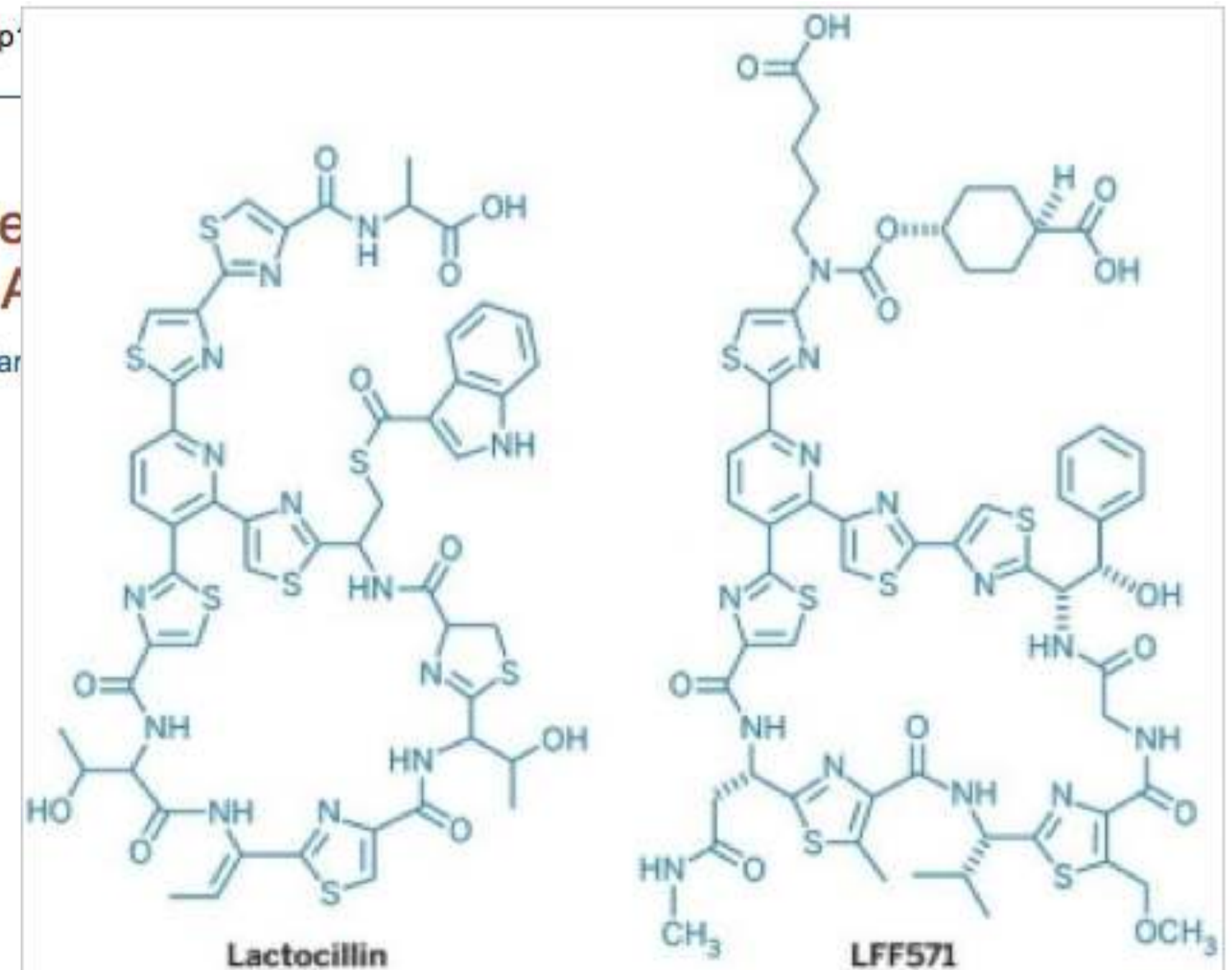
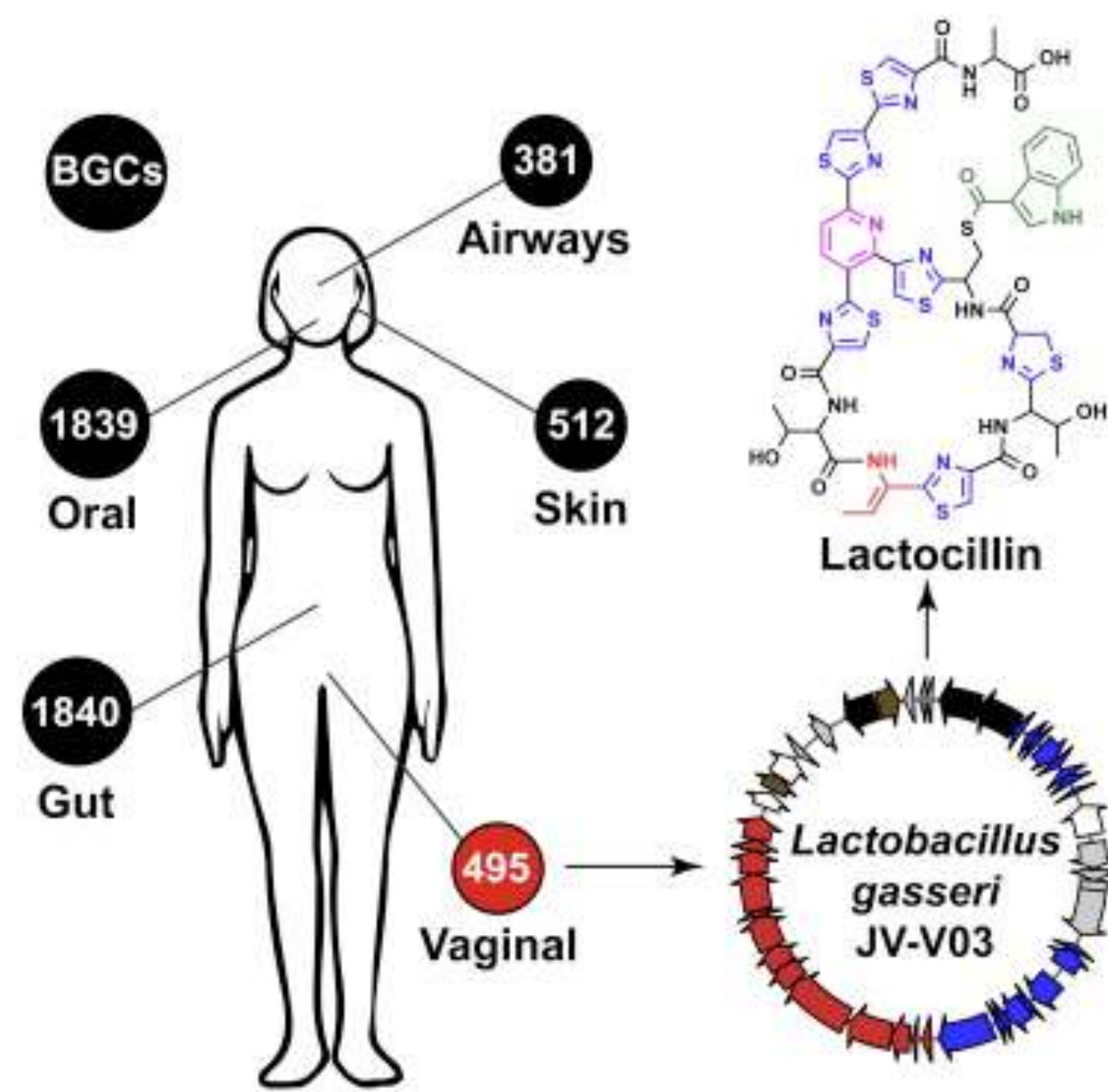
[< Previous Article](#)

Volume 158, Issue 6, p

Article

A Systematic Analysis of Biosynthetic Gene Microbiome Reveals a Common Family of A

Mohamed S. Donia, Peter Cimermanic, Christopher J. Schulze, Laura C. Wielar, Jon Clardy, Roger G. Linington, Michael A. Fischbach  



SELF-MEDICATING

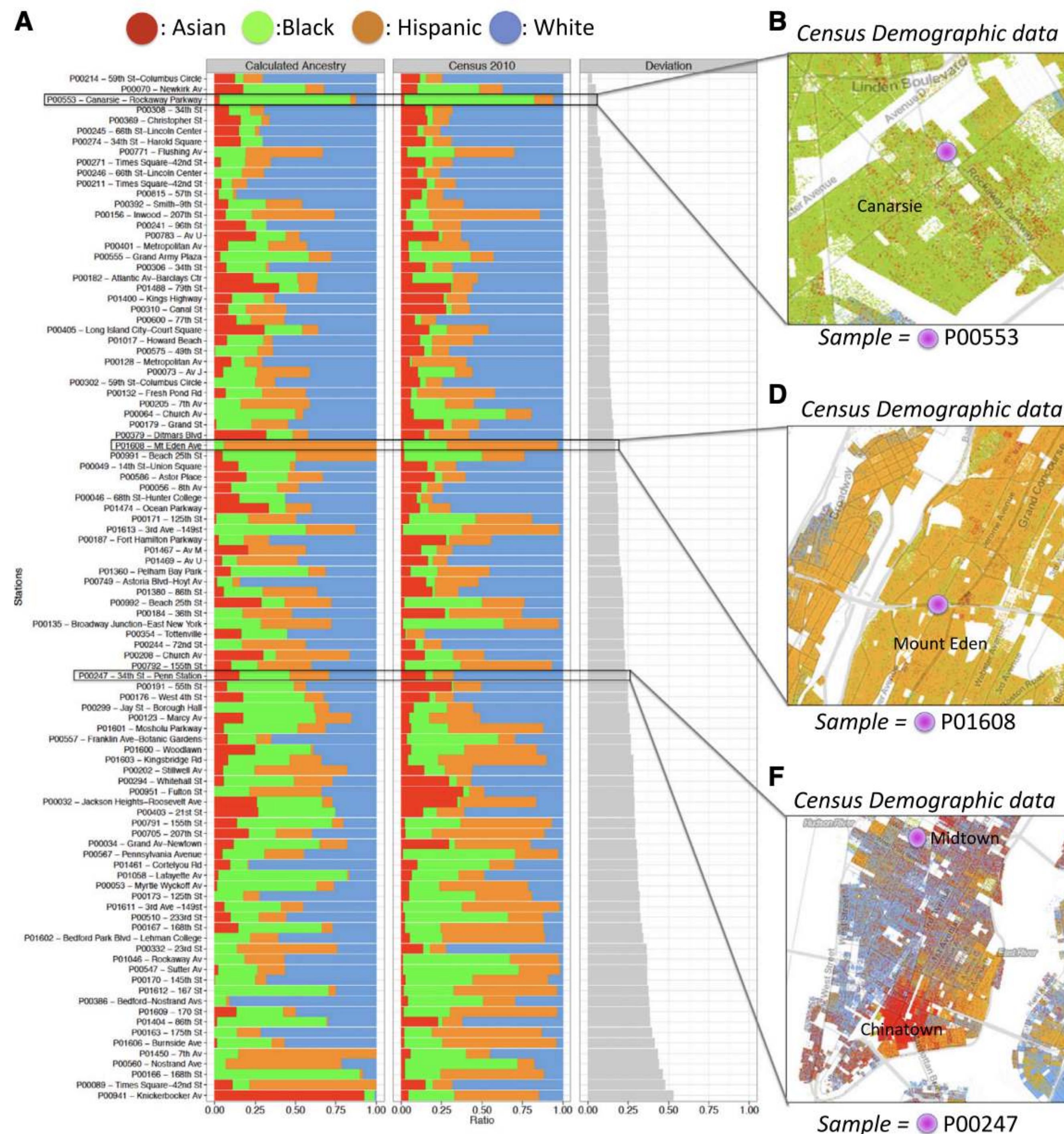
A peptide (left) made by a human microbe bears a resemblance in form and function to a semisynthetic drug candidate (right).

<http://cen.acs.org/articles/92/i39/Mining-Microbiome-Therapeutics.html>

Human Ancestry Predictions from Subway Metagenomes Mirror Census Data

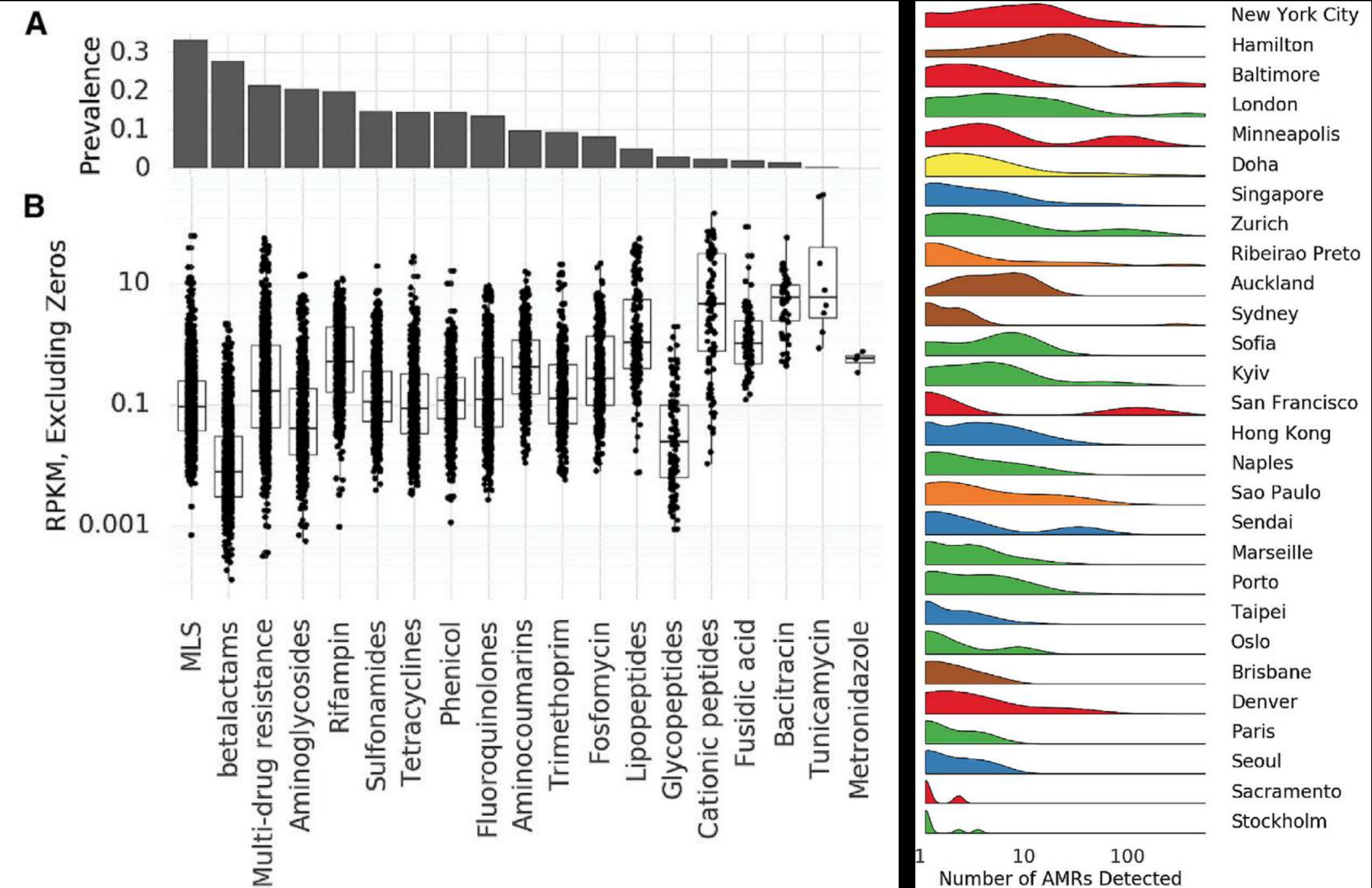
Population-level signals in environmental DNA can reveal where communities may face different disease risks or exposures.

Afshinnkoo, Ebrahim et al. "Geospatial Resolution of Human and Bacterial Diversity with City-Scale Metagenomics." Cell systems vol. 1,1 (2015): 72-87. doi:10.1016/j.cels.2015.01.001

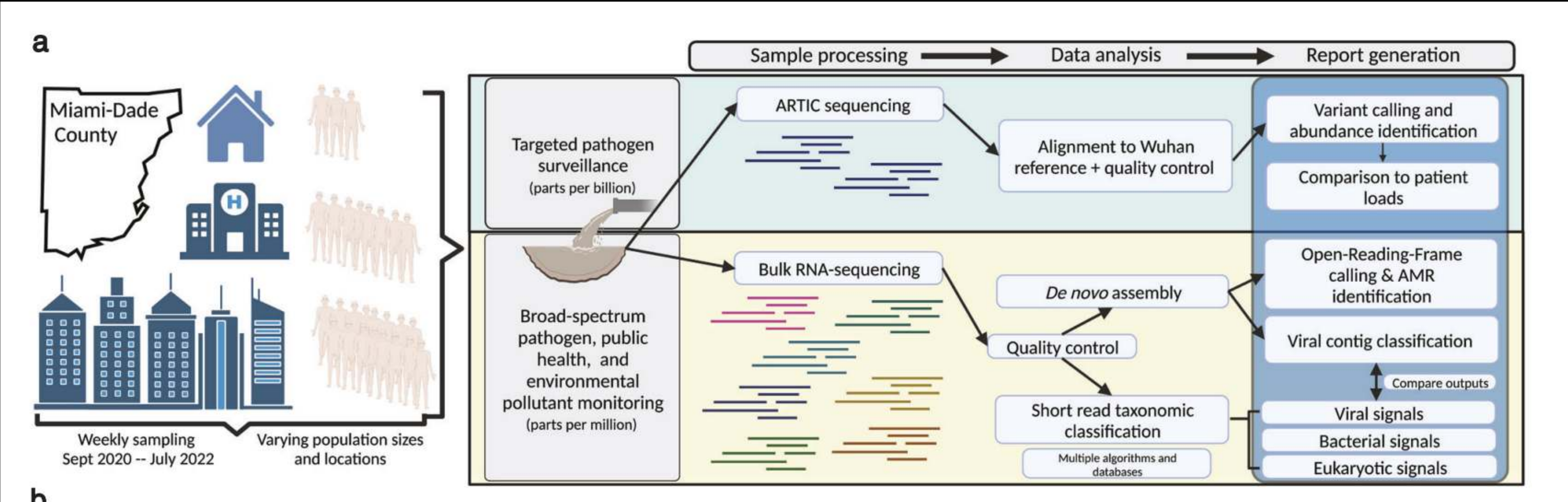


Prevalence of Antimicrobial Resistance (AMR) Genes Across Major Drug Classes

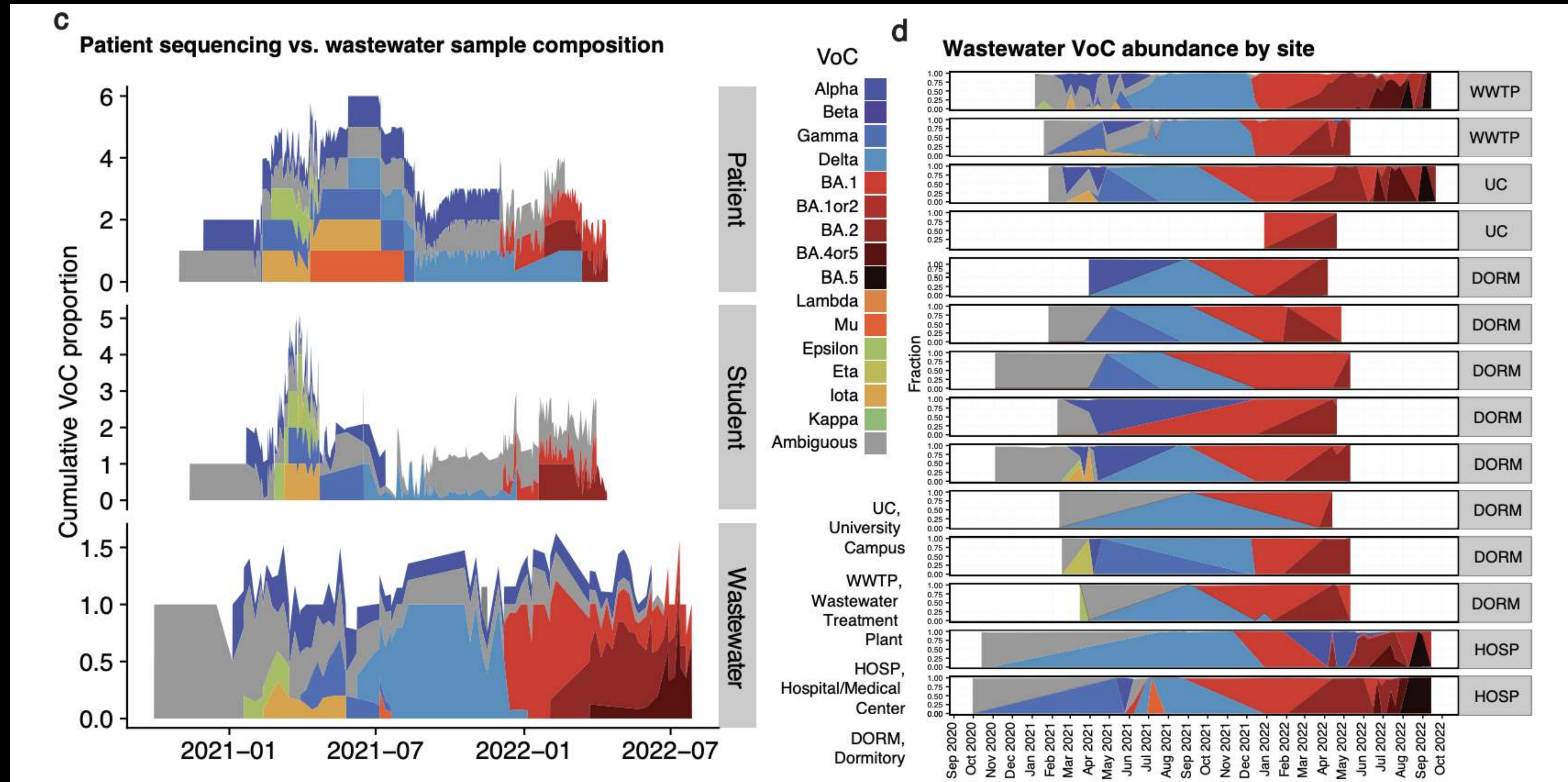
This analysis examines the distribution of AMR genes conferring resistance to particular antimicrobial drug classes.



Urban sampling collected in weekly intervals between 2020 and 2022 from 34 sites within Miami-Dade County.



Variation in wastewater SARS-CoV-2 VoCs across time in different sampling sites



Wastewater Monitoring in the U.S.

*CDC's Wastewater Program collects and displays wastewater (sewage) data from communities across the United States.
(Image: January 2026)*

Wastewater Data Snapshots

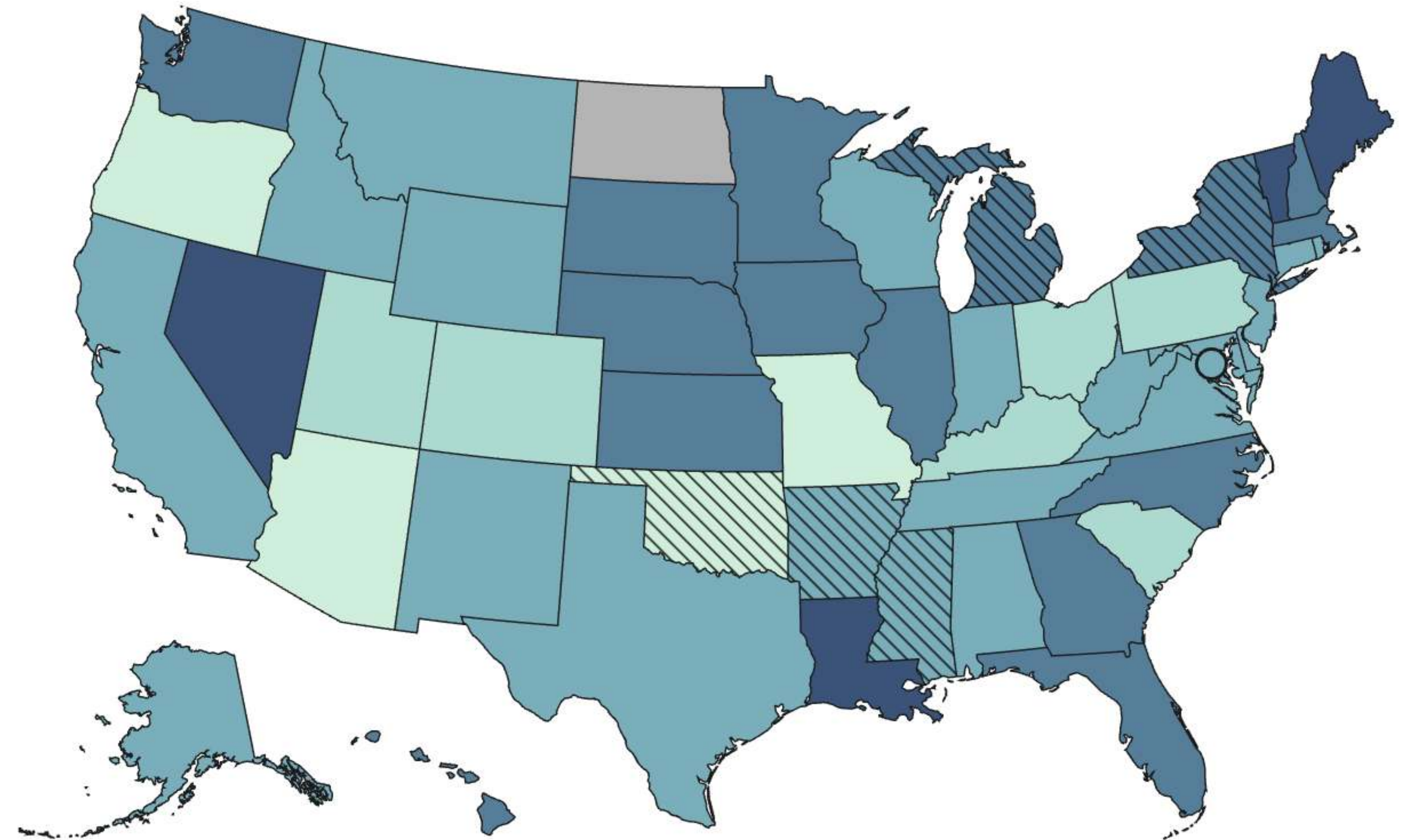
Explore the most recent data for common respiratory pathogens.

Covid-19 Influenza A RSV

Wastewater Viral Activity



*Limited Coverage



U.S. territories

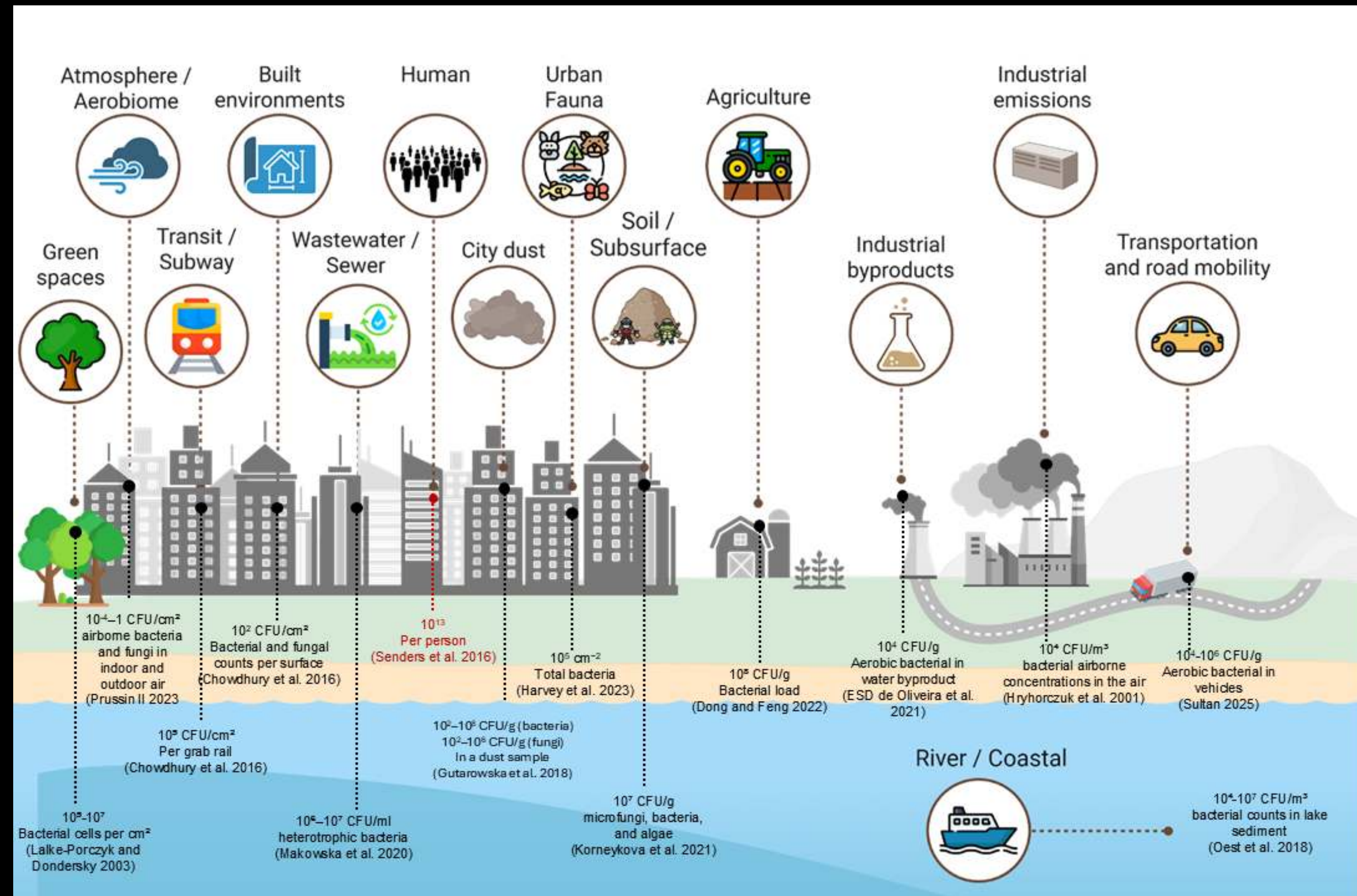


Mapping the Living City



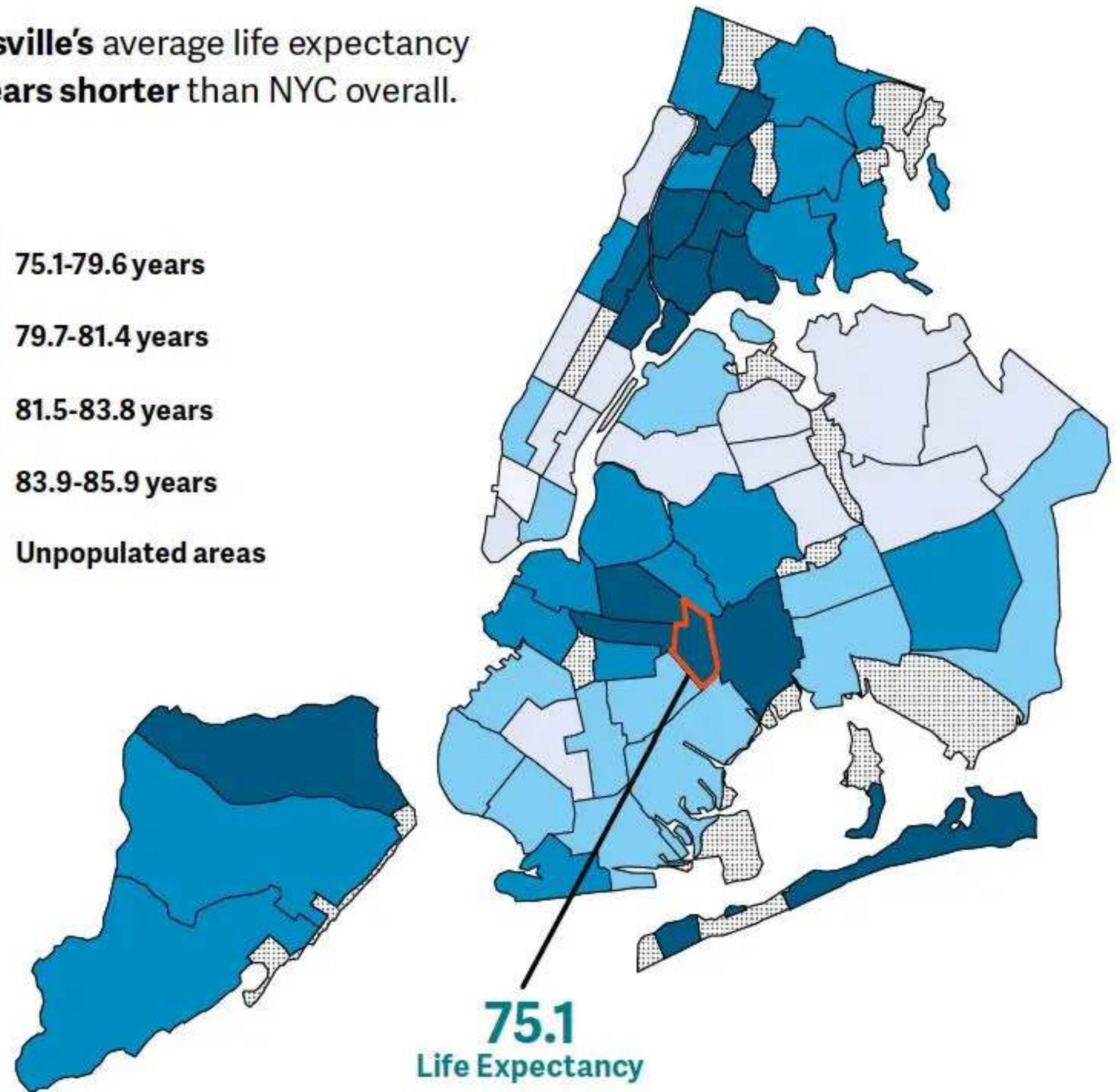
Harnessing the urban microbiome for the future of cities

Cities of the future will be judged not just by air and water quality, but by how well they sustain healthy urban microbiomes



“Urban microbiomes are consequential, measurable, and inseparable from the health of the populations that live within them.”

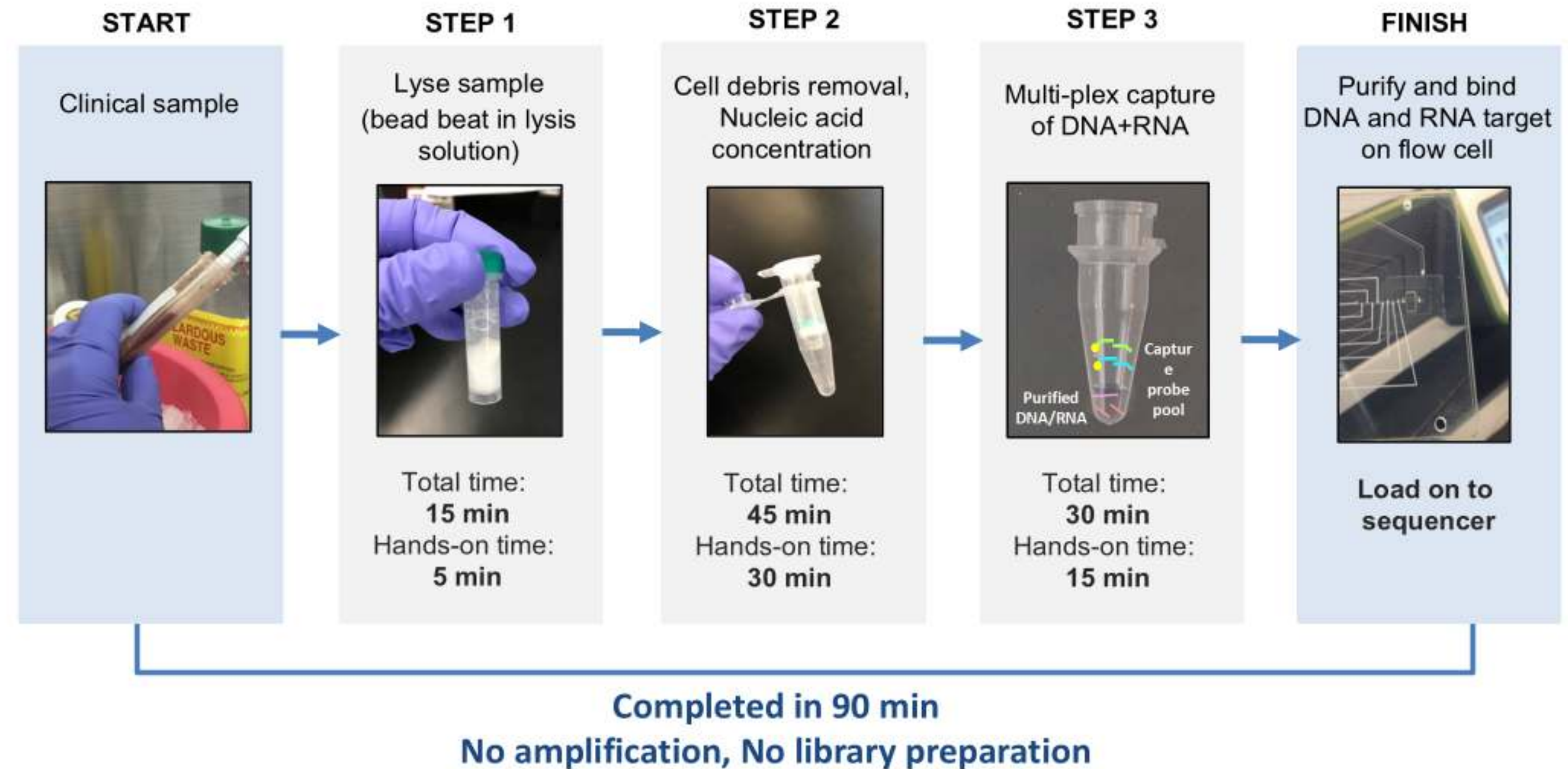
Brownsville’s average life expectancy is **6.1 years shorter** than NYC overall.



Source: NYC DOHMH,
Bureau of Vital Statistics, 2006-2015

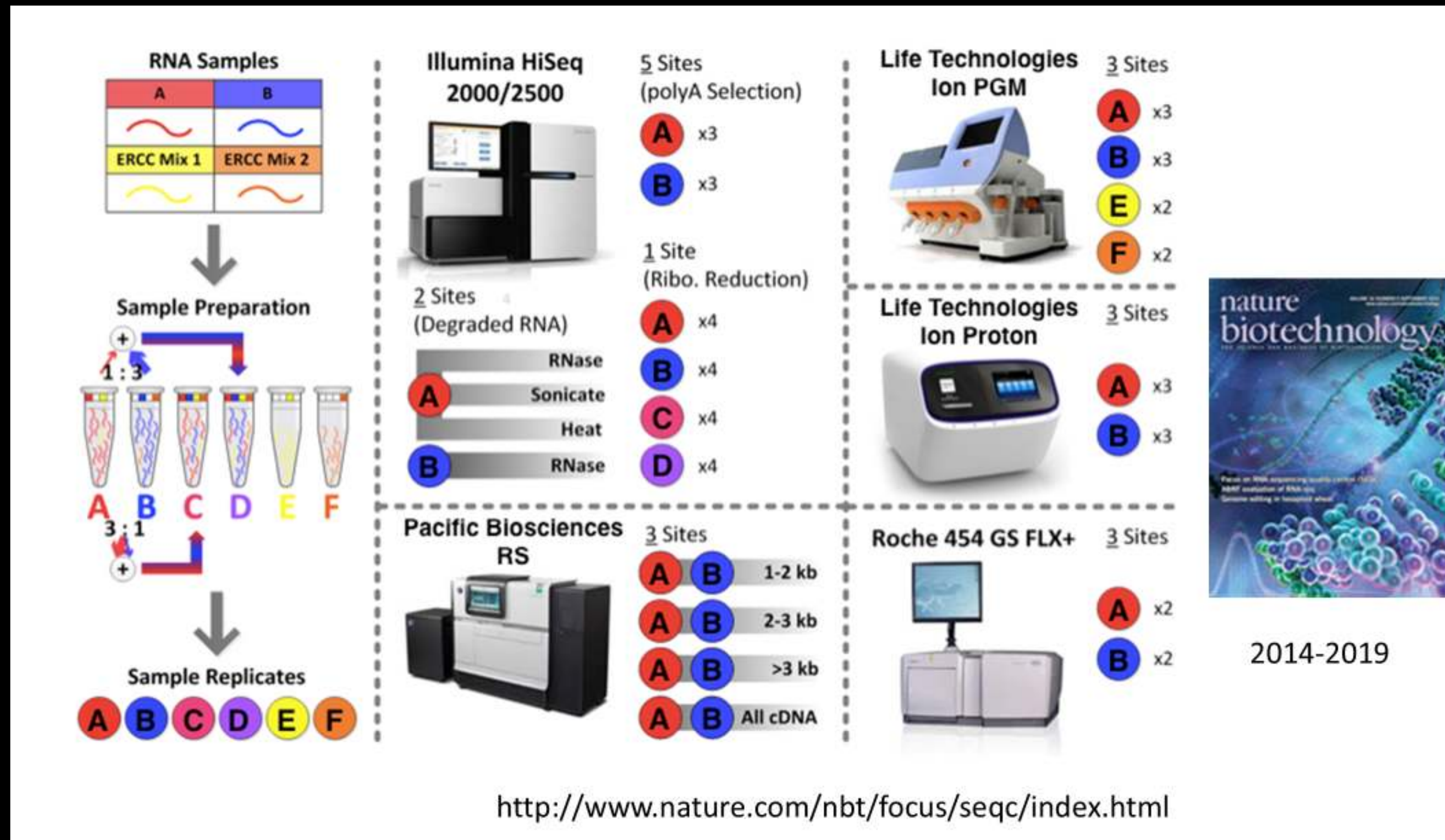
Learning from Clinical Pipelines

Applying high-throughput, standardized clinical sample processing to accelerate and scale microbiome studies.

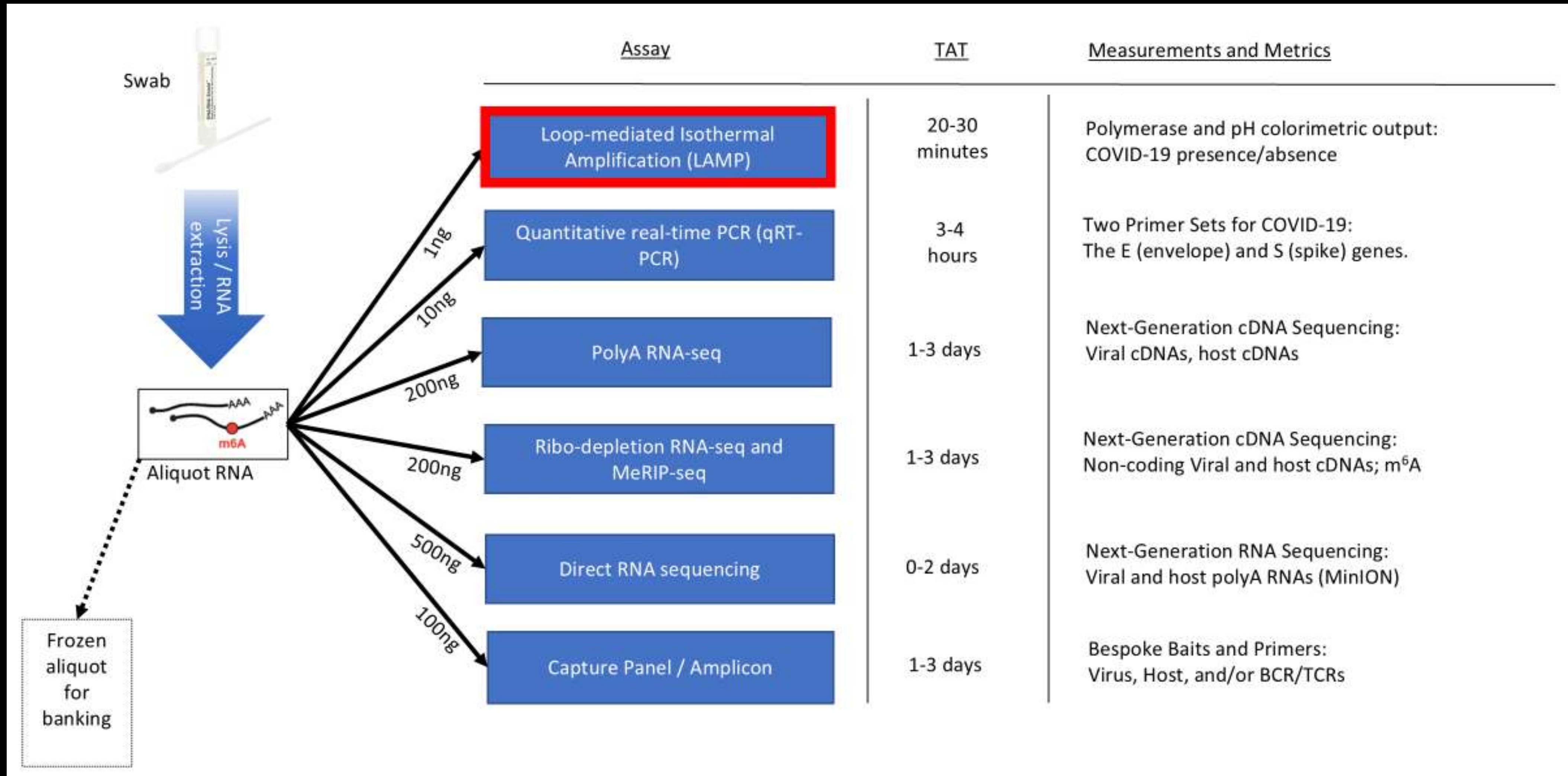


Many Paths to Sequencing

Samples can be profiled using diverse sequencing strategies, each offering different trade-offs in resolution, cost, and biological insight



A Spectrum of Molecular Techniques



Planetary Genomics



Mars Sample Return

Mars Sample Return would be NASA's most ambitious, multi-mission campaign that would bring carefully selected Martian samples to Earth for the first time.

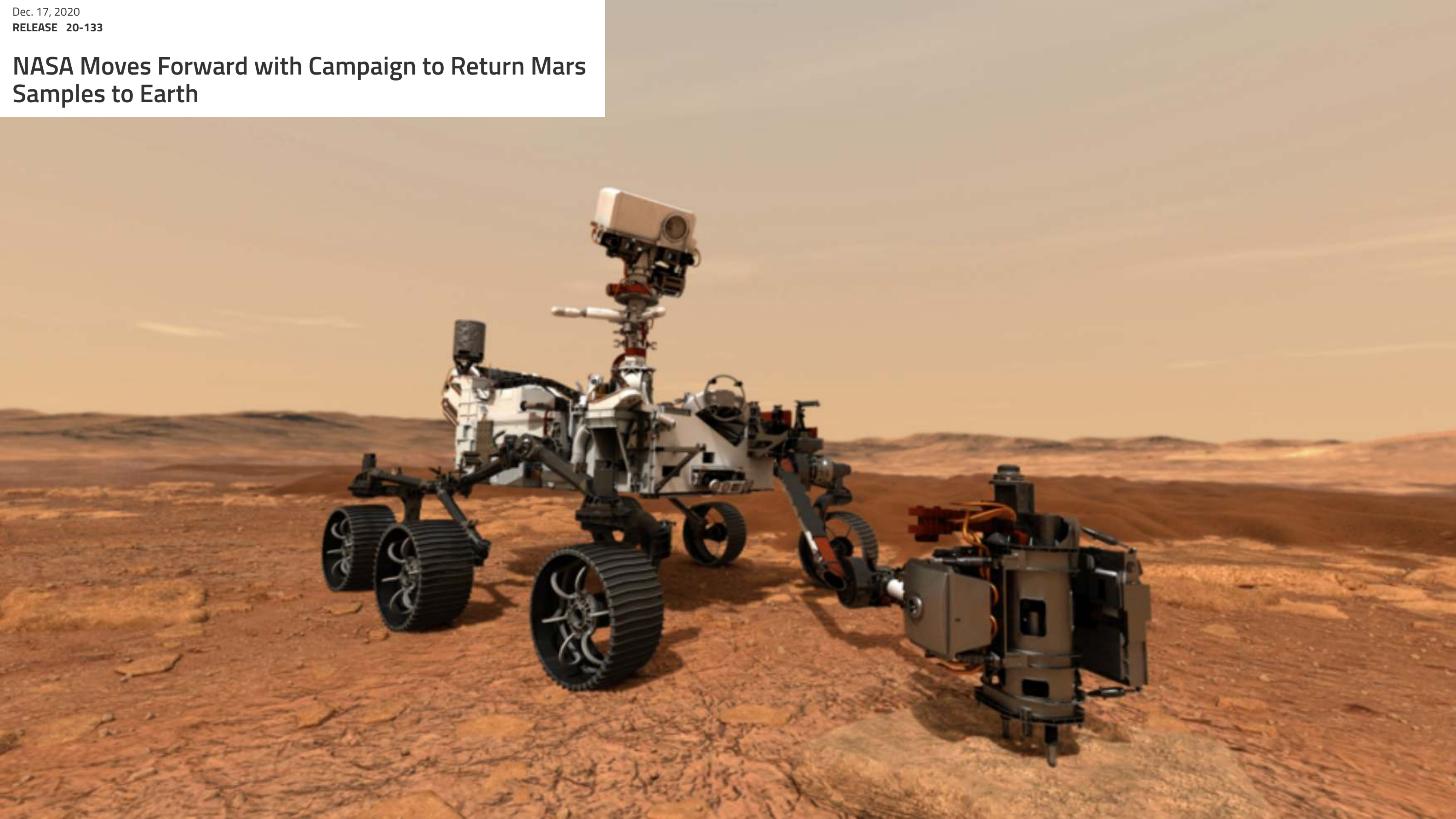
● FUTURE MISSION

[Mission Concept](#) 

[Mars Rock Samples](#) 

[NASA to Explore Two Landing Options](#) 

NASA Moves Forward with Campaign to Return Mars Samples to Earth





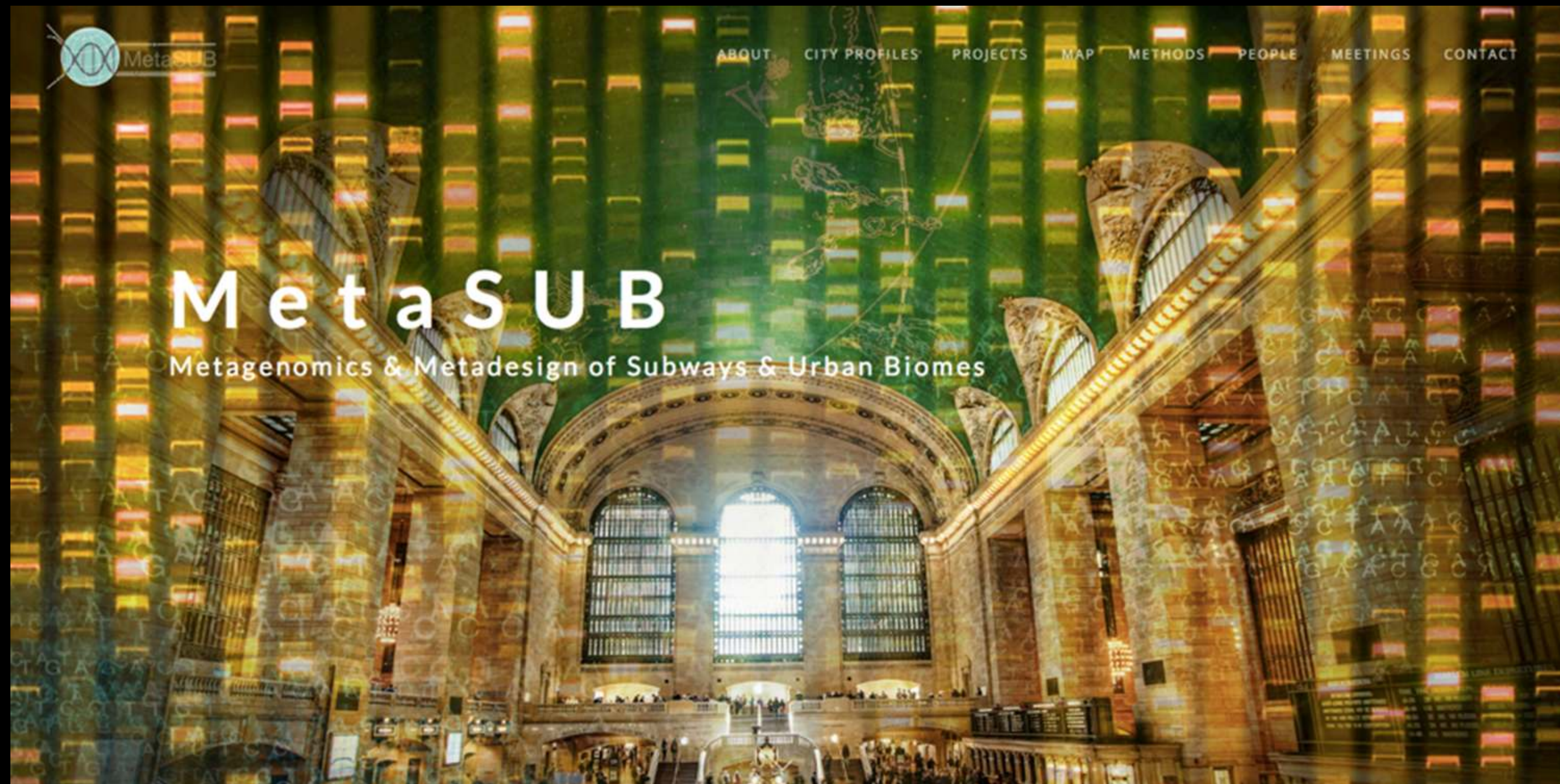
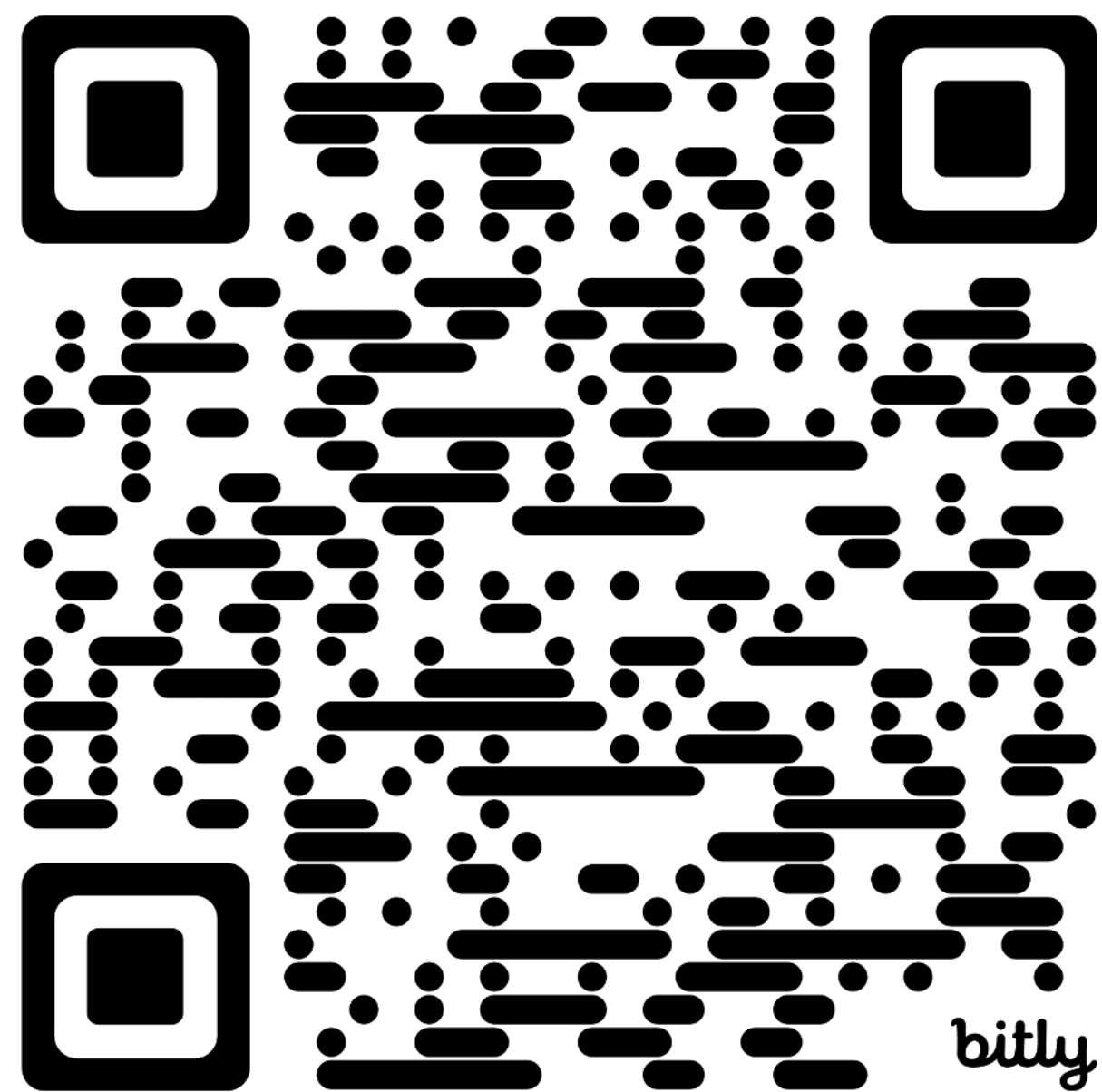
“All science is either physics or stamp collecting”

Ernest Rutherford

Awarded Nobel Prize in 1908 for Chemistry, not Physics.

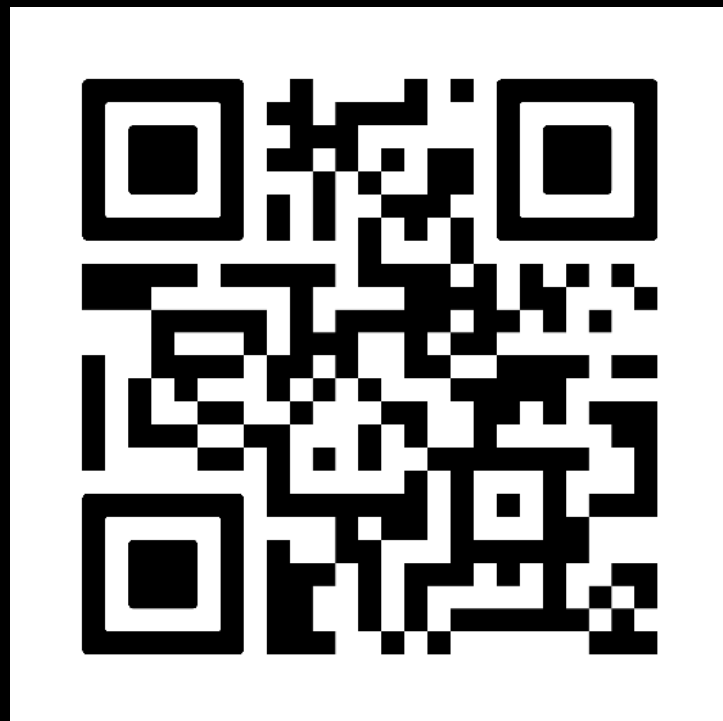
Get involved! Visit metasub.org

<https://bit.ly/MetaSUB>



2026 Global Metagenomics Summit

Registration



Abstract submission



Additional ongoing projects: metasub.org/projects

METACATS: DOMESTIC CAT CORONAVIRUSES & MICROBIOME

The MetaCATS Project aims to study the prevalence of SARS-CoV-2 and other coronaviruses in domestic cats, along with their microbial communities or microbiomes. Studies suggest that cats may be more permissive to the SARS-CoV-2 virus infection, however lack of field studies leave a lack of understanding to the transmissibility of the virus in these animals.

2020 – Present

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OLYMPIOME: OLYMPIC GAMES SAMPLING

As a part of a global consortium to monitor cities, the aim of the project is to pioneer the first-ever, city-scale collection and measure of the Olympics, dubbed the Olympiome. The project will track the localization, transit, and persistence of these visitors' metagenomes and determine where they colonize and change the local urban metagenome of the host city, including the presence and the fluctuations of medically relevant entities such as anti-microbial resistance markers (AMRs) and phages.

2016, 2021

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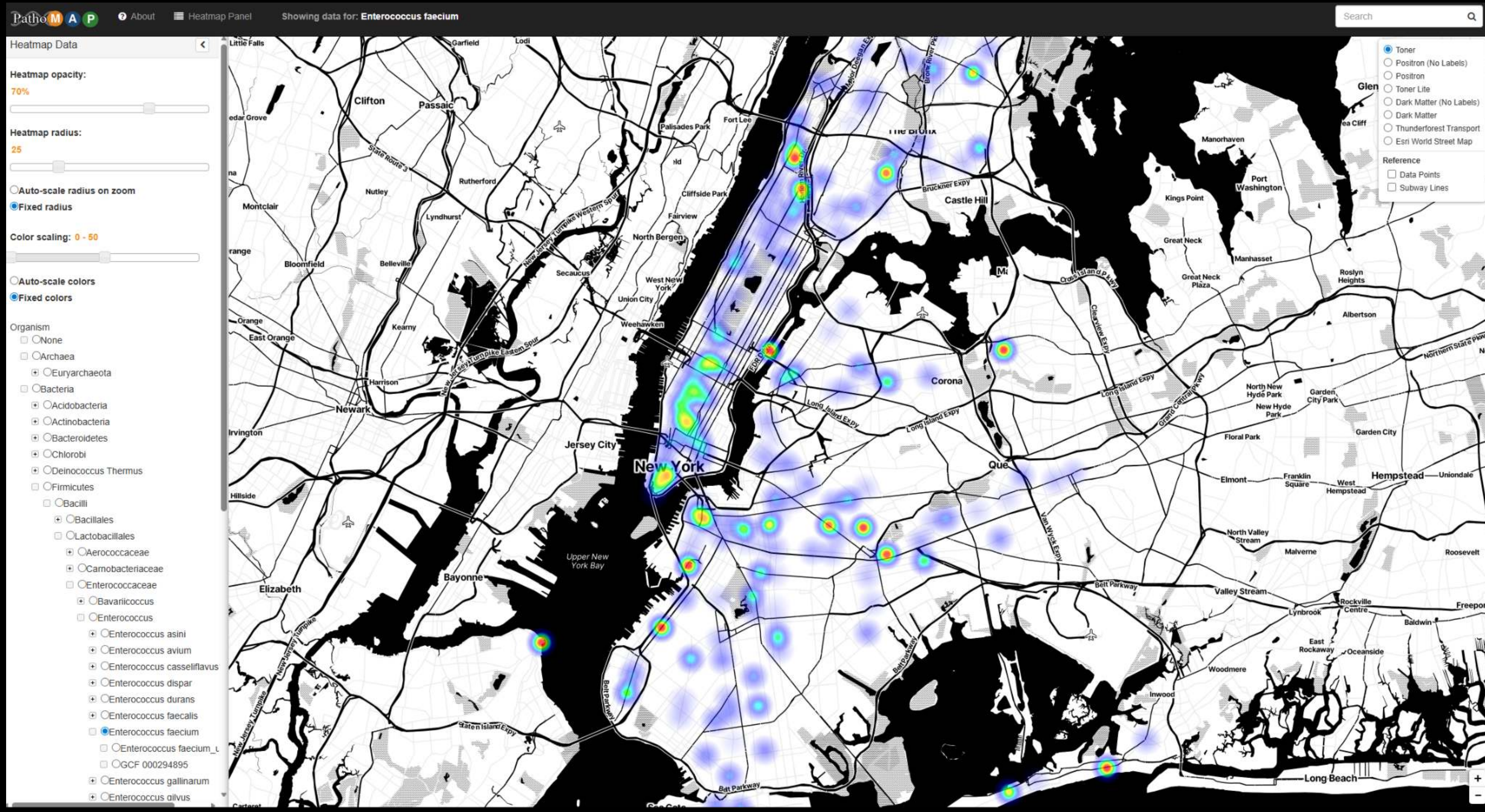
THE EXTREME MICROBIOME PROJECT

The Extreme Microbiome Project (XMP) is a project launched by the Association of Biomolecular Resource Facilities Metagenomics Research Group (ABRF MGRG) that focuses on whole genome shotgun sequencing of extreme and unique environments using a wide variety of biomolecular techniques.

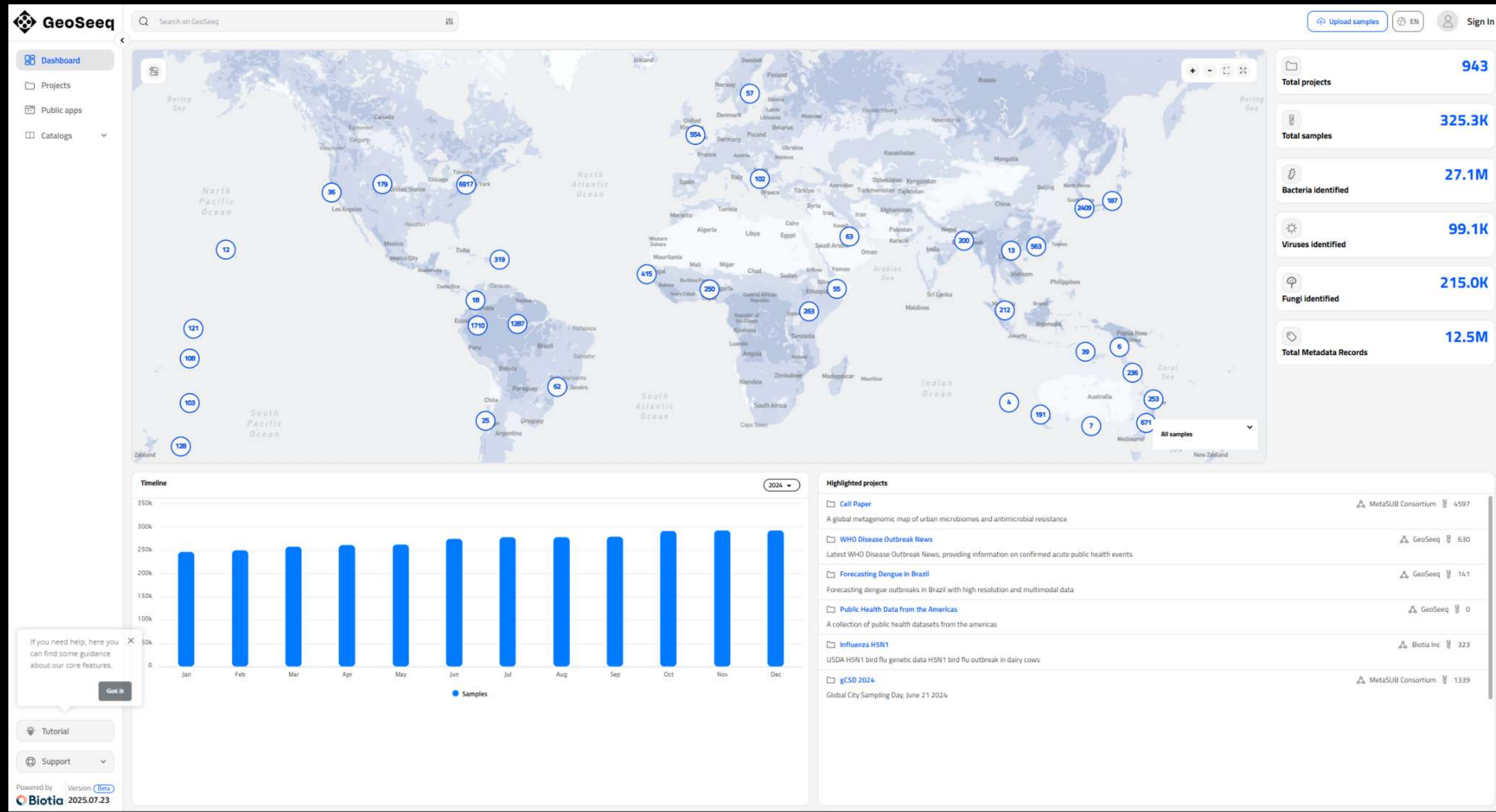
2014 – Present

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Resources (Interactive data): pathomap.org/map

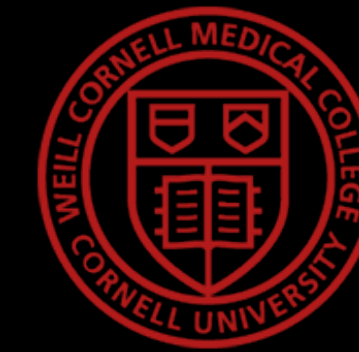


portal.geoseeq.com

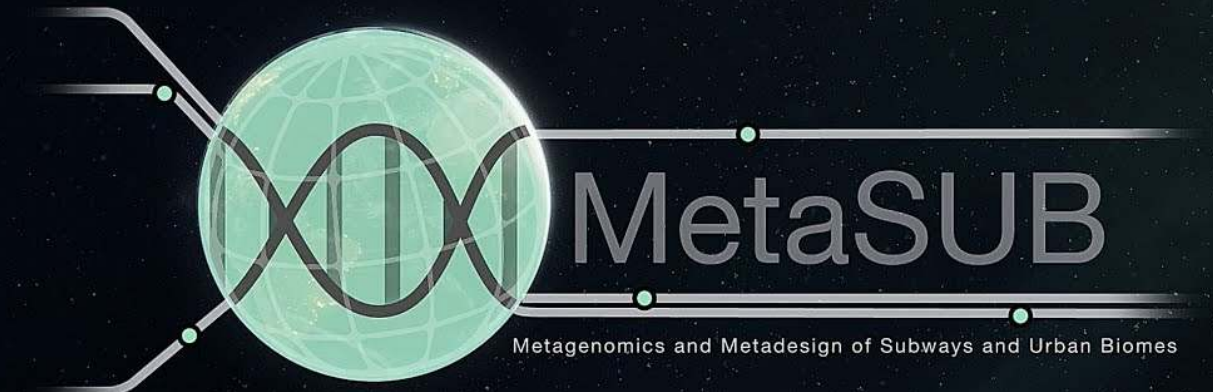


Thank you to all members of the Mason lab!

masonlab.net



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Questions?

Email:

agl4001@med.cornell.edu